

RENANCE

FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE

BIO 104 — Past Questions & Materials

by Renance

Course: BIO 104

Institution: Federal University of Technology, Akure

Sources in this pack: 10

Includes past questions and lecture notes

This study pack was compiled and restructured by Renance for BIO 104 students at Federal University of Technology, Akure. It brings together 10 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

Study online: renance.app

Get the app: Google Play — [dev.renance.renance](https://play.google.com/store/apps/details?id=dev.renance.renance)

Generated by Renance AI

Paper 1 of 10 — 2nd Semester BIO104 Past Questions BIO 104 CBE MCQ Questions with Correct Answers Marked BY

PastQuestion · Google Drive harvest

BIO 104 CBE – MCQ Questions with Correct Answers Marked

1. In an ocean floor where there is little or no penetration which cause darkness, the organisms adapt through?

- A) a large mass
- B) a small body size
- C) well developed sense organs
- D) quick movements

2. Which is not a primary consumer?

- A) grasshopper
- B) lizard
- C) rabbit
- D) snake

3. Thermistor is used to measure _____ in an aquatic environment

- A) pressure
- B) current
- C) depth
- D) temperature

4. Fish belongs to class _____

- A) Pisce
- B) Pisces

5. Nuchal crest is _____ when frog is in an aggressive state

- A) lowered
- B) strong
- C) weak
- D) raised

6. Poison on toad are

- A) relatively few on the gland
- B) abundant on all the glands
- C) there is no poison gland

7. Egg of lizard is covered by _____

- A) coat
- B) layer
- C) shell
- D) gel

8. Snakes mostly live on

- A) Terrestrial
- B) Aquatic
- C) Terrestrial and Aquatic
- D) None

9. Is an example of Mastigophora

Answer: Euglena

10. What's the function of Phytoplankton

- A) Tertiary consumer
- B) Secondary consumer
- C) Primary consumer
- D) Primary producer

11. Does Snake have Teeth?

Answer: Yes

12. What type of Dentition does Lizard have.....

- A) Homodont Dentition
- B) Heterodont Dentition

13. The Hind Limb of Toad is

- A) Short and stout
- B) Long and lanker
- C) Long and stout

14. What do reptiles use for respiration

- A) Lungs ☐
- B) Nostril
- C) Gills
- D) None

15. Diagram of Euglena

Answer: Euglena

16. Diagram of Paramecium

Answer: Paramecium

17. Diagram of Amoeba

Answer: Amoeba

18. The tympanum in frog is located where

- A) Between the head and the body gland
- B) Between the parotoid gland and hind limb
- C) Between the wart
- D) Between the eyes and parotid gland

19. Tilapia Fish reproduce by.....

- A) Asexual reproduction

- B) Sexual reproduction
- C) Egg laying
- D)

20. Which of the following organisms are cartilaginous?

- A) Skates
- B) Rays
- C) Mackerel
- D)

21. Secchi disc is used to check?

- A) Turbidity
- B) Temperature

22. In an ocean floor where there is little or no penetration which cause darkness, the organisms adapt through?

- A) a large mass
- B) a small body size
- C) well developed sense organs
- D) quick movements

23. Lizards are hermaphrodite.

True / False (False)

24. Phytoplanktons serve as..... in aquatic food chain.

Answer: Primary producer

25. Tilapia is an example of - animal

- A) Exoskeleton
- B) Hydrostatic
- C) Vertebrate

26. Which of the following has a slippery skin?

- A) Frog
- B) Toad
- C) Newt
- D) None of the above

27. Fish reproduce through

- A) Sexual reproduction
- B) Asexual reproduction
- C) Egg laying
- D) None of the above

28. The role of phytoplankton in the aquatic food chain?

- A) Secondary consumer
- B) Primary consumer
- C) Primary Producer

D) Decomposer

29. The immature stage of a toad is

A) Terrestrial like the adult

B) Terrestrial with tail

C) Aquatic without tail

D) Aquatic with tail

30. The black spot on a toad is called

Answer: Wart

31. What is the scientific name of Earthworm

A) Lumbricus terrestris

B) Lumbricus gigantea

C) Actina

D) Nereis

32. The poison on toad

A) Relatively few in glands

B) Abundant in glands

C) Absent in glands

D) No poison glands in toads

33. The nuchal crest in lizard is responsible for

A) Movement of head

B) Reproduction

C) Locomotion

34. Diagram of Euglena - what is the function of part a (flagellum)

A) Reproduction

B) Respiration

C) Locomotion

D) Irritability

35. Diagram of Amoeba - what protozoan is this

A) Paramecium

B) Amoeba

C) Euglena

D) Chlamydomonas

36. Diagram of Paramecium - what is the labelled part I

Answer: Anal pore

37. Which of the following is true about snakes

A) Do not have legs for walking

B) Body is streamlined for walking

C) Have legs for walking

38. Which of the following is a rhizopoda

- A) Amoeba
- B) Paramecium
- C) Plasmodium
- D) Euglena

39. What is the specimen in the diagram above

- A) Garden snail
- B) Slug
- C) Holothuroid
- D) Actina

40. Tympanum in toad is located

- A) On the hind limb
- B) Between the forearm and arm
- C) Between the eyes and parotid gland
- D) On the head

41. Which of the following feather is abundant in birds

- A) Contour
- B) Down
- C) Filoplume
- D) Quill

42. Function of Gular Fold in Lizard?

Answer: Display / defense (attack enemies)

43. Habitat where snakes mostly reside?

Answer: Terrestrial

44. Natural habitat of Euglena?

Answer: Aquatic

45. Function of phytoplankton in the sea?

Answer: Producers

46. Function of crown in birds?

Answer: Thermoregulation

47. What phylum does Euglena belong to?

Answer: Protista

48. Streamline body of earthworm is adapted to what kind of habitat?

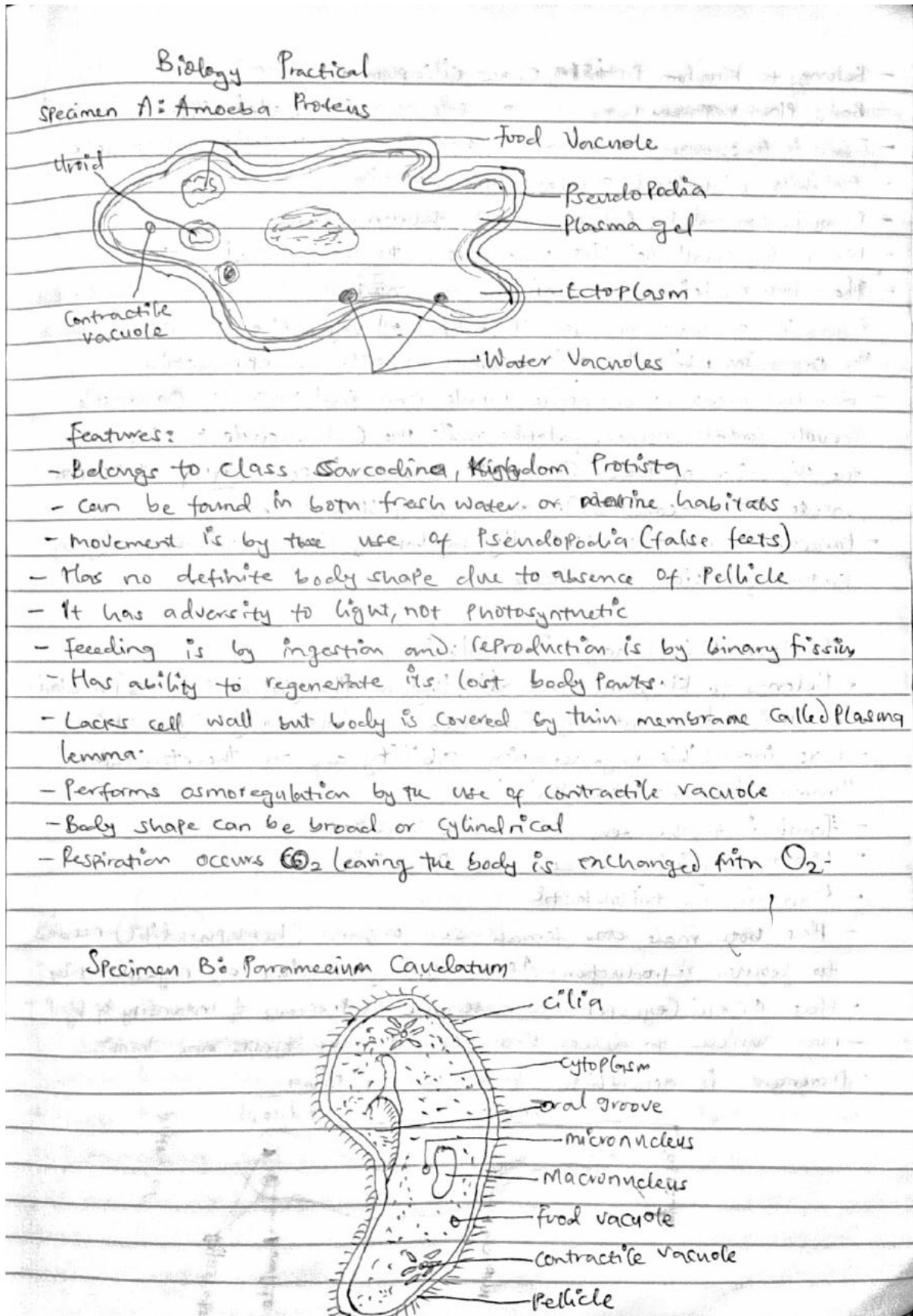
Answer: Narrow burrows and soil

49. Which of the following is an example of Mastigophora?

- A) Amoeba
- B) Paramecium
- C) Plasmodium
- D) Euglena

Paper 2 of 10 — 2nd Semester BIO104 Past Questions BIO 104 Exam.pdf.BY PHOTOCHEM-1.pdf

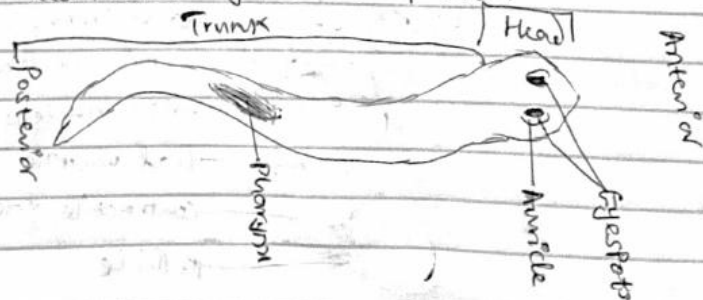
PastQuestion · Google Drive harvest



- Belongs to Kingdom ~~Protista~~ ^{Animalia}, Class Ciliophora
- Body Plan ~~resembles~~ resembles the sole of a shoe (called slipper animalcule)
- Found in freshwater environment and is a single-celled eukaryotic
- Exhibits high-level cellular differentiation
- Body is covered by pellicle and a gelatinous substance
- Moves by small hair-like projections on the body called cilia
- Has two nuclei: macronucleus and micronucleus. The macronucleus is like vegetative nucleus because it controls all vegetative activities/processes in the organism while the micronucleus controls its reproduction
- Has two vacuoles: contractile vacuule and food vacuole. Contractile vacuule controls osmoregulation while the food vacuole participates in the digestion of foods. Contractile vacuule is contractile but the food vacuole is non-contractile and roughly spherical
- Paramecium reproduce asexually by binary fission and sexually by temporary fusion of gametes

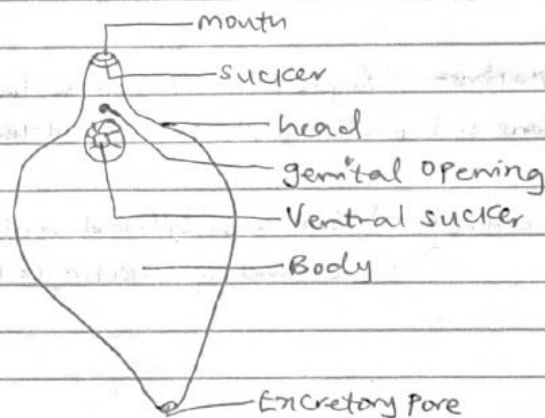
Specimen C: Planaria Turbellaria

- Belongs to Kingdom Animalia, Phylum Platyhelminthes, Class Turbellaria
- Body is dorsoventrally flattened and bilaterally symmetric
- Has incredible regeneration ability and are therefore called "Immortal under the knife"
- Found in freshwaters or marine habitat
- Most Planarian species are Parasitic
- Planaria is triploblastic
- Has both male and female sex organs (hermaphroditic) needed for sexual reproduction. Asexually, they reproduce by regeneration
- Has ocelli (eyespot) to determine the presence & intensity of light
- Has auricles to detect water currents, solid objects and chemicals
- Movement is accomplished by the use of cilia



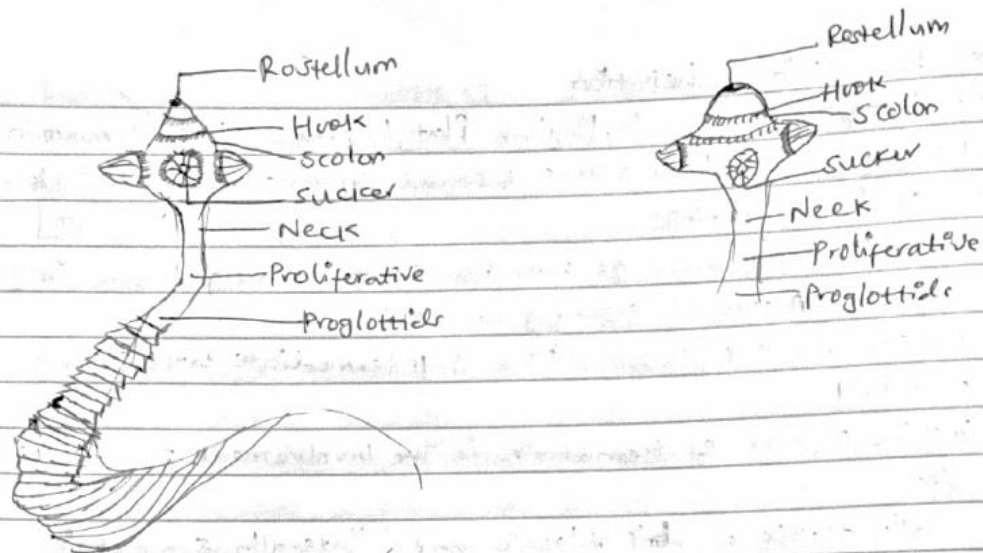
Specimen D: *Fasciola hepatica*

- Belongs to Kingdom Animalia, Phylum Platyhelminthes, Class Trematoda
- Is parasitic and causes liver fluke diseases known as fascioliasis in the liver of human beings [liver rot]
- Occurs in water bodies, such as river banks, slow-moving streams, irrigation channels and affects all grazing animals
- Has 4 complete metamorphosis: egg, intermediate host, young fluke, the adult
- One distinctive feature of trematodes is the involvement of molluscs in their life span.
- dorso-ventrally flattened, triploblastic and bilaterally symmetrical
- unsegmented and acoelomates
- endoparasitic with two suckers and no hooks



Specimen E: *Taenia Solium*

- Kingdom Animalia, Phylum Platyhelminthes, Class Cestoda
- endoparasitic flatworm with pseudometameric segmentation
- Lacks digestive, or nervous tissue
- Maintains attachment & anchorage to host by scolex (hooker & sucker)
- Each body segment is called proglottids
- Each proglottids contains male and female reproductive organs
- Intermediate hosts are cattle or pigs.



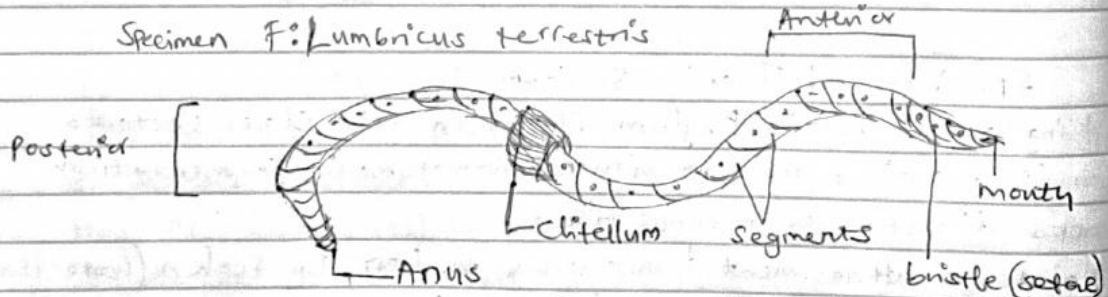
Taenia solium (Pork Tapeworm)

Adaptations

(a) Morphological adaptations → degeneration of organs: (loss of some organs or body parts such as organs of locomotion, tropic organs (nutrition organs), nervous and sensory organs)

(b) Attainment of new organs → shape of body (round or dorsoventrally flattened)
→ development of protective covering

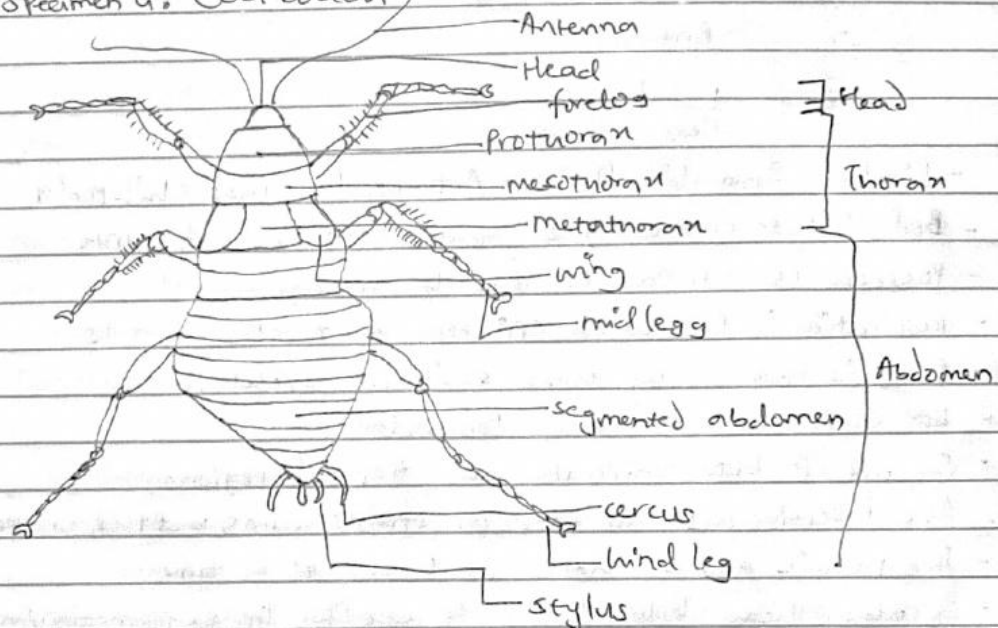
(c)



- Kingdom Animalia, Phylum Annelida, Class Oligochaeta
- Body is metamerically segmented and coelomate
- Body is triploblastic and exists at organ-system level of organization
- Cylindrical body possess two pair of muscle that aids its movement
cylindrical and longitudinal muscles
- Possesses hydrostatic skeleton that aids in burrowing

- bristle hair-like projections found on the body is called setae and helps greatly in anchorage to the soil
- Monoecious, hermaphroditic - Can reproduce sexually or asexually
- Excretion is through metanephridia or protonephridia
- Poor cephalization features and lacks sensory appendages
- called "farmer's friend" because it strongly influence soil fungi
- feeding is detritivorous and feeds on dead leaves in plant soil
- Has a complete digestive system and a closed circulatory system
- Respiration is by diffusion, through the moist skin

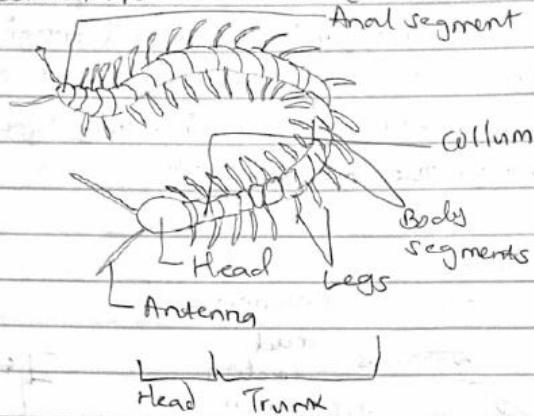
Specimen G: Cockroach



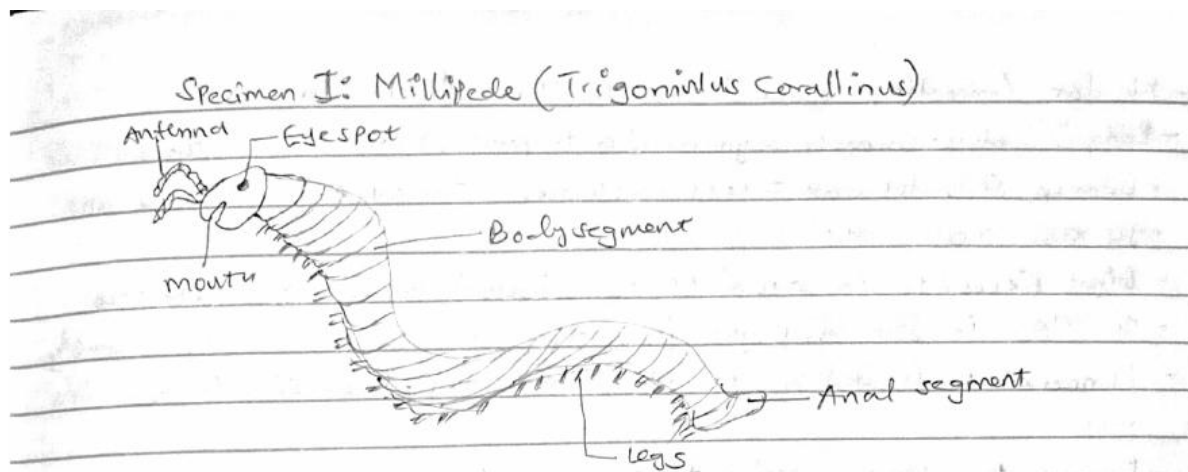
- Kingdom Animalia, Phylum Arthropoda, Class Insecta
- Possesses thick and hard exoskeleton made of chitin
- Body is heteronomously segmented into head, thorax & abdomen
- Mouth parts include Labrum, mandible, maxillae
- Compound eyes to spot predators, antennae to detect chemicals and ocelli
- Are social insects and can be found on almost every habitat
- Open blood circulatory system. Has hemocoel where visceral organs live
- Reproduction is sexual and female lay eggs
- Possess pair of jointed appendages
- Body size is small or reduced to enable faster escape

- Possession of wings for flight
- Respiration is through trachea, gills or lung bodies
- Exoskeleton prevents water loss or desiccation

Specimen H: Centipede (*Scolopendra* species)

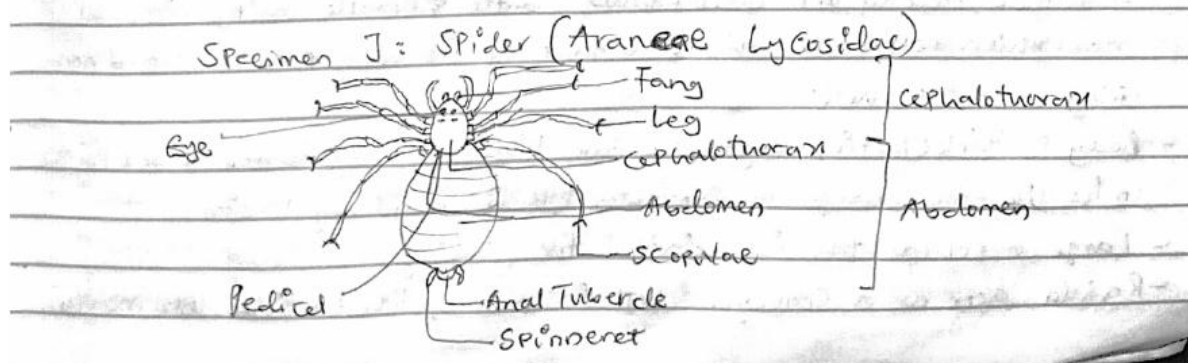


- Kingdom Animalia, Phylum Arthropoda, Class Chilopoda
- Body is heteronomously segmented into head & trunk
- Possesses 15-191 pair of legs in average adult
- Respiration is by paired spiracles on trunk segments
- Body is covered by hard chitinous skeleton
- ~~And~~ Centipedes does have poison claw
- Occurs in both temperate and tropical regions
- Are insectivorous and feed on spiders, worms, beetles, crickets etc.
- Are prolific and can have 60 babies at a time.
- Circular head holds a pair of eye (for lineage formation) and antenna for sensitivity
- Have a simple digestive system and excretion is through a single pair of malpighian tubules
- The first pair of leg are modified into forcipules for injecting venom into prey and the last pair of leg is modified for sexual reproduction. They're called "ultimate legs" and play roles in mating rituals
- Sexual reproduction does not involve copulation, but male drops eggs & ~~female~~ female to keep and nurture



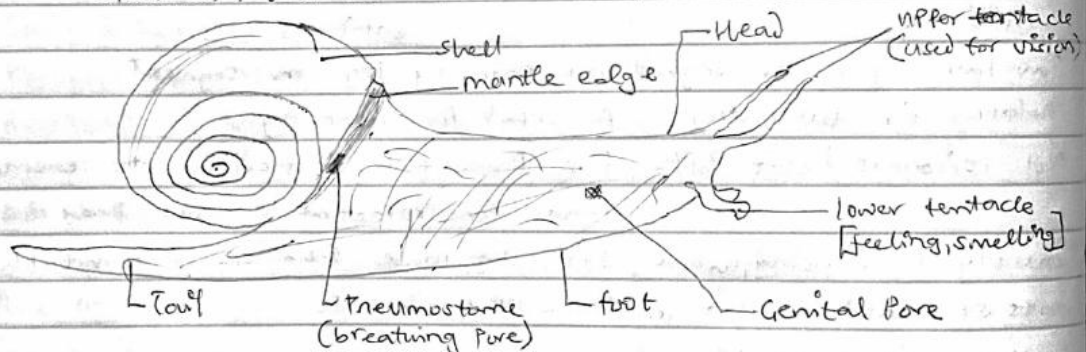
- Kingdom Animalia, Phylum Arthropoda, Class Diplopoda
- Body is segmented into head & trunk and is cylindrical
- Body is covered by chitinous hard exoskeleton
- Are not poisonous and do not bite
- legs can be up to 100 or 750 pieces

Millipede	Centipede
- Two Pairs of leg per segment	one pair of leg per segment
- Adapted for burrowing	Adapted for running
- Not Poisonous, doesn't bite	first pair of leg is modified into venomous fangs for protection against prey
- Sexually reproducing, male inserts spermatophore into female using his gonopods	Sexually, male produces spermatophore
- legs exist below body	legs exist by the body side
- body is cylindrical	- body is flattened
- Herbivorous or detritivorous	- Carnivorous
- black or dark reddish-brown	yellowish grey or brown



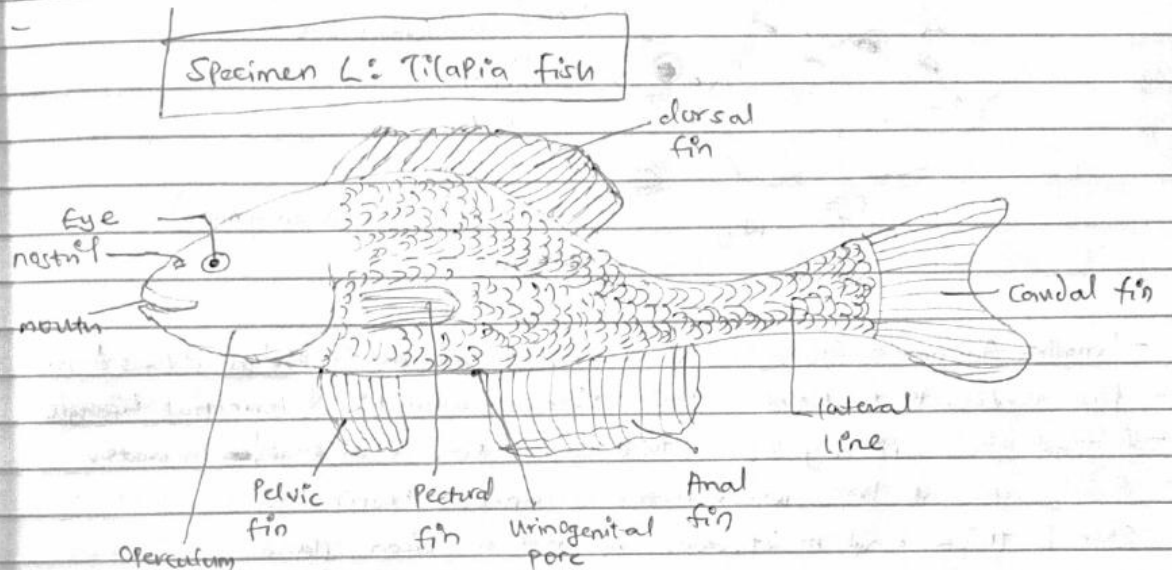
- Kingdom Animalia, Phylum Arthropoda, Class Arachnida
- Body is heteronomously segmented or tagmatized into cephalothorax and abdomen. Cephalothorax = Cephalo (head) + thorax and bears the legs, eyes and mouthparts
- Eight pieces (two in a row) of eyes, body is covered by exoskeleton
- The legs is jointed, eight in number and used for walking & climbing
- Spinneret is located on the abdomen and produces silk for building web
- Possesses two fangs (similar to antenna) located near the mouth
- Feeds on smaller insects (Insectivorous or carnivorous)
- Although possess eye, sensitivity is by perceiving vibrations on its web
- Posses extracellular digestion - releases digestive enzyme on captured prey

Specimen K: Giant African Land Snail



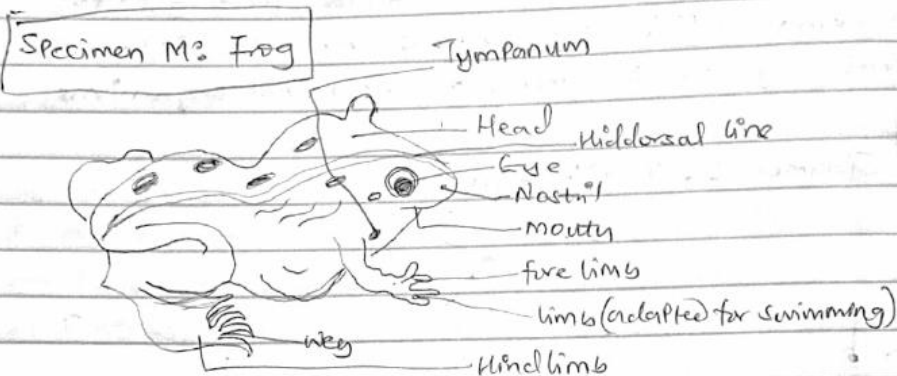
- Kingdom Animalia, Phylum Mollusca, Class Gastropoda (belly foot)
- defining feature is the possession of hard carbonated shell which houses the visceral organs and used for protection.
- Body is heteronomously tagmatized into a large muscular foot, a twisted feeding belt called radula, mantle & mantle cavity, visceral mass which represents delicate organs of its body plus a hard non-living calcareous shell.
- Body is triploblastic, unsegmented and bilaterally segmented at some point in its life time. Changes in symmetry type is caused by torsion
- Large muscular foot is adapted for gliding
- Radula acts as a conveyor belt for moving food particles into mouth

- scientific name is Cornu asperum
- Forwarding of the shell brings the gills (respiratory organ) to the front for better gaseous exchange, makes room for quick retraction and puts the anus over the head, increasing chances of fouling the head
- Pulmonate is the respiratory organ - ends/opens to the Pneumostome
- Show herbivory, filter feeding and Carnivorous feeding Patterns
- Snail is hermaphroditic and elicits (are also protandrous)
- Snails, particularly terrestrial species, are nocturnal organisms

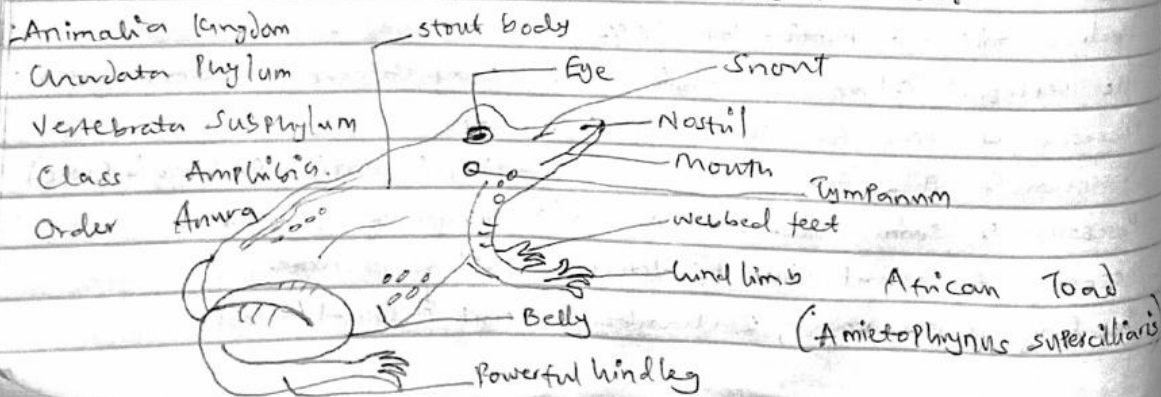


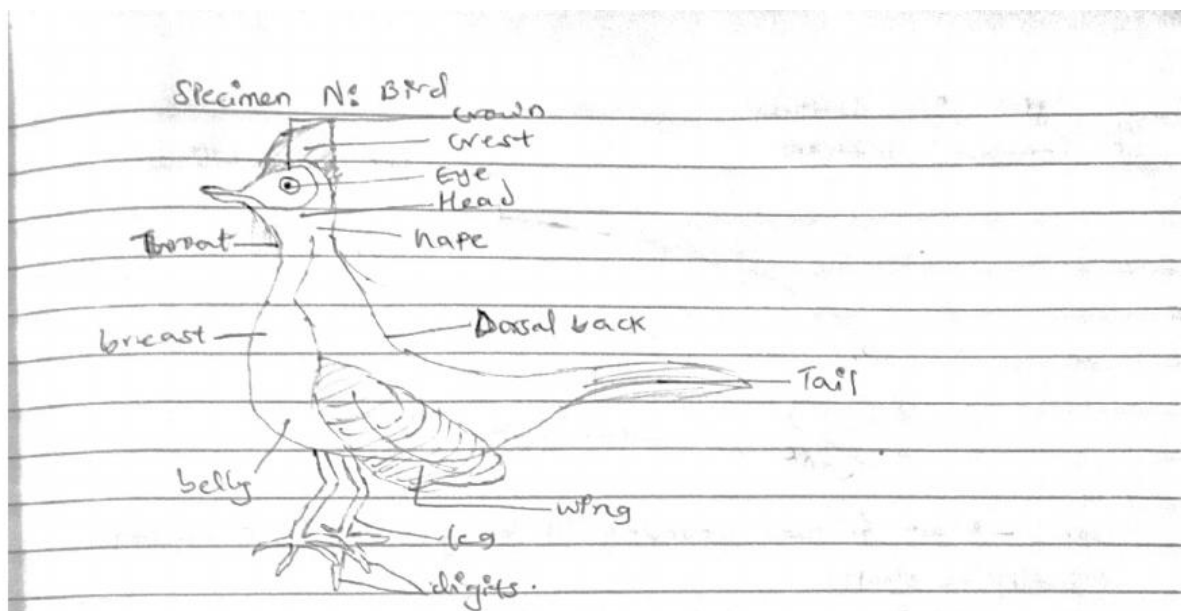
- Kingdom ~~Animalia~~, ~~Subphylum~~ ~~Vertebrata~~ or ~~Craniata~~, ~~Phylum~~ ~~Pisces~~
- ~~Class~~ ~~Osteichthyes~~
- Kingdom Animalia, Subphylum Vertebrata, Phylum Pisces, Superclass Gnathostomata, Class Osteichthyes (bony fishes)
[Jawed fishes]
- Found, lives, exists & reproduce in aquatic lifestyle
- Body is tubular, unsegmented and epidermic secretes slimy mucus which reduces friction in motion by 66%
- The vertebral column is made up of bony calcareous skeletal system
- Possession of fins for swimming in water
- Possession of gills for aquatic respiration, circulation of blood is closed
- Possession of swim bladder to maintain buoyancy in water
- Possession of lateral lines to detect vibrations in water
- Feeding is carnivorous, 2 chambered heart & cold-blooded

- Possession of Paired fins (pelvic & pectoral) and median fins (dorsal, caudal & anal)
- Pair of immovable eyes without eyelid
- Excretion of nitrogenous waste is through the kidney
- Have elaborate mating traditions, reproduction is sexual, copulation is absent, eggs are laid (oviparous) and fertilization is external



- Kingdom Animalia, Subphylum Vertebrata, Phylum Amphibia, Class Anura
- first vertebrate to exhibit dual existence: aquatic & terrestrial lifestyle
- Adapted with respiratory organs that ensure survival on land & in water
- Small inefficient lungs, would prefer cutaneous respiration
- Skin is thick and moist, does not have a poison gland
- Skeletally strengthened for jumping & hopping on land
- can respire through lung, mouth, gill or skin
- 3 chambered heart: 2 atria & 1 ventricle, are cold-blooded
- Sensitivity is by free nerve ending on the skin
- Excretion of ammonia or urea waste is by the kidney while salt and water are excreted through the skin
- Survive winter by hibernation & summer by estivation
- Reproduction is sexual and fertilization is external





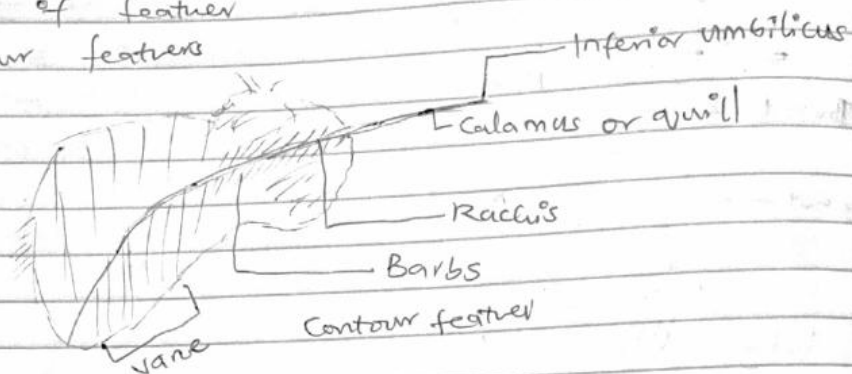
- Animalia (kingdom), Phylum Chordata, Subphylum Vertebrata, Class Aves, Order Carinata (flying birds)
- The entire anatomy of bird is built around its ability to fly.
- Body is covered by fur and wings are modified fore limbs
- Possession of hollow, light, delicate but sturdy bones allows for flight
- Spindle-like body is metameromously segmented into head, neck, trunk and tail
- Possession of Carinate breastbone which allows attachment of flight muscles
- Beak is the main feeding structure, and is adapted to the food type
- Possession of ~~an~~ oil gland (Uropygial gland) at the base of the tail. It secretes a mixture of waxes, lipids, etc for the bird to preen.
- Is a voracious feeder due to high metabolic rates.
- 4 chambered heart and warm-blooded
- Shows great parental care. Reproduction is sexual, fertilization is internal to produce/lay fertilized eggs (oviparity).
- Lives in nests on trees or in crevices

Economic Importance of feathers

- Are used for flight, covers the whole body of aves
- Lightweight, possesses remarkable toughness and tensile strength
- Used for flight, in courtship display, in camouflaging, to provide warmth, to insulate the bird and to keep dry.

Types of feather

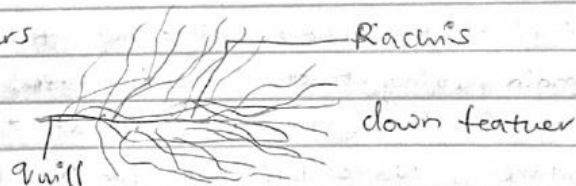
(a) Contour feathers



Uses: - main feather covering the body (giving it a smooth spindle-like shape)

- gives bird its characteristic visual colouring
- provides a first-level defence against physical objects and elements of weather
- are also called flight feathers and the loss of them means the bird won't be able to fly again
- contour feathers are insulated with oil from the preening gland
- also aids in thermoregulation

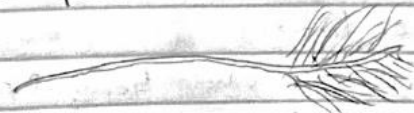
(b) Down feathers



Uses: - traps air needed for insulation against heat loss

- contributes to buoyancy of water birds (e.g. duck, goose etc)
- helps the bird to feel comfort and relaxed
- extremely elastic, lightweight and is three-dimensional

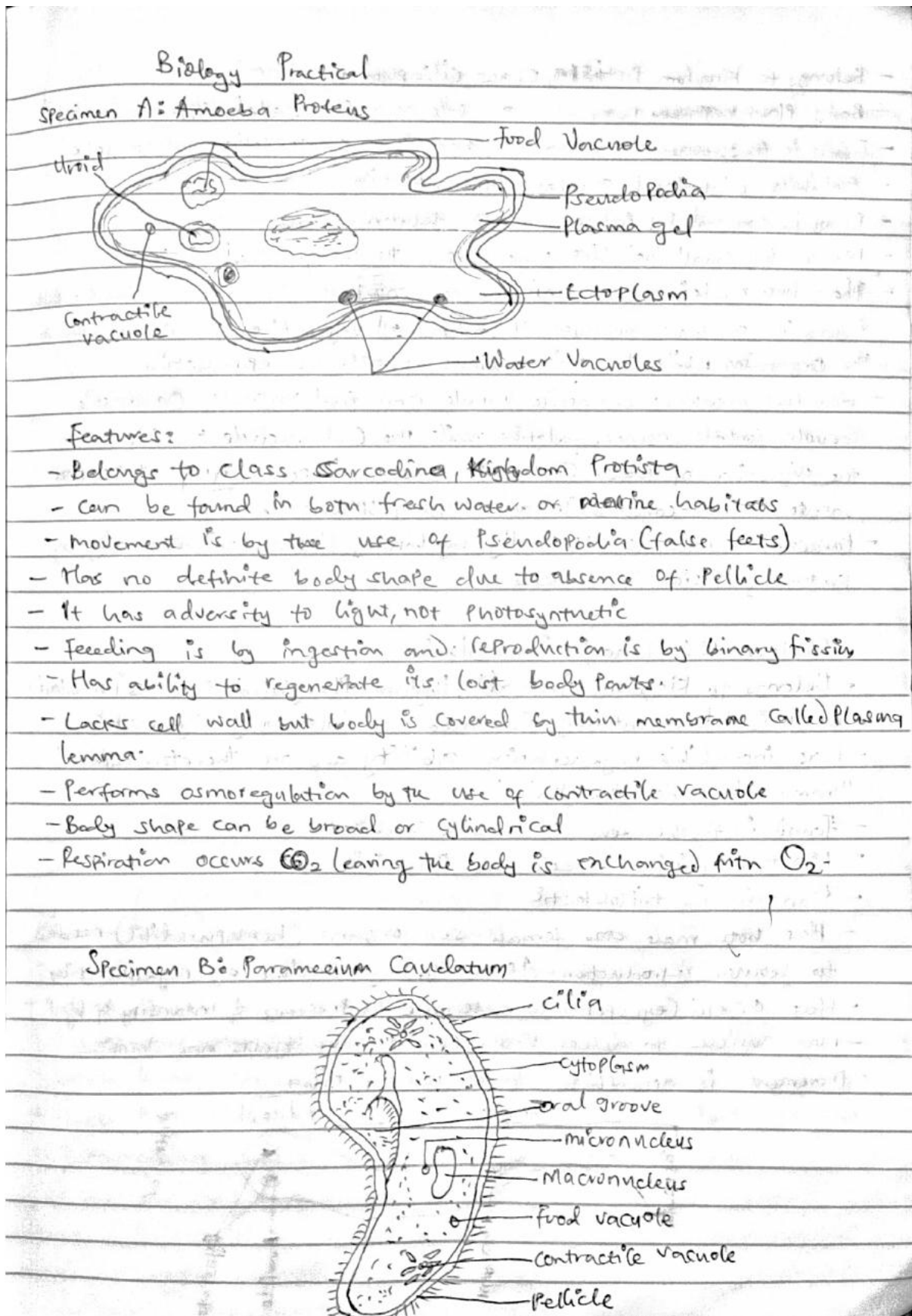
(c) Filoplume feather



- serves as aerodynamic devices
- are hair-like with a few soft barbs near the tip
- associated with contour feathers and may be sensory or decorative

Paper 3 of 10 — BAMBOY 100L 2nd semBIO 104BAMBOY BIO 104 Exam.pdf

Lecture/Notes · Google Drive harvest



- Belongs to Kingdom ~~Protista~~ ^{Animalia}, Class Ciliophora
- Body Plan ~~resembles~~ resembles the sole of a shoe (called slipper animalcule)
- Found in freshwater environment and is a single-celled eukaryotic
- Exhibits high-level cellular differentiation
- Body is covered by pellicle and a gelatinous substance
- Moves by small hair-like projections on the body called cilia
- Has two nuclei: macronucleus and micronucleus. The macronucleus is like vegetative nucleus because it controls all vegetative activities/processes in the organism while the micronucleus controls its reproduction
- Has two vacuoles: contractile vacuole and food vacuole. Contractile vacuole controls osmoregulation while the food vacuole participates in the digestion of foods. Contractile vacuole is contractile but the food vacuole is non-contractile and roughly spherical
- Paramecium reproduce asexually by binary fission and sexually by temporary fusion of gametes

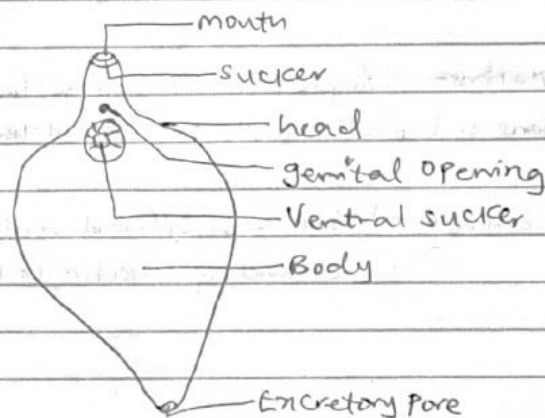
Specimen C: Planaria Turbellaria

- Belongs to Kingdom Animalia, Phylum Platyhelminthes, Class Turbellaria
- Body is dorsoventrally flattened and bilaterally symmetrical
- Has incredible regeneration ability and are therefore called "Immortal under the knife"
- Found in freshwaters or marine habitat
- Most Planarian species are Parasitic
- Planaria is triploblastic
- Has both male and female sex organs (hermaphroditic) needed for sexual reproduction. Asexually, they reproduce by regeneration
- Has ocelli (eyespot) to determine the presence & intensity of light
- Has auricles to detect water currents, solid objects and chemicals
- Movement is accomplished by the use of cilia



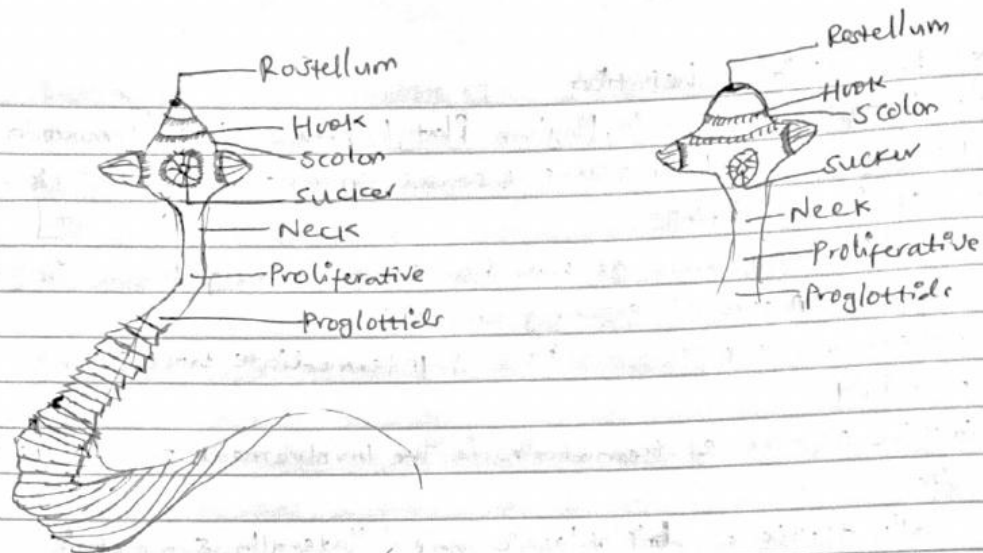
Specimen D: *Fasciola hepatica*

- Belongs to Kingdom Animalia, Phylum Platyhelminthes, Class Trematoda
- Is parasitic and causes liver fluke diseases known as fascioliasis in the liver of human beings [liver rot]
- Occurs in water bodies, such as river banks, slow-moving streams, irrigation channels and affects all grazing animals
- Has 4 complete metamorphosis: egg, intermediate host, young fluke, the adult
- One distinctive feature of trematodes is the involvement of molluscs in their life span.
- dorso-ventrally flattened, triploblastic and bilaterally symmetrical
- Unsegmented and acoelomates
- endoparasitic with two suckers and no hooks



Specimen E: *Taenia Solium*

- Kingdom Animalia, Phylum Platyhelminthes, Class Cestoda
- endoparasitic flatworm with pseudometameric segmentation
- Lacks digestive, or nervous tissue
- Maintains attachment & anchorage to host by scolex (hooker & sucker)
- Each body segment is called proglottids
- Each proglottids contains male and female reproductive organs
- Intermediate hosts are cattle or pigs.



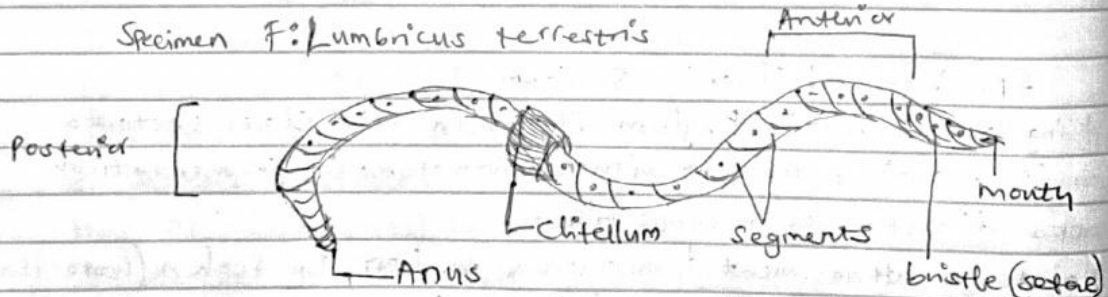
Taenia solium (Pork Tapeworm)

Adaptations

(a) Morphological adaptations → degeneration of organs: (loss of some organs or body parts such as organs of locomotion, tropic organs (nutrition organs), nervous and sensory organs)

(b) Attainment of new organs → shape of body (round or dorsoventrally flattened)
→ development of protective covering

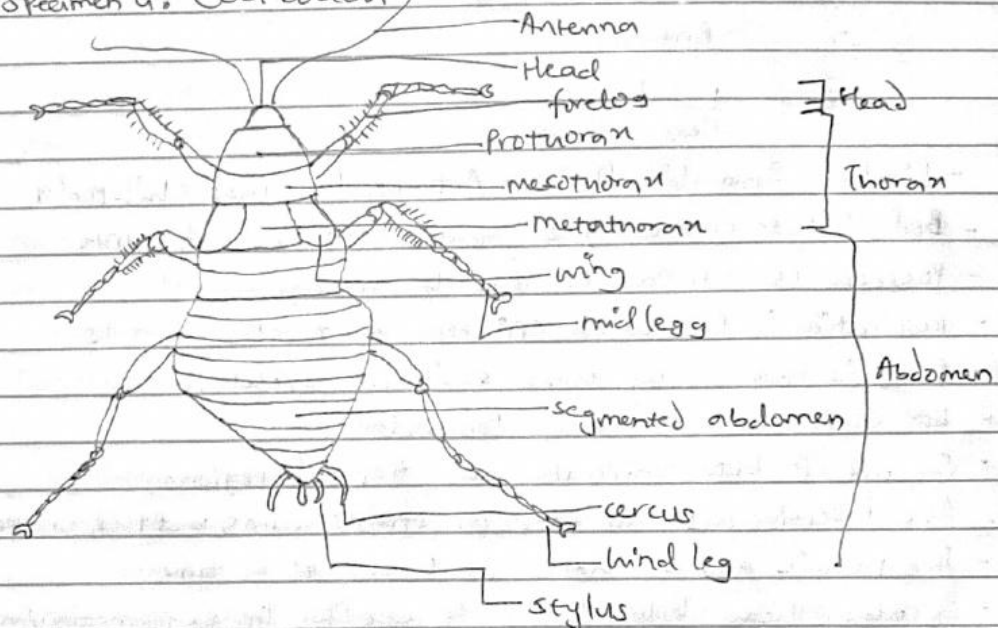
(c)



- Kingdom Animalia, Phylum Annelida, Class Oligochaeta
- Body is metamerically segmented and coelomate
- Body is triploblastic and exists at organ-system level of organization
- Cylindrical body possess two pair of muscle that aids its movement
cylindrical and longitudinal muscles
- Possesses hydrostatic skeleton that aids in burrowing

- bristle hair-like projections found on the body is called setae and helps greatly in anchorage to the soil
- Monoecious, hermaphroditic - Can reproduce sexually or asexually
- Excretion is through metanephridia or protonephridia
- Poor cephalization features and lacks sensory appendages
- called "farmer's friend" because it strongly influence soil fungi
- feeding is detritivorous and feeds on dead leaves in plant soil
- Has a complete digestive system and a closed circulatory system
- Respiration is by diffusion, through the moist skin

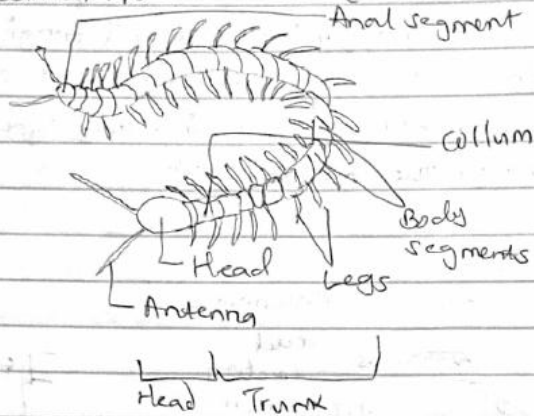
Specimen G: Cockroach



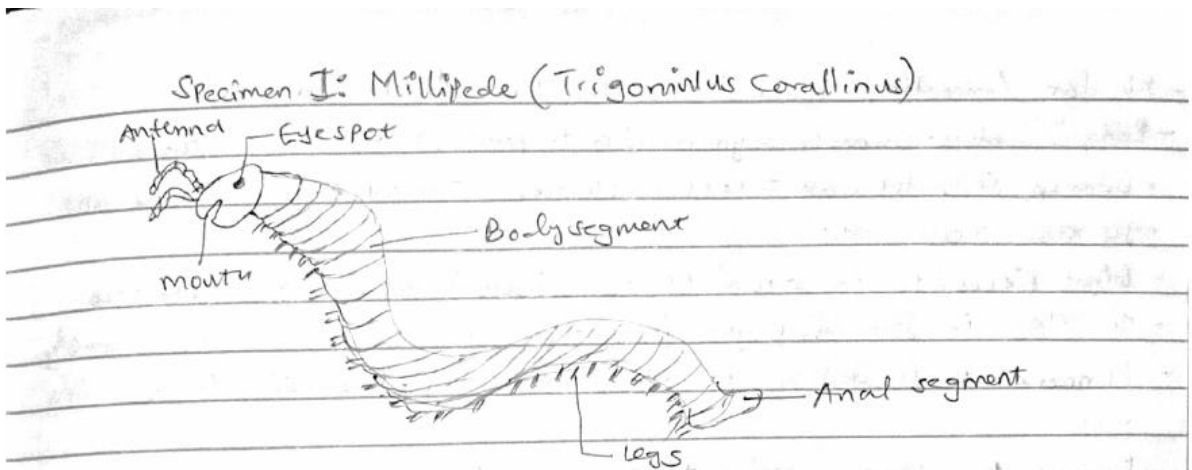
- Kingdom Animalia, Phylum Arthropoda, Class Insecta
- Possesses thick and hard exoskeleton made of chitin
- Body is heteronomously segmented into head, thorax & abdomen
- Mouth parts include Labrum, mandible, maxillae
- Compound eyes to spot predators, antennae to detect chemicals and ocelli
- Are social insects and can be found on almost every habitat
- Open blood circulatory system. Has hemocoel where visceral organs live
- Reproduction is sexual and female lay eggs
- Possess pair of jointed appendages
- Body size is small or reduced to enable faster escape

- Possession of wings for flight
- Respiration is through trachea, gills or lung bodies
- Exoskeleton prevents water loss or desiccation

Specimen H: Centipede (*Scolopendra* specie)

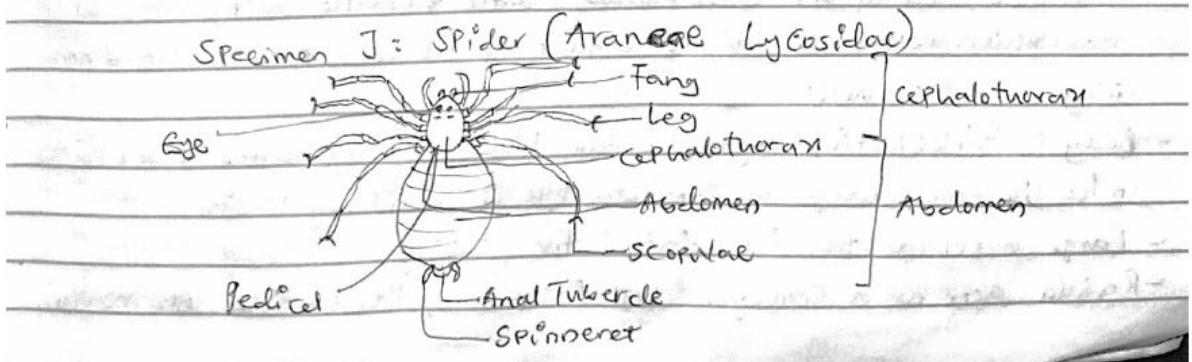


- Kingdom Animalia, Phylum Arthropoda, Class Chilopoda
- Body is heteronomously segmented into head & trunk
- Possesses 15-191 pair of legs in average adult
- Respiration is by paired spiracles on trunk segments
- Body is covered by hard chitinous skeleton
- ~~And~~ Centipedes does have poison claw
- Occurs in both temperate and tropical regions
- Are insectivorous and feed on spiders, worms, beetles, crickets etc.
- Are prolific and can have 60 babies at a time.
- Circular head holds a pair of eye (for lineage formation) and antenna for sensitivity
- Have a simple digestive system and excretion is through a single pair of malpighian tubules
- The first pair of leg are modified into forcipules for injecting venom into prey and the last pair of leg is modified for sexual reproduction. They're called 'ultimate legs' and play roles in mating rituals
- Sexual reproduction does not involve copulation, but male drops eggs & ~~female~~ female to keep and nurture



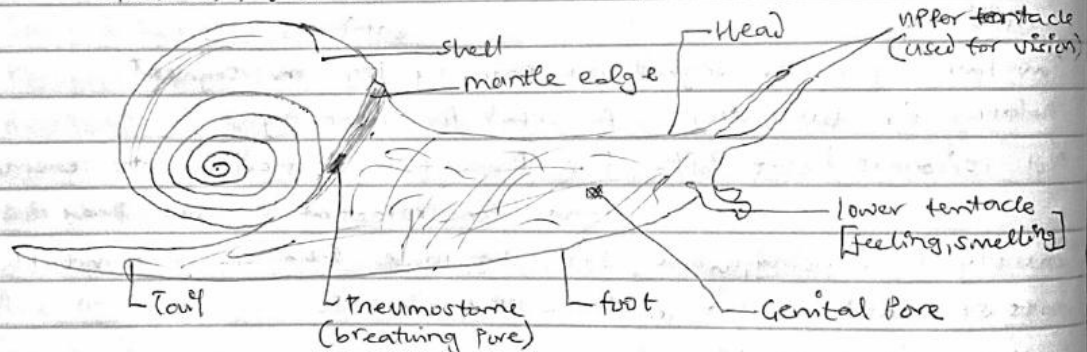
- Kingdom Animalia, Phylum Arthropoda, Class Diplopoda
- Body is segmented into head & trunk and is cylindrical
- Body is covered by chitinous hard exoskeleton
- Are not poisonous and do not bite
- legs can be up to 100 or 750 pieces

Millipede	Centipede
- Two Pairs of leg per segment	one pair of leg per segment
- Adapted for burrowing	Adapted for running
- Not Poisonous, doesn't bite	first pair of leg is modified into venomous fangs for protection against prey
- Sexually reproducing, male inserts spermatophore into female using his gonopods	Sexually, male produces spermatophore
- legs exist below body	legs exist by the body side
- body is cylindrical	- body ^{body} is flattened
- Herbivorous or detritivorous	- Carnivorous
- black or dark reddish-brown	yellowish grey or brown



- Kingdom Animalia, Phylum Arthropoda, Class Arachnida
- Body is heteronomously segmented or tagmatized into cephalothorax and abdomen. Cephalothorax = Cephalo (head) + thorax and bears the legs, eyes and mouthparts
- Eight pieces (two in a row) of eyes, body is covered by exoskeleton
- The legs is jointed, eight in number and used for walking & climbing
- Spinneret is located on the abdomen and produces silk for building web
- Possesses two fangs (similar to antenna) located near the mouth
- Feeds on smaller insects (Insectivorous or carnivorous)
- Although possess eye, sensitivity is by perceiving vibrations on its web
- Posses extracellular digestion - releases digestive enzyme on captured prey

Specimen K: Giant African Land Snail

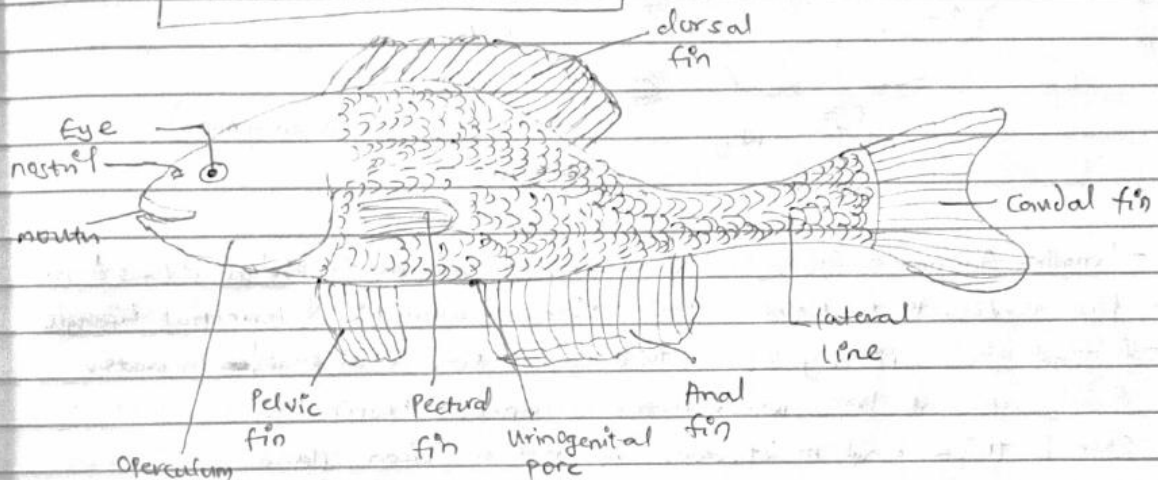


- Kingdom Animalia, Phylum Mollusca, Class Gastropoda (belly foot)
- defining feature is the possession of hard carbonated shell which houses the visceral organs and used for protection.
- Body is heteronomously tagmatized into a large muscular foot, a twisted feeding belt called radula, mantle & mantle cavity, visceral mass which represents delicate organs of its body plus a hard non-living calcareous shell.
- Body is triploblastic, unsegmented and bilaterally segmented at some point in its life time. Changes in symmetry type is caused by torsion
- Large muscular foot is adapted for gliding
- Radula acts as a conveyor belt for moving food particles into mouth

- scientific name is Cornu asperum

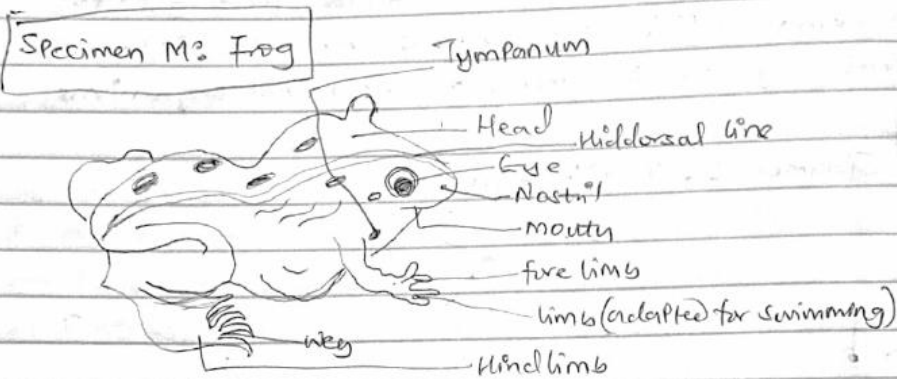
- Forwarding of the shell brings the gills (respiratory organ) to the front for better gaseous exchange, makes room for quick retraction and puts the anus over the head, increasing chances of fouling the head
- Pulmonate is the respiratory organ - ends/opens to the Pneumostome
- Show herbivory, filter feeding and carnivorous feeding patterns
- Snail is hermaphroditic and elicits (are also protandrous)
- Snails, particularly terrestrial species, are nocturnal organisms

Specimen L: Tilapia fish

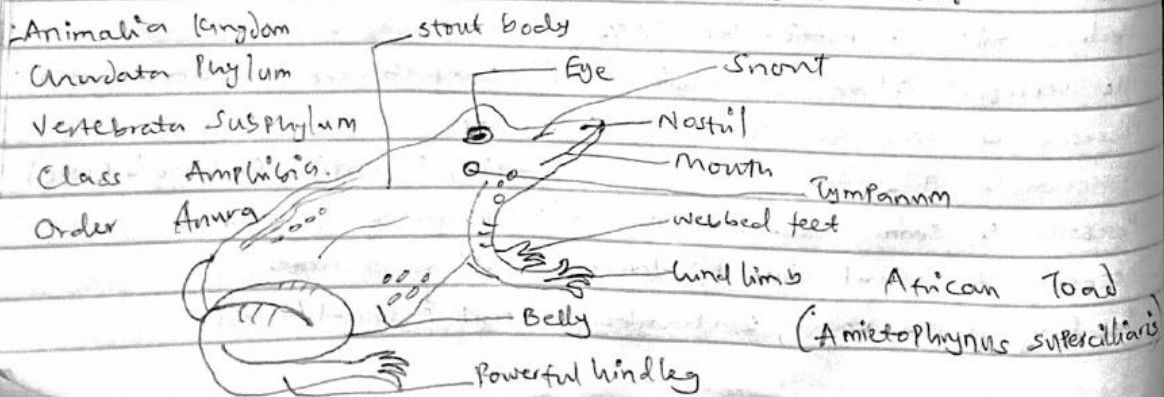


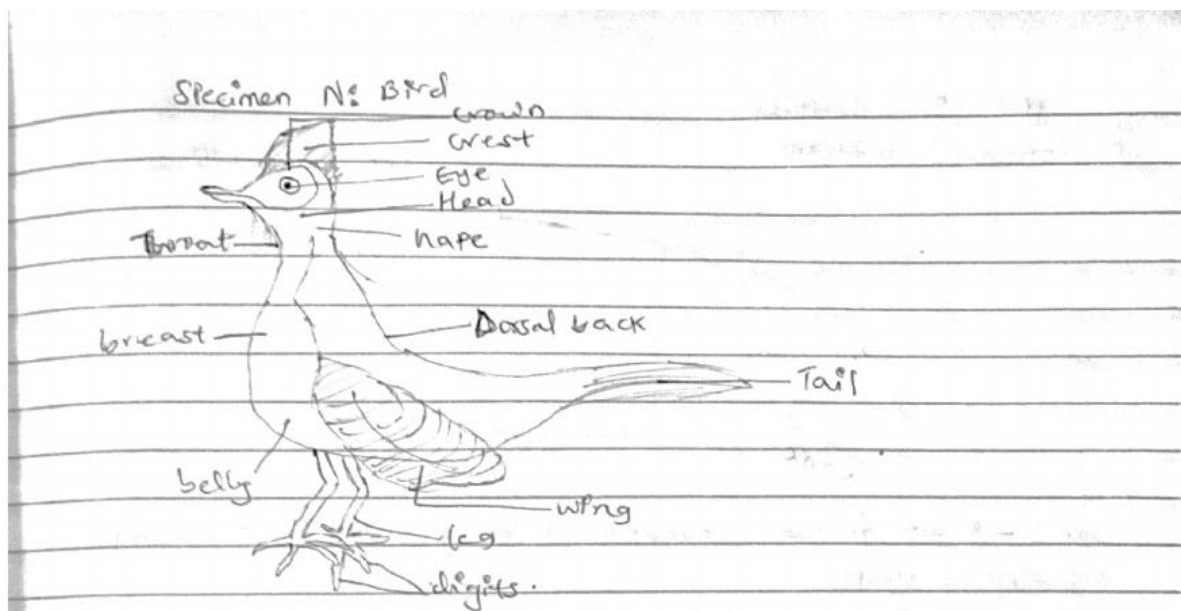
- Kingdom ~~Animalia~~, ~~Subphylum~~ Vertebrata or Craniata, Phylum ~~Pisces~~ ~~Pisces~~
- ~~Kingdom~~ ~~Animalia~~
- Kingdom Animalia, Subphylum Vertebrata, Phylum Pisces, Superclass Gnathostomata, Class Osteichthyes (bony fishes)
[Jawed fishes]
- Found, lives, exists & reproduce in aquatic lifestyle
- Body is tubular, unsegmented and epidermic secretes slimy mucus which reduces friction in motion by 66%
- The vertebral column is made up of bony calcareous skeletal system
- Possession of fins for swimming in water
- Possession of gills for aquatic respiration, circulation of blood is closed
- Possession of swim bladder to maintain buoyancy in water
- Possession of lateral lines to detect vibrations in water
- Feeding is carnivorous, 2 chambered heart & cold-blooded

- Possession of Paired fins (Pelvic & Pectoral) and median fins (dorsal, caudal & anal)
- Pair of Immovable eyes without eyelid
- Excretion of nitrogenous waste is through the kidney
- Have elaborate mating traditions, reproduction is sexual, copulation is absent, eggs are laid (oviparous) and fertilization is external



- Kingdom Animalia, Subphylum Vertebrata, Phylum Amphibia, Class Anura
- first vertebrate to exhibit dual existence: aquatic & terrestrial lifestyle
- Adapted with respiratory organs that ensure survival on land & in water
- Small inefficient lungs, would prefer cutaneous respiration
- Skin is thick and moist, does not have a poison gland
- Skeletally strengthened for jumping & hopping on land
- can respire through lung, mouth, gill or skin
- 3 Chambered heart: 2 atria & 1 ventricle, are cold-blooded
- Sensitivity is by free nerve ending on the skin
- Excretion of ammonia or urea waste is by the kidney while salt and water are excreted through the skin
- Survive winter by hibernation & summer by estivation
- Reproduction is sexual and fertilization is external





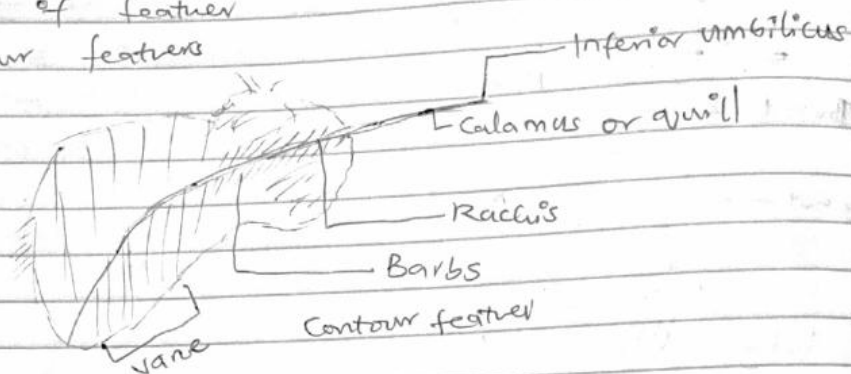
- Animalia (kingdom), Phylum Chordata, Subphylum Vertebrata, Class Aves, Order Carinata (flying birds)
- The entire anatomy of bird is built around its ability to fly.
- Body is covered by fur and wings are modified fore limbs
- Possession of hollow, light, delicate but sturdy bones allows for flight
- Spindle-like body is metameromously segmented into head, neck, trunk and tail
- Possession of Carinate breastbone which allows attachment of flight muscles
- Beak is the main feeding structure, and is adapted to the food type
- Possession of ~~an~~ oil gland (Uropygial gland) at the base of the tail. It secretes a mixture of waxes, lipids, etc for the bird to preen.
- Is a voracious feeder due to high metabolic rates.
- 4 chambered heart and warm-blooded
- Shows great parental care. Reproduction is sexual, fertilization is internal to produce/lay fertilized eggs (oviparity).
- Lives in nests on trees or in crevices

Economic Importance of feathers

- Are used for flight, covers the whole body of aves
- Lightweight, possesses remarkable toughness and tensile strength
- Used for flight, in courtship display, in camouflaging, to provide warmth, to insulate the bird and to keep dry.

Types of feather

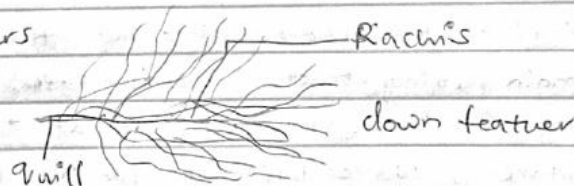
(a) Contour feathers



Uses: - main feather covering the body (giving it a smooth spindle-like shape)

- gives bird its characteristic visual colouring
- provides a first-level defence against physical objects and elements of weather
- are also called flight feathers and the loss of them means the bird won't be able to fly again
- contour feathers are insulated with oil from the preening gland
- also aids in thermoregulation

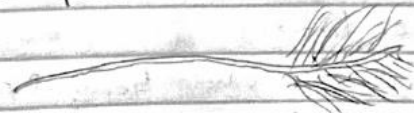
(b) Down feathers



Uses: - traps air needed for insulation against heat loss

- contributes to buoyancy of water birds (e.g. duck, goose etc)
- helps the bird to feel comfort and relaxed
- extremely elastic, lightweight and is three-dimensional

(c) Filoplume feather



- serves as aerodynamic devices
- are hair-like with a few soft barbs near the tip
- associated with contour feathers and may be sensory or decorative

Paper 4 of 10 — BAMBOY 100L 2nd semBIO 104BAMBOY BIO 104 Toad Fish Mollusc.pdf

Lecture/Notes · Google Drive harvest

External Features of a Toad:

The body of a toad is divisible into head, trunk and limbs. Head and trunk are not separated by a flexible neck; this condition is very suitable for aquatic (aqua = water) life. The adult has no tail.

The toad is a bilaterally symmetrical animal; that is, if the animal is cut through the middle of its length, the parts on the right half will be similar to those on the left half. For the purpose of description, the back of the animal is spoken of as the dorsal and the underneath as the ventral surface; the part of the body which is foremost during locomotion is the anterior end and the rear is the posterior end.

Scanned by CamScanner

The part of a limb which lies nearest to the trunk is proximal and the opposite is distal. Each side of the body is said to be lateral and structures on the middle line of the body are medial.

1. Head:

JOMO

The head is nearly semicircular in outline with the bluntly rounded snout in front. The mouth is a wide gaping slit which extends from side to side along the anterior end of the snout; it is meant for the ingestion of food.

On the dorsal surface of the head, there are two small openings near the anterior end. These are the nostrils or external nares which lead into the nasal passages and serve as the inlet or outlet of air. On either side of the head, there is a large protruding eye which can be closed by means of eyelids.

Scanned by CamScanner

Each eye is provided with three eyelids:

- (i) A thick and prominent upper eyelid;
- (ii) A rudimentary lower eyelid; and
- (iii) A transparent third eyelid or nictitating membrane which is attached below to the lower eyelid and can be stretched up to cover the eyeball. There are no eyelashes. The bulging eyes enable the animal to see all rounds, and the nictitating membrane protects the eyeball from dirt. Behind each eye there is a circular patch of tightly stretched greyish membrane. This is the eardrum or tympanum for the purpose of hearing.

There is no pinna of the ear in the toad. A little behind the eardrum, there is a round elevated structure called parotid or parotid gland. These extend behind into the trunk and when the toad apprehends any danger, the glands secrete a pungent sticky juice for repelling the enemy. Parotoid glands, therefore, are organs of offence and defence.

Scanned by CamScanner

The ventral surface of the head forms the floor of the mouth cavity. Its rise and fall along with breathing can be observed in a living specimen. The skin in this part of the body is loose and may, in male toads, be inflated by a dark-coloured vocal sac. The female has no vocal sac.

2. Trunk:

The trunk is short and flattened from above down wards. There is nothing to indicate the boundary between the head and the trunk. The dorsal surface is covered with innumerable rough warts which indicate the position of the poison glands.

The ventral surface is comparatively smooth and free from warts. The posterior end of the trunk is broadly rounded. The vent cloaca aperture lies dorsally at the posterior end, between the bases of the two hind legs; it serves for the exit of faeces, urine and germ cells.

Scanned by CamScanner

A toad, when disturbed, often passes water through the vent. There is a widely spread rumour that handling of toads causes warts on human hands; this is a false superstition.

3. Limbs:

Attached to the trunk are a pair of forelimbs and a pair of hind limbs. The hind limbs are kept in a folded state when the animal is at rest.

i. Fore Limbs:

The forelimbs are short and divisible into segments by movable joints. The proximal part, by which it joins the trunk, is known as the arm or brachium. To this is joined the forearm or ante-brachium.

Distal to the forearm is an insignificant wrist or carpus followed by a hand or manus. There are four digits or fingers corresponding to those on our own hand; but the thumb is absent. The male toad develops a cushion-like swelling, called thumb pad, at the bases of the index and middle fingers.

Scanned by CamScanner

ii. Hind Limbs:

The hind limbs are longer and stronger than the forelimbs. Each hind limb consists of a proximal segment called thigh or femur which is followed by the shank or crus. Distal to the shank is a long ankle or tarsus, a foot or pes, and five slender digits or toes of which the fourth toe is the longest. The toes are united by a thin fold of skin called web.

The limbs are very suitable for the purpose of hopping on land as well as for swimming in water. A toad, when disturbed, jerks off its strong hind limbs, leaps into the air, and finally lands on the ground at a considerable distance.

This peculiar method of locomotion is known as hopping. When taking the hop, the long ankles and feet are pressed against the ground and used as levers. The animal swims by alternately bending and straightening its powerful hind limbs, the webbed feet being used as paddles.

4. Integument:

Scanned by CamScanner

4. Integument:

The skin is rough but moist. It does not bear scales, feathers, hairs or claws and is, therefore, said to be naked. Toads are almost of the same colour as the earth; the dorsal surface of the animal is greyish-black and the ventral surface is yellowish-grey. They are thus very difficult to locate when not moving.

Toads have the power of changing the colour of their skin, so that they may match with the background and thus become invisible to the enemy. This phenomenon is known as protective colouration. The tree frog *Hyla* is better than toad in this respect and can change its colour very quickly.

The skin forms a cover for the entire body. It protects the underlying soft parts from injury and excludes disease germs. It serves as an additional respiratory organ by absorbing oxygen and giving out carbon dioxide. Toads never drink water; they freely absorb water by their skin.

Scanned by CamScanner

The skin is composed of two distinct layers:

- (i) An outer epidermis, and
- (ii) An inner dermis.

The epidermis consists of several layers of epithelial cells. The superficial epidermal cells are thin, scale-like and devoid of nuclei. They form a horny layer called stratum corneum which is shed from time to time as one piece. This process is known as ecdysis or moulting. The deeper epidermal cells are tall and polygonal. They form the stratum germinativum which produces new cells to replace the old horny layer.

The dermis is much thicker than the epidermis. It is composed of glands, pigment cells, involuntary muscle cells, blood vessels and nerves, all enclosed and held together by connective tissue.

Scattered in the superficial part of the dermis are spider-like pigment cells or chromatophores which can change the

Scanned by CamScanner

scattered in the superficial part of the dermis are spider-like pigment cells or chromatophores which can change the colour of the animal by simply shifting their position. The chromatophores are either black or yellow, but never green. The glands produce the rough warts on the outer surface.

There are two kinds of glands:

(i) Poison glands and

(ii) Mucous glands.

Jomo

They manufacture useful fluids or secretions which pass out by narrow ducts and moisten the surface of the skin. The poison glands are larger and more numerous. They secrete a thick whitish fluid with burning taste which repels the enemy. The mucous glands secrete a thin watery fluid which keeps

Scanned by CamScanner

The mucous glands secrete a thin watery fluid which keeps the skin moist and shiny.

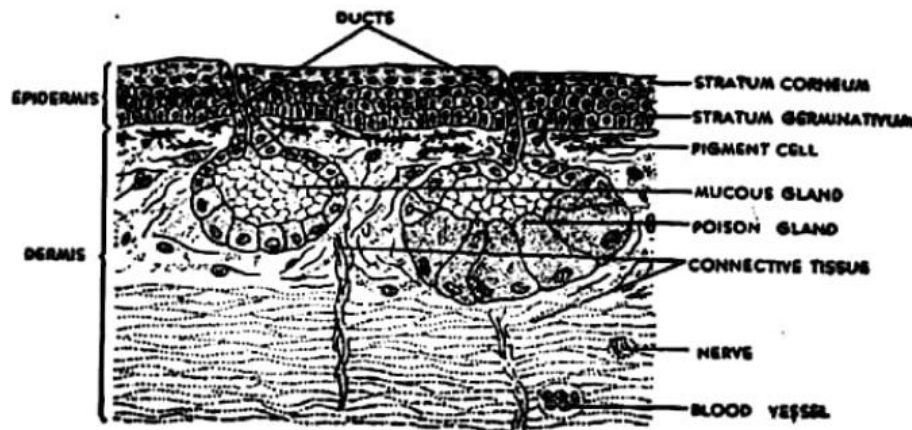


Fig. 6. Internal structure of toad's skin.

Body Wall and Body Cavity of Toad:

The toad's body is essentially a double tube. The outer tube is formed by the body wall and the inner tube by the digestive canal. The two tubes are separated from one another by the body cavity or coelom.

The body wall is composed of the following structures:

- (i) Skin,
- (ii) Subcutaneous connective tissue,

Scanned by CamScanner

following structures:

- (i) Skin,
- (ii) Subcutaneous connective tissue,
- (iii) Muscle layers, and
- (iv) Peritoneum.

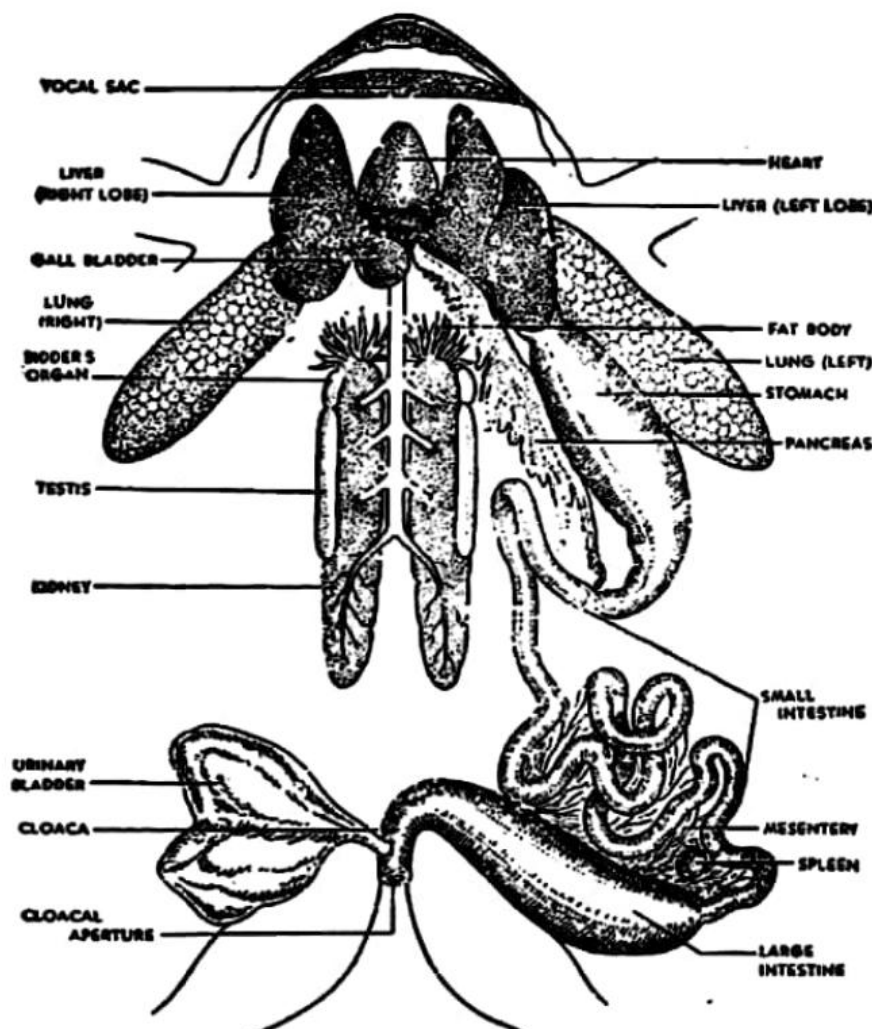
The skin is loosely attached to the underlying muscles by a thin layer of yellowish white connective tissue. There are large spaces enclosed between the skin and the muscular layer. As these are filled with a colourless fluid called lymph, they are known as the subcutaneous lymph spaces. The peritoneum lies beneath the muscular layers.

It is a thin glistening membrane which forms an internal lining for the whole of the body cavity or coelom. Most of the internal organs of toad lie in the coelom and are exposed by cutting down the different layers of the body wall through an incision

Scanned by CamScanner

The peritoneum not only lines the body cavity but it also encircles the digestive tube and other organs, suspending them to the body wall by fan-shaped folds called mesentery. The relative position of the various internal organs or viscera is shown in Fig. 7.

The integumentary system, including the skin and its various functions.



Scanned by CamScanner

PISCES –(FISHES)

CLASSIFICATION:

PISCES-CLASSIFICATION

The super class Gnathostomata includes craniates, in which one pair of the visceral arches is modified into this jaws. They have internal ear with three semi-circular canals and provided with paired 4, appendages (fins or limbs). Sexes are separate. The super class is divided into, two groups.

1) Pisces and 2) Tetrapoda.

Pisces Include three classes.

.1) Placodermi,

.2) Elasmobranchi (Chondrichthyes),

3) Osteichthyes.

Muller, Goodrich. Berg, Romer etc, classified the Pisces. There is no agreement among many authors with respect to the classification of fishes.

Class 1 Placodermi (Aphisthoids)

4/11/2023, 11:06 AM

Scanned by CamScanner

Class 1 Placodermi (Aphstohyoids)

- 1) It includes extinct fishes.
- 2) They are all armored fishes. Their exoskeleton is in the form of bony plates or shields.
- 3) Their endoskeleton is bony:
- 4) The hyoidean gill-slits are complete. It is not reduced.
- 5) The autodiastylic jaw suspension is seen in these fishes.
- 6) Heterocercal caudal fin is seen.
- 7) Hyoid arch will not support the jaws.
- 8) Primitive jaws are seen.
- 9) They survived up to Permian period of Paleozoic era.

Ex Climatius, Bothriolepis.

Scanned by CamScanner



Jomo

Class 2. Chondrichthyes or Elasmobranchs

Scanned by CamScanner

7) Heterocercal caudal fin is seen.

8) Males show claspers for copulation.

9) Air-bladder is absent in these fishes.

The Class Elasmobranchi is divided into two sub-classes.

1. Sub-class Selachi :-

In these fishes the pectoral fin has cartilagenous rods. Fins are well developed. The caudal fin is hetetocercal. This subclass has four orders in which only two are living.

Order 1. Proto selachi

1) Jaws show many pointed teeth.

2) Nasal openings are paired.

3) Hyostylic or Axnphistylic jaw suspension.

It is represented by few living species.

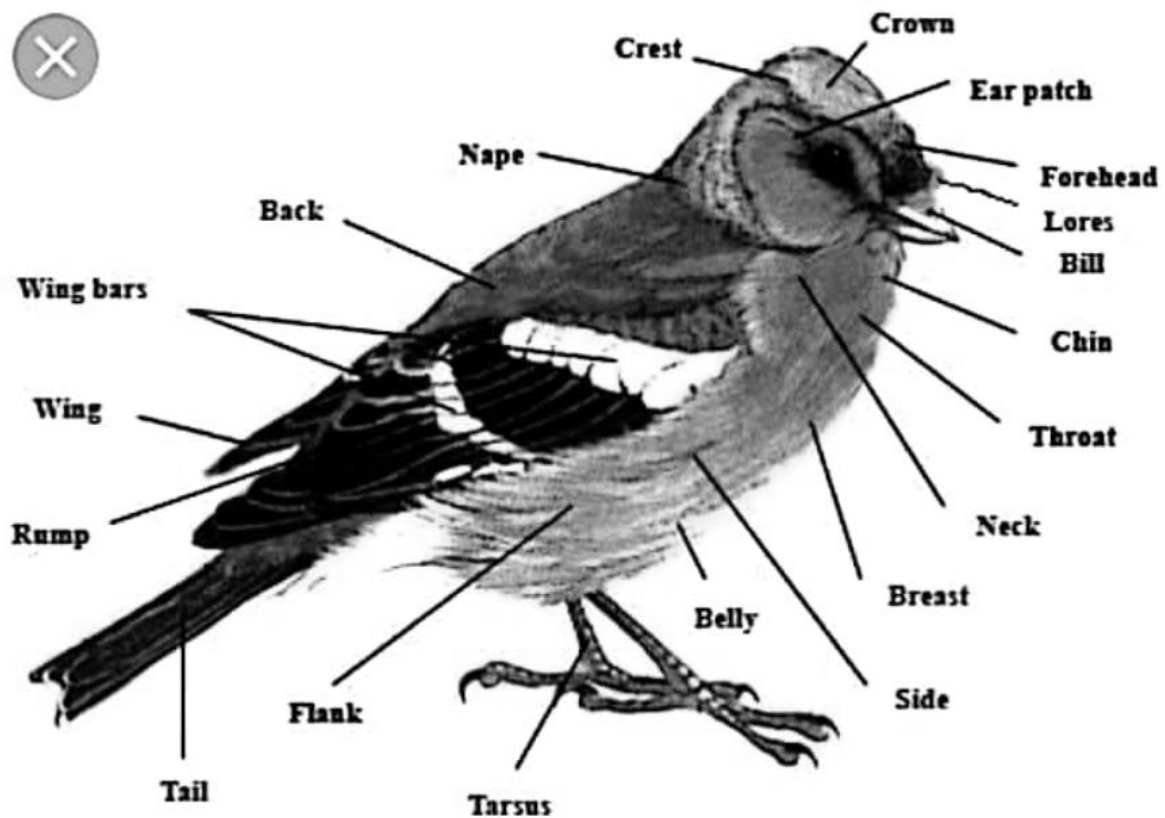
Ex :Heteroloatas

Order 2. Euselachi

1) Skin contains placoid scales.

2) These fishes are exclusively marine.

Scanned by CamScanner



Scanned by CamScanner

External Bird Anatomy

- **Beak:** A bird's beak serves many purposes -- such as eating, grooming, and of course, singing! The beak is an extension of the bird's jaw bone, and is covered in keratin, the same substance that makes up our fingernails. The top part of the beak is called the cere, and is where the bird's nostrils, or nares, are located.
- **Eye:** Whoever came up with the phrase "eagle eye" wasn't joking -- birds have extraordinarily accurate vision. The eye holds scores of

A 2

Scanned by CamScanner

receptor cells, called rods and cones, that translate whatever the bird looks at to the image that it sees. To give an idea of how sharp their sight is, humans typically have around 200,000 of these cells per millimeter inside of their eyes. Some birds, particularly birds of prey, have five times that many!

- **Wings:** A bird's wings are constructed of a series of small thin bones similar to miniature versions of the bones in human arms. Externally, the wings are home to several different kinds of feathers: the Primary Flight Feathers, the Secondaries, the Main and Lesser

Scanned by CamScanner

Coverts, the Tertials, and the Alula.

- **Foot:** The feet and legs of birds vary greatly depending on the species. Generally the legs, feet, and claws are structured to allow a bird to take off, land, climb, and grasp with them. Since birds spend most of their lives perching, the feet and legs are covered with a tougher skin than the skin on the rest of the bird's body.

Scanned by CamScanner

- **Tail:** During flight, a bird's tail acts much like the tail of an airplane -- it's used like a rudder to help the bird steer. The muscles of the tail also aid in helping the bird expand its lungs to take in extra air when needed.

- **Anus:** The anus is the external opening through which the bird passes waste.

Scanned by CamScanner

4. Characteristic Features of Molluscs:

1. Majority of the members aquatic and a few terrestrial.
2. Bilaterally symmetrical body except some gastropods and cephalopods become secondarily asymmetrical.
3. Body soft and un-segmented in the adult stage.
4. Exoskeleton, in the form of calcareous shell, present in most of them except aplousobranchs, nudibranchs and most octopuses. The shell is mostly located externally, although a few forms with internal shell.
5. Cephalization well-marked in gastropods and cephalopods but in other classes the head is small or poorly differentiated.
6. Visceral mass is enclosed by a thick muscular fold of the body wall, called mantle or pallium, which secretes spicules or shell. It is the unique feature of all molluscs.

Scanned by CamScanner

7. A ventral muscular foot is present which helps in locomotion and may be secondarily modified in some forms.

8. A radula is present in the oesophagus which is a ribbon-like structure with recurved chitinous teeth stretched over a cartilaginous base helps as a scraper in feeding; absent in bivalves.

9. Respiratory organs are of comb-like structures known as molluscan gills or ctenidia or branchiae lodged in the mantle cavity, found in aquatic forms. In most pulmonate gastropods the mantle cavity is transformed into a pulmonary sac, helps in aerial respiration.

10. A chemoreceptor or tactile receptor organ, called the osphradium, generally located adjacent to the ctenidium. Other sense organs are eyes, tentacles and statocysts in most of the groups.

11. Blood-vascular system is a haemocoel, usually in the form of several large connected sinuses and serves as a hydrostatic skeleton in the locomotion of some species. The two-chambered heart is

Scanned by CamScanner

situated in the haemocoelomic cavity. The type of circulation is open type. In open type the capillaries are absent and the blood bathes tissues directly.

12. Respiratory pigment is usually haemocyanin, a oxygen-carrying blood pigment. Haemoglobin is found in a number of species (e.g., Solen, Arca, Planorbis).

13. Excretory organs in most cases include kidneys or called metanephridia.

14. A well-defined nervous system with a primitive brain in most cases, consists of a circumoesophageal nerve ring with a pair of pedal nerve cords and a pair of visceral cords.

15. Sexes usually separate (gonochoristic or dioecious) but hermaphrodite (monoecious) in few groups. Mostly non-marine gastropods, some bivalves and cephalopods are hermaphrodites.

16. Fertilization may be external or internal. External fertilization takes place in many marine gastropods and most bivalves, but

Scanned by CamScanner

gastropods.

17. Development through spiral cleavage.

18. In marine and freshwater forms the development involves by two larval stages—the multiciliated trochophore is the first larval stage and the veliger is the second larval stage.

19. Direct development mainly in Gastropoda and Cephalopoda.

5. Classification of Molluscs:

Classification in Outline (Up to Orders):

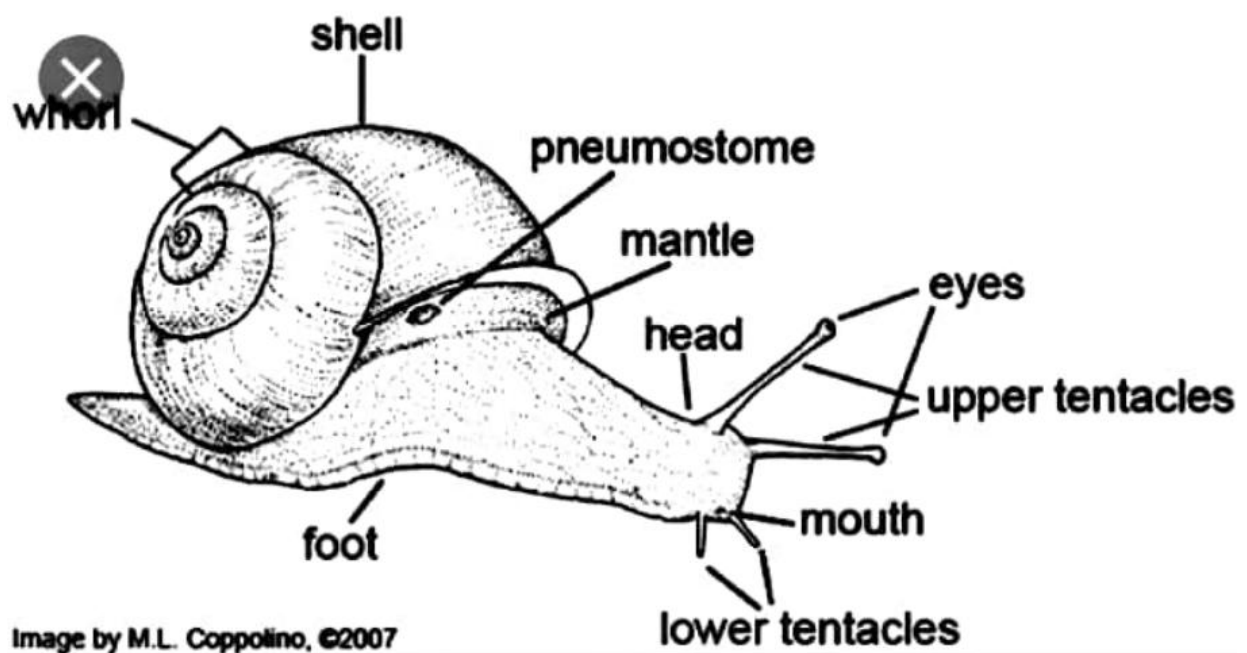
The classificatory scheme followed in the present text is based largely on the plan laid down by E.E. Ruppert and R.D. Barnes (1994) in their book “Invertebrate Zoology”.

Phylum Mollusca:

Class 1. Aplacophora

Subclass Chaetodermomorpha (=

Scanned by CamScanner



Scanned by CamScanner



Fig. 16.1 : Toad (*Bufo melanostictus*)

Scanned by CamScanner

Similarities:

- Mollusca is very similar to the Annelids due to their common ancestor
- Mollusca and Annelids both remove metabolic wastes through nephridia
- Blood is pumped by heart in both Mollusca and Annelids
- Mollusks and Cnidarians are both marine animals
- Mollusks (Cephalopods and Gastropods) feed on food including detritus and some Annelids also feed on detritus
- Mollusks, Annelids, Platyhelminthes and Nematoda may be hermaphrodites under desperate conditions when they need to reproduce

Scanned by CamScanner

Differences:

- Mollusca have one heart while Annelids have five hearts
- Mollusca can have open circulatory system but Nematoda and Platyhelminthes have closed circulatory system
- Mollusks are marine creatures while Nematoda and Platyhelminthes are land creatures
- Mollusks have a more complex system than any other phyla
- Mollusks have shells, either externally or internally, but phylum such as Cnidarians, Annelids, Nematoda and Platyhelminthes do not

Scanned by CamScanner

Paper 5 of 10 — BAMBOY 100L 2nd semBIO 104BAMBOY bio104.pdf

Lecture/Notes · Google Drive harvest

PRACTICAL I—ROTOZOANS AND COELENTERATES

Introduction:

PHYLUM PROTOZOA

1. They are unicellular, some colonial and some with multicellular stages in their life cycles.
2. They are mostly microscopic, although some are large enough to be seen with naked eye.
3. Locomotion by pseudopodia, flagella, cilia and direct cell movements.
4. Nutrition of all types-autotrophic, heterotrophic, saprozoic.
5. Free-living, mutualism, commensalism, parasitism are all represented in the group.

Examples: *Ameoba*, *Entamoeba*, *paramecium*, *Vorticella*, *Monocystic*, *Eimeria*, *Plasmodium*, *Toxoplasma*, *Babesia*, *Opalina*, *Trichomonas*, *Trypanosoma*, *Leishmania* etc.

Unicellular, eukaryotic

Class: Mastigophora (Flagellate) e.g. *Euglena* sp, *Chlamydomonas* sp

Class: Rhizopoda (Sarcodina) e.g. *Ameoba* sp

Class: Ciliophora e.g. *Paramecium* sp

Class: Sporozoa e.g. *Plasmodium* sp

THE COELENTERATES

This group comprises the cnidarians and ctenophorans among others. They are aquatic, radially symmetrical, diploblastic and have no blood system e.g. Hydra, sea anemone, hydroids, jelly fish.

PHYLUM-CNIDARIA

1. Two basic types of individuals: polyp and medusa.
2. Radial symmetry or biradial symmetry around a longitudinal axis with oval and arboreal ends.
3. Asexual reproduction by budding (in polyps) or sexual reproduction by gametes (in Medusae and some Polyps).
4. Presence of special stinging cell organelles called nematocysts in either epidermis or gastrodermis or in both.
5. Body with two layers, epidermis and gastrodermis with mesogloea (Diploblastic).

Example: Hydra, Obelia, Tubularia, Aurelia, Acropora, Tubipora, etc.

PHYLUM-CTENOPHORA

1. Symmetry biradial
2. Nematocysts absent (except in one species), but adhesive cells (colloblasts) present
3. No polymorphism
4. Luminescence common

5. Ectoderm, endoderm and a mesoglea (ectomesoderm) with scattered cells and muscle fibres; may be considered triploblastic.

Examples: Beroe, Ctenophora, Mnemiopsis.

Procedures:

- a) Put a drop of the luminal fluid obtained from the rectum of toad on a cover slip and observe the wet preparation for protozoans especially ciliates. Add a drop of dye and observe at high power objective. Draw and label the organisms you see.
- b) Mount a drop of the pond water provided on the slide. Observe under the microscope and draw as many different types of cells as possible.

PRACTICAL II – WORMS AND MOLLUSCS

Introduction:

PHYLUM PLATYHELMINTHES

1. Bilateral symmetry, definite polarity of anterior and posterior ends.
2. Three germ layers (triploblastic).
3. No internal body space other than digestive tube (Acoelomate).
4. Presence of simple sense organs-eyes/spots in some.
5. Class Turbellaria are mostly free living. Classes Monogenea, Trematoda and Cestoda are entirely parasitic.

Examples: *Planaria*, *Digenia*, *Gyrodactylus*, *Dactylogymus*, *Fasciola*, *Clonorchis*, *Schistosoma*, *Paragonimus* (lung flukes), *Taenia solium*, *Taenia sp.*, *Echinococcus multilocularis* etc.

PHYLUM NEMATODA

1. Unsegmented round worms with slender cylindrical bodies.
2. They possess pseudocoel.
3. They are mostly fluid feeders.
4. They have no circulatory or respiratory system.
5. Sexes are usually separate.

Examples: *Ascaris lumbricoides* (roundworm), *Enterobius*, *Ancylostoma* and *Necator* (hookworms), *Onchocerca*, *Dracunculus* (guinea worm), *Trichinella* etc.

PHYLUM ANNELIDA

1. Body metamerically segmented, symmetry-bilateral.
2. Chitinous setae often present, Setae absent in leeches.

3. Coelom (schizocoel) well developed and divided by septa except in leeches.
4. Hermaphroditic or separate sexes.
5. Larvae if present are trocophore type.

Example: Earthworms, *Lumbricus*, *Stylaria*, *Nereis*, Leech or *Hirudo*, *Macrobdella* etc.

PHYLUM MOLLUSCA

1. Body bilaterally symmetrical (bilateral asymmetry in some).
2. Body unsegmented often with definite head.
3. Coelom limited mainly to area around the heart and perhaps lumen of gonads and part of kidneys.
4. Ventral body wall specialized as a muscular foot, variously modified but used chiefly for locomotion.
5. Sensory organs to touch, smell, taste, equilibrium and vision (in some): eye highly developed in Cephalopods.

Examples: *Neophina*, *Chitons*, *Dentalium*, Snails and slugs such as *Archachatina marginata*, *A. Fulica*, *Helix*, *Aplysict*, *Anodonta*, *Venus*, *Tagelus*, *Teredo*, *Anadara senilis*, *Egaria*, *Octopus*, *Sepia* etc.

Exercises:

1. **The Platyhelminthes (flatworms)**
You are provided with prepared slides of *Taenia piriformis*. Note the section provided and draw. What is the economic importance of *Taenia sp*?
2. **The Nematodes (roundworms)**
 - Draw and labels the prepared slides of *Ascaris sp* provided.
 - Note the section/part.
 - What is the main differences between a nematode and a plantyhelminth?
3. **The Annelids**
 - Draw and label the specimen of *Lumbricus sp.* (The earthworm) provided.
 - What are the differences between a round worm and an annelid?
4. **Molluses**
Observe and draw the supplied specimen of *Archachatina Maginata*.

PRACTICAL III-THE ARTHROPODS

Introduction:

PHYLUM-ARTHROPODA

1. They have bilateral symmetry and metameric segmentation.
2. They are triploblastic.
3. Possession of jointed appendages.
4. Highly cephalised, exoskeleton of cuticle containing protein, lipid.
5. Possession of jointed appendages.
6. Highly cephalised, exoskeleton of cuticle containing protein, lipid, chitin and often calcium carbonate secreted by underlying epidermis and shed at intervals.
7. Coelom is reduced in adult, most of body cavity consisting of haemocoel (sinuses or spaces in the tissue) filled with blood.
8. Sexes are separate, respiration by body surface, gills, trachea (air tubes) or book lungs.

- a) Examples Subphylum – Chelicerata
Class Arachnida e.g. Horseshoe crabs, scorpions, spiders, ticks, aquatic chelicerates like *Limulus* etc.
- b) Subphylum-Crustacea e.g. prawn, crayfish, crab, shrimps, waterfleas etc.
- c) Subphylum-Uniramia (insects and Myriapods)
 - (i) Class – Diplopoda (Millipedes) e.g. *Julus*, *Spirobolus* etc.
 - (ii) Class – Chilopoda (Centipede) e.g. *Cermatia*, *Lithobius*, *Geophilus*, etc.
 - (iii) Class – Pauropoda e.g. *Pauropus*
 - (iv) Class – Insecta e.g. Beetles, Butterflies, dragonflies, bugs, flies, termites, bees, wasps, etc.

Exercises:

Observe, draw and label all the arthropod specimens supplied. Some of them are millipedes, centipedes, scorpions, mites, spiders, mosquitoes, bugs, beetles, flies, ants, crab, crayfish, cotton strainers, butterflies, etc.

PHYLUM-ECHINODERMATA

1. Body unsegmented (non-metameric) with radial, pentamerous symmetry
2. No head or brain
3. Locomotion by tube feet
4. Coelom is extensive
5. Sexes are separate (except a few hermaphrodites) with large gonads.

Examples: Sea stars, *Asterias*, *Piaster*, Brittle Stars, Basket Stars, Starfish, Sea urchins, Sea biscuits, Sand dollars, Sea Cucumbers, Sea lilies, Feather Stars, *Arbacia*, *Mellita*, *Parastichopus*, *Cucumaria*, etc.

PRACTICAL IV – THE PROTOCHORDATES AND FISHES

Introductions:

PHYLUM – CHORDATA

1. Bilateral symmetry, segmented body, three germ layers, well developed coelom.
2. Notochord present at some stage in life cycle
3. Possession of single, dorsal, tubular nerve cord.
4. Pharyngeal pouches present at some stage in life cycle.
5. Presence of post-anal tail.

Protochordates

These are the evolutionary links between the two main groups of animals, the vertebrates and the invertebrates. They bear close affinities to the invertebrates and also share certain characters with the chordates.

- (i) Presence of gill slits/visceral cleft
- (ii) Possession of hollow dorsal nerve cord.
- (iii) Presence of notochord
- (iv) Possession of post-anal tail.

A. SUB-PHYLUM: HEMICHORDATA

1. Soft bodied and wormlike in appearance
2. Body divided into proboscis, collar and trunk
3. Sexes are separate
4. Class Enteropneusta: free moving and have burrowing habits; Class Pterobranchia; sessile, mostly colonial, living in secreted tubes.

Examples:

- a) Class Enteropneusta e.g. *Balanoglossus* and *Saccoglossus*
- b) Class Pterobranchia e.g. *Cephalodiscus* and *Rhabdopleura*.

B. SUB-PHYLUM: EUROCHORDATA (TUNICATA)

1. Body surrounded by tough, non-living tunic or test.
2. Mostly sessile at adult stage, some free-living.
3. Solitary or colonial
4. Notochord and nerve cord in free swimming larva only.

Examples: *Ciona* sp. Sea squirt.

C. SUB-PHYLUM: CEPHALOCHORDATA

1. Notochord and nerve cord found along entire of body and persist throughout life.
2. They are fish-like in appearance.
3. Sexes are separate.
4. Closed circulatory system and the blood is devoid of erythrocytes and haemoglobin i.e. blood is colourless.

Examples: Lancelets (*Amphioxus* or *Branchiostma*).

D. SUB-PHYLUM: VERTEBRATA

1. Bony or cartilaginous vertebrae surrounding spinal cord.
2. Notochord only in embryonic stages, persisting in some fishes.
3. Highly differentiated brain
4. Body plan consists of head, trunk, and post-anal tail.
5. Neck present in some especially terrestrial forms.

FISHES

1. All are aquatic.
2. They have two pairs of appendages.
3. They have lateral lines.
4. They possess exoskeleton in form of scales.

- i) Superclass: Agnatha (Cyclostomata) or Jawless fishes e.g. Hagfish, Lampreys
- ii) Superclass: Gnathostomata (jawed fishes)

There are two main groups:

1. Bony fishes (Teleosts) e.g. Tilapia, *Oreochromis niloticus* (formerly *Tilapia nilotica*), Cod, Mackerel, lung fishes, *Clarias* highly diversified and most modern.
2. Cartilaginous fishes (Elasmobranchs): Skates, Rays, Dog fishes are all living but others are fossil e.g. the Monstrous fish – *Dinichthys*.

The Lamprey (*Petromyzon fluviatilis*)

External Features

Note that the skin is everywhere smooth and devoid of scales. The division of the muscles into myotomes is usually apparent through the skin. The skin is slimy, due to the presence of numerous glands. In the middle line the skin is produced into two median dorsal fins, and one round the tail. These fins are stiffened by rods of cartilage: fin rays are absent. At the front end will be seen the circular mouth. Inside the rim of the mouth, which is used for attaching the animal to stones, will be observed a number of ectodermal horny teeth. The paired eyes have no eyelids. On the dorsal surface of the middle line is a single median opening which leads into the hypo-physical cavity and the nasal sac. Behind this is a transparent area of skin above the pineal "eye". Behind the eyes on either side are the seven pairs of gill openings. The cloaca lies rather far back in the mid-ventral line. In most specimens there will be seen protruding from the mouth of the cloaca the slender urogenital papilla.

Note: The absence of paired fins and the absence of a lateral line.

Exercises:

A. Study the demonstration put out which show:

Amphioxus – Typical example of a protochordate.

Lamprey – living example of the early or lower vertebrate, others are all extinct.

- a) Draw and label the protochordate. Specify the vertebrates and invertebrates parts in the drawing

- b) Draw and label the ammocoete larva and adult lamprey supplied.
- B. Observe a living fish in a tank of water. Identify head, trunk, and tail regions.
- Note:
- Movement of mouth and the operculum
 - Lateral line along the length of the trunk
 - Jaws, mouth, eyes, shape of the body
 - Paired and unpaired fins
- C. You are provided with a preserved specimen of Tilapia and Dogfish.
- i) Feel the body from the head to the tail and the other way round. Comment on the direction of scales. Remove a scale from the body of the Tilapia and make a wet mount, then make a detailed drawing under the microscope.
- ii) Draw detailed structure of the fin.
- iii) Make clearly labeled large drawings of the Tilapia and Dogfish.
- iv) Remove an operculum, count number of gills, cut out one and draw.

PRACTICAL V – AMPHIBIANS

Introduction:

1. They are Ectothermic tetrapods.
2. Live in water and also adapted for life on land.
3. Skin moist and slimy (containing mucous glands) and lack scales.
4. Mouth usually large with small teeth in upper or both jaws, two nostrils open into anterior part of mouth cavity.
5. Blood vascular system consists of a three chambered heart, two atria and one ventricle, and double circulation through the heart.
6. Respiration by lungs, skin and gills in some; either separately or in combination.

Examples: Toad (*Bufo regularis*), frogs, Salamanders, Caecilians, etc.

Basically two major types

- (1) The limbless type of the caecum caecilian – worm-like and
- (2) Those with limbs which can be subdivided into four (4) groups:
 - Early/Extinct amphibians (Rachitons, cacops)
 - Urodeles/Canadata (Newts and Salamanders)
 - Apodans/Shark-like
 - Anura/Salienta (Frog and Toads)

Exercises:

- a) Make a drawing of the specimen of a live toad/preserved toad and label parts fully.
- b) Examine eggs and larva (tadpole). Draw external gill stage and the internal gill stage.
- c) Dissect a toad/frog and draw the alimentary canal, respiratory system and reproductive organs.

PRACTICAL VI – REPTILES

Introduction:

1. Body varied in shape, compact in some and elongated in others.
2. They possess shelled eggs, they have no gills and larva stage.
3. Sexes are separate and fertilization is internal.
4. They have dry skin, their body is covered with an exoskeleton of horny dermal scales with addition sometimes of bony dermal plates.
5. They are ectothermic and many behaviourally thermoregulate.
6. Heart consists of three chambers; Crocodiles have four chambered heart.

Examples: Turtles, Tortoise, Lizard (eg. *Agama agama*), Snakes, Crocodiles, etc

- Three (3) main groups of the living or modern day reptiles.
- Chelonian = Tortoise and Turtles.
 - Crocodilian = Alligators and Crocodiles
 - Squamata = Lizards and Snakes

A fourth group known as the Rhynchocephalia consists of mainly extinct members which were lizard-like.

Exercises:

- a) Examine the features of the Lizard, *Agama agama*.
- b) Draw the side view/dorsal lateral view and label.
- c) Recall the structure of the fishes and the toad. Note any feature(s) of resemblance or difference to the fish and the toad.

THE LIZARD (*Lacerta viridis*)

EXTERNAL FEATURES

NOTE: That the general appearance of the body is very much like that of the salamander. The limbs are a little longer but hardly raise the belly off the ground. The surface of the body is covered with horny scales arranged in a regular pattern. On the head the scales are large and closely applied to the roofing bones of skull, but the position of these scale does not correspond with that of the underlying bone. The scales on the under surface are larger than the scales of the upper surface. The body behind the head region is slightly constricted to form the neck. The tail is long and tapering, being about twice the length of the head trunk together. The mouth is tennial as in the salamander. The paired external nares lie on either side at the tip of the snout. The eyes lie on either side of the head and are provided with movable eyelids which can completely cover over the eye. A nictitating membrane freely exposed on the surface. (In the skin the tympanic membrane is covered by a fold of skin bearing scales). Open the mouth and observe the small teeth on the upper and lower jaws. On the floor of the mouth lies the bifid tongue which is attached posteriorly. On the fore and hind limb there are five long digits, each ending in strong claws. The fourth digits on the hind limb is much longer than the rest just behind the hind limb lies the cloaca which appears as a transverse slit. A large transverse scale, the preanal plate, lies in front of the cloaca.

GRASS SNAKE (*Tropidonotus natrix*)

EXTERNAL FEATURES

Observe the elongation shape of the body and note the entire absence of limbs. This elongation of the body is produced by drawing out the trunk region, the tail region behind the cloaca is no more developed than it is in Lacertilia. Find the cloaca a short distance from the posterior end of the body. It is a transverse slit, bounded in front by two scales with fold over it and make the opening rather inconspicuous. The general surface of the body is covered by epidermal scales arranged in a regular pattern. The scales on the dorsal surface are small except in the head region where they are large and fit

closely to the skull. As in the Lizard, these large scales on the head do not correspond in arrangement with the underlying bones of the skull. The scales on the ventral surface from the throat to the cloaca are large, overlapping posteriorly and extending completely across the ventral surface. The ribs are connected to these scales and some snakes move by raising these scales and depressing them on one after another in succession from the front backwards. This enables the snake to glide over the ground without executing serpentine movements. The scales behind the cloaca and the one immediately in front of it on the ventral surface are paired and do not stretch completely across the ventral surface. The mouth is terminal. Open the mouth and feel the teeth which are on the premaxilla, maxilla, pterygoid, and palatine bone in the upper jaw and on the dentary in the lower jaw. The internal nares can be seen in the roof of the mouth. In the floor of the mouth note the median prominence which is the upper jaw and on the dentary in the lower jaw. The internal nares can be seen in the roof of the mouth. In the floor of the mouth note the median prominence which is the larynx. Find the glottis in the middle of this prominence and insert down it a probe. Feel the cartilage supporting the larynx either side.

Just anterior to the larynx will usually be seen the forked end of the tongue protruding from its pouch. In some specimens it will be possible to catch hold of the tongue and pull it forward to its extreme extent which is about one and half or two inches. The external nares are situated on either side of the snout. The eyes are on either side of the head. The eyelids are permanently fused over the eyes but form transparent "spectacles."



PRACTICAL VII-THE BIRDS

Introduction:

CLASS: AVES

1. They are warm blooded
2. The body is usually spindle shaped with four divisions: head, neck, trunk, and tail. Neck disproportionately long for balancing and food gathering.
3. Sexes are separate, females with left ovary and oviduct only.
4. Possession of fully ossified skeleton with air cavities.
5. Epidermal covering of feathers and leg scales, thin integument of epidermis and dermis, no sweat glands.
6. Limbs paired with the fore limbs usually modified for flying.
7. Examples: Ostrich (*Struthio camelus*), Eagle, Hawk, Vulture, Falcon, Condors, Buzzards, Quail, Grouse, Pheasant, Turkey, Domestic Fowl, Cranes, Rails, Woodcocks, Sand pipers, Pigeon, Dove, Parrot, Parakeet, Roadrunners, Owl, Goatsuckers, Night Hawks, Mouse birds, King fishers, Hornbills, Wood peckers, Honey guides, etc.

There are two main types of the modern birds:

1. Flightless birds e.g. Ostrich, Emu, Kiwi, etc.
2. Flying birds e.g. Eagle, Dove, Vulture, etc.

The Pigeon: External Features

General

The general shape of the body of a bird is entirely different from any vertebrate hitherto studied. The modifications are due to the adoption of bipedal gait and the habit of flight. The body is carried well off the ground as the animal stands on the two hind limbs. When the bird is standing upright, the center of gravity tends to be well in front of the two hip joints but over the toes.

The head is separated from the trunk by well-marked neck. The trunk is deep dorsoventrally. The tail region is very short. The mouth is situated at the anterior end of the body. The horny covering of the jaws form the beak.

The external nares lie at the posterior end of the upper beak. Just above the nares is a swollen patch of skin, usually white in colour called the cere.

Immediately behind the mouth are the eyes. This is provided with two eye lids; there is also a flap of transparent skin which can be drawn over the eye from the upper anterior border, the nictating membrane can easily be pulled down by a pair of forceps.

The external opening of the ear lies about half an inch below the posterior border of the eye. The opening is easily seen by raising the feathers in this region. Stretch out one of the wings, the bones of the wings can be identified by feeling for them through the skin. A comparison should be made with skeleton of the wing prepared with the feathers attached.

The surface of the wings is greatly creased by the feathers which are attached to the hand and fore arm. These feathers are composed of large quills, the remiges, divisible into the primaries and secondaries. The primaries are the longer feathers attached to the fused second and third digits. The shorter secondaries are attached to the ulna bone of the arm. The primaries and secondaries merge into each other so gradually in the region of the wrist that they cannot be distinguished by their size alone. It will probably be necessary to look at the prepared skeleton in order to be certain of the point of division. At their base, the primaries and secondaries are overlain by the wing converts. The bastard wing is a small group of feathers attached to the pollex.

The long tail feather or rectrices, are of the same nature as the primary and secondary feathers of the wing. They are also overlain at their bases by tail converts. Raise the tail converts and immediately behind the insertion of the rectrices will be seen a prominent papilla at the tip of which is the opening of the oil gland. It secretes a substance which is spread over the feathers when the animal preens itself. The cloaca lies at the base of the tail on the ventral surface. As in Reptiles, it opens by a transverse slit. The leg is bare of feathers below the tarsal joint and the skin is here covered with feathers. These are specially modified epidermal structures and they are one of the most characteristic features of the bird. The features are arranged in regular tracts, called pterylae. These pterylae are not so distinct in the pigeon as in some other birds and can only be distinguished with difficulty.

Structures of a feather

Pull out one of the large primaries from the wings. The feather consists of a central stem (a quill or rachis) bearing on either side vanes. The rachis consists of a stiff solid portion between the vanes and a hollow portion between them. At the base of the rachis, is a small hole, the inferior umbilicus, leading into the hollow portion of the rachis, and at the junction of the solid rachis and the hollow base there is another small hole, the superior umbilicus.

The vane is made up of a large number of barbs on either side of the rachis. These barbs in turn have a number of barbules on either side. The barbules are provided with hooks interlocking with the barbules of the barbs on either side. In order to see the barbs and barbules it will be necessary to cut off a piece of the feather, mount it in alcohol and examine it under the microscope.

Varieties of feather

The contour feathers cover the general surface of the body. They are smaller than the wings or tail feathers, and the hooks on the barbules are not so well developed, especially at the basal portion which has a downy appearance.

The down feathers are found chiefly on the breast. In this the barbules and hooks are hardly developed

all, so that the
of barbs at

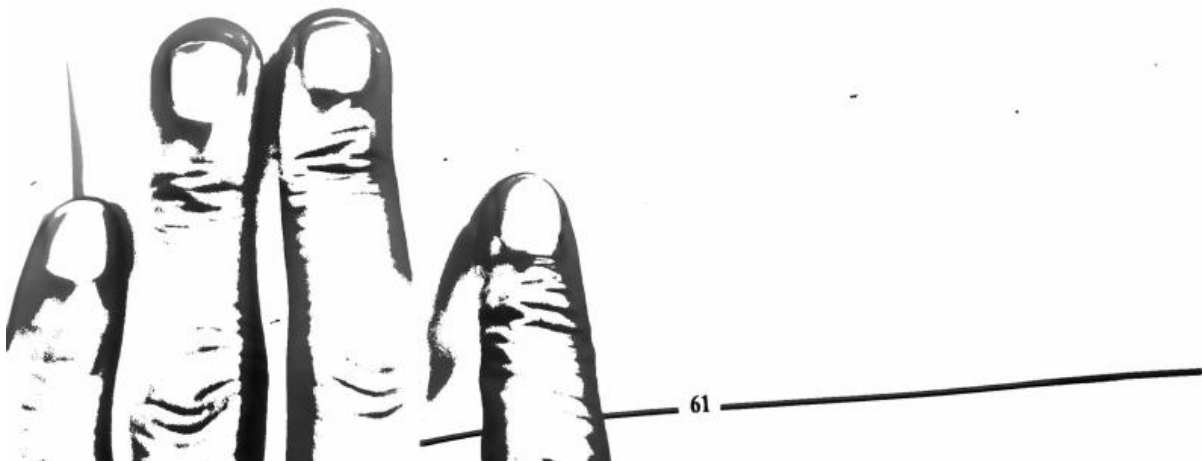
Exercises

- Examine
- Note the eyelid
- Draw on the
- Remove vane
- Draw

at all, so that the feather has no stiff vein. The filoplumes are minute feathers resembling hairs with a tuft of barbs at the tip. They are easily seen remaining in the skin after the feather is plucked.

Exercises

- a) Examine the features of the familiar bird provided.
- b) Note the peculiar structures on the head and trunk regions: beak, nostrils, scaly feet, eyes with eyelids and nictitating membrane, ear opening with feathers, neck, feather types and location on the body.
- c) Draw this bird showing the above features and any other relevant ones.
- d) Remove and examine the following types of feathers from the bird specimen given below: vane or contour feather, pheasant feather, filoplumes and down feathers.
- e) Draw and label fully the feathers listed in (d) above.



PRACTICAL VIII – THE MAMMALS

Introduction:

CLASS: MAMMALIA

1. Body covered with hair
2. Possession of movable eyelids and fleshy external ears
3. Brain highly developed especially neocerebrum
4. Young nourished by milk from mammary gland
5. Mouth with diphyodont teeth (milk teeth replaced by permanent set of teeth).

Examples: Duck-billed platypus, spiny anteater, Kangaroos, Anteaters, Armadillo, Pangolin, Elephant Shrew, Rabbits, Hares, Rats, Squirrels, Hedgehog, Mole flying lemur, Monkey, Ape, Humans, Bush babies, Chimpanzee, Elephants, Hyrax, Horses, Ass, Zebra, Rhinoceros, Swine, Camel, Deer, Antelopes, Hippotamus, Goat, Whales, Dolphins, etc.

Procedure:

RAT (*Rattus rattus*)

General Instructions for Dissection

1. Pin the rat on a board with its ventral surface upper most.
2. Lift the skin midway between the mouth and anus and cut through it.
3. Cut forward in the mid-ventral line to the jaw and backwards to the anus, cutting round the penis or openings of urethra and vagina.
4. Loosen the skin, using fingers or scapel and pin it out so that the elbow and knee joints and outline of the trunk are exposed, taking care not to injure the blood vessels in the axilla and groin.
5. In the female rat, not the mammillary glands forming transparent masses of fatty tissue, and remove these attached to the skin.
6. Note their blood supply in the neck, thoracic, axillary and inguinal regions. Notice also the pair of preputial glands on either side of the urethra opening.

The Abdomen

1. Lift the abdominal wall and cut forward in the midline until the xiphoid cartilage (at the posterior end of the sternum) is exposed and backwards to the base of the penis or clitoris.
2. Cut laterally from the xiphoid cartilage along the edges of the thorax and pin out the flaps of the abdominal wall.
3. Turn back the large sheaths of fat usually present in the posterior part of the abdomen.
4. Without moving the visceral, identify the liver, stomach, spleen, left kidney, coils of intestine, large caecum and in the female, uterus.

Solar Plexus

1. Gently move the visceral to the animal's right side, exposing the left veins and the origin of the celiac and anterior mesenteric arteries (this may be enclosed in fat).

2. Pour on a small amount of 70% alcohol to make nerves opaque.
3. Very carefully dissect away fat and lymphoid tissues, exposing the aorta and solar plexus.
4. Identify the celiac and anterior mesenteric sympathetic ganglia (close together and between the arteries of the same names), the cardiac sympathetic ganglion (by the aorta), and the greater and lesser splanchnic nerves (this arise from the ganglia of the sympathetic chains).

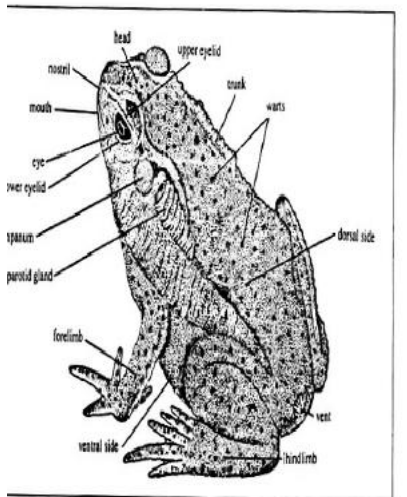
Alimentary canal

Separate the rectum from the duodenum and manipulate the alimentary canal looking for branches of the celiac and anterior mesenteric arteries and of the hepatic portal vein. The liver has four lobes; the left is large and undivided, the median or cystic has a deep fissure, the right is large and partly divided and the caudate or speigelian lobe is wrapped round the oesophagus. The oesophagus enters the kidney-shaped stomach which is divided into two regions: the white part contains no digestive glands whereas the pink, highly vascular part secretes pepsin, acid, and rennin and leads through the pyloric sphincter to the duodenum. The bile-duct enters the first part of the duodenal loop (there is no duodenal end of the duct). The pancreas is diffuse and pinkish in colour with many ducts entering the duodenum; it lies within the duodenal loop and also extends round the stomach to the spleen. The ileum (rest of the small intestine) is very much coiled so that its total length is about six times the length of the head and trunk of the animal. An ileocecal valve is present at the junction of ileum, caecum and colon and there are masses of lymphoid tissue. The blind end of the caecum forms a poorly defined appendix. The colon (large intestine) is short and the diagonal markings on its walls correspond to internal folds. The colon leads into the longer narrower rectum which may contain faecal pellets and opens at the anus.



Paper 6 of 10 — BAMBOY 100L 2nd semBIO 104BAMBOYBIOLOGY 104(1).pdf

Lecture/Notes · Google Drive harvest



Showing an adult Bufo in a sitting posture. Note the warty skin, parotid gland, bulging eye and elongated hindlimbs.

Jomo

The class amphibians

Toad

Adaptive features of a toad

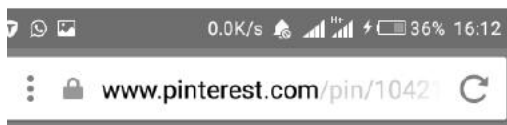
Presence of vascularise moist skin for gaseous exchange in water.

Presence of webbed fore and hind limbs for swimming

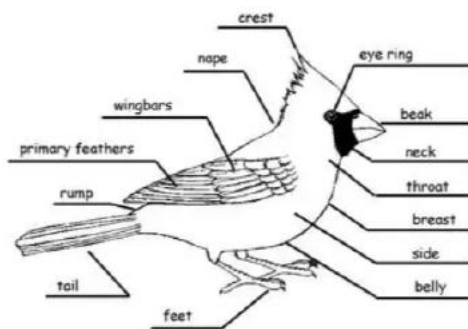
Presence of long sticky tongue for catching prey

Presence of buccal cavity for buccal respiration

Presence of tympanum membrane for



Name the Parts of a Bird



©Sheri An

The class aves

Adaptive features of a bird

Presence of streamline body for flight

Presence of a pair of wings with feathers for flying

Body covered with feathers and legs covered with scale to prevent loss of heat and water

Fore limbs modified into wings hind limbs modified into leg for working

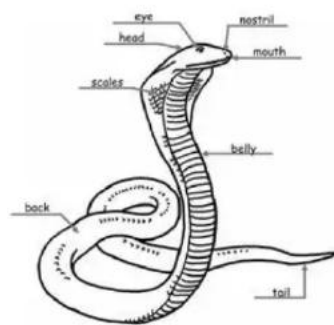
Beak and leg modified for feeding adaption

See More

0.05K/s 36% 15:54

Animal Diagram: Snake (labeled a

Name _____ Date _____
Parts of a Snake (reptile)



©2009 abcteach.com

The class reptilia

Adaptive features of a snake

Absence of limbs but presence of flexible body that aid locomotion

Presence of long tongue for catching prey

Presence of venom for escape and defense mechanism

Mimicry attitude as a means of catching prey

Presence of elastic jaw for for allowing different size of prey

1 of 2



The class Pisces

Presence of lateral line to detect vibration

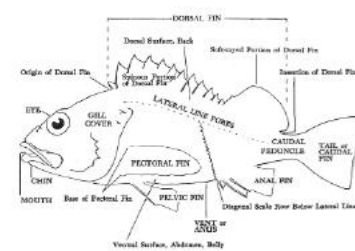
Streamlined body for easy movement in water

Presence of Gills for respiring in water

Body covered with scales to prevent loss of heat and water

0.1K/s 36% 16:17
www.meer.org/ebook/generalize

Generalized fish body see also:
Fishes, bony and cartilaginous
Fishes, bony and cartilaginous
Fishes, bony and cartilaginous
Fishes, bony and cartilaginous
Fishes, bony and cartilaginous
Fishes, bony and cartilaginous



A spiny-rayed fish, Salmo, showing fins and general body areas.

ADAPTED FROM BARNARD AND LEE, 1972, THANKS TO THE COURTESY OF MATHWORLD.COM

Presence of caudal finally for balancing

Presence of dorsal fins for swimming

Jomo

The class mammalian

Presence of nose for detecting odour or aroma

Presence of pinna/external ear for hearing and sound detection

Presence of whiskers for detecting vibration

Presence of hair or fur on the body to prevent loss of heat and water

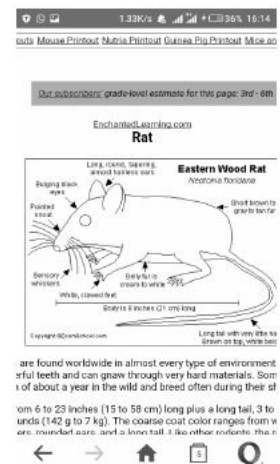
Presence of long tail for balancing

Presence of flexible body for easy movement in channel and holes

Presence of fore and hind limbs for locomotion and escape from danger

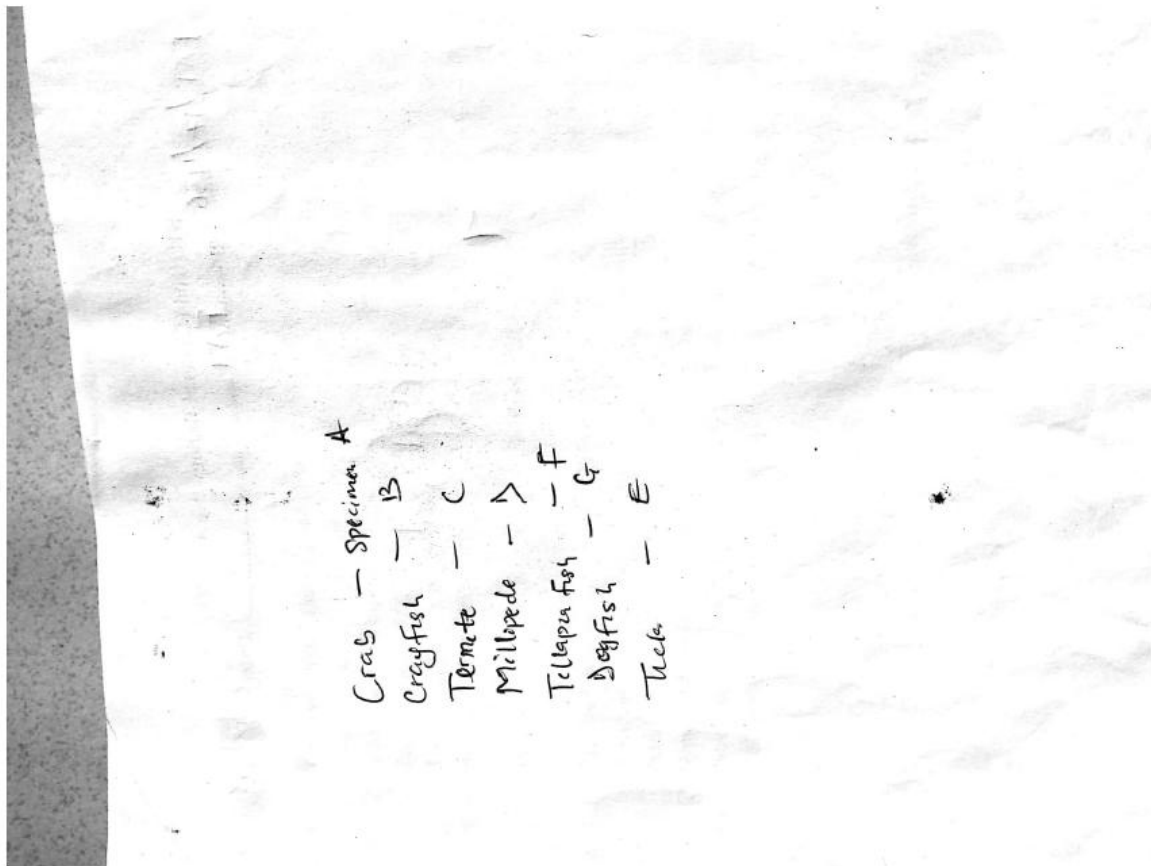
Mouth parts modified for biting and chewing

Best of luck.....

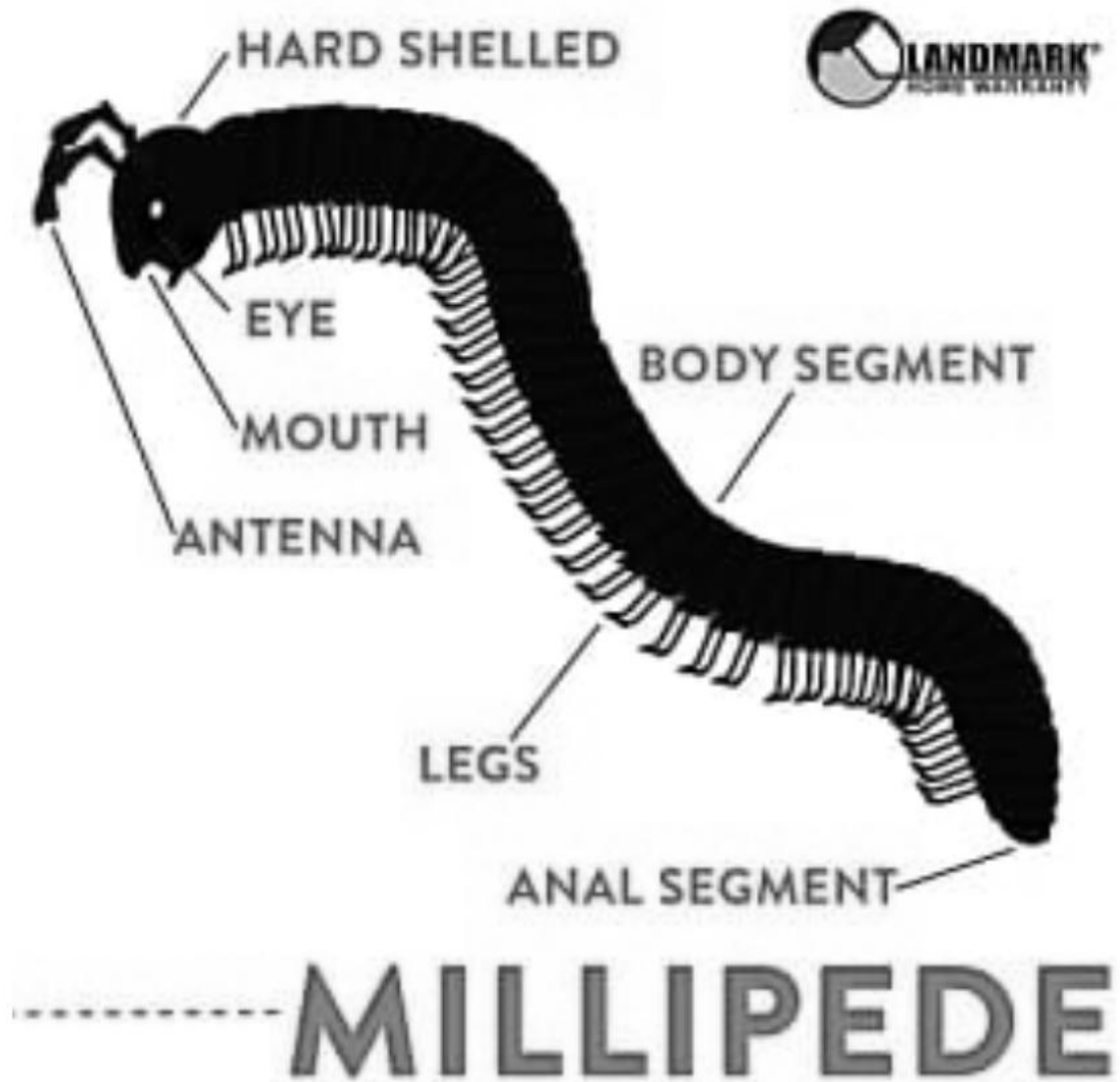


Paper 7 of 10 — BAMBOY 100L 2nd semBIO 104BIO 104 by BAMBOY_20.pdf

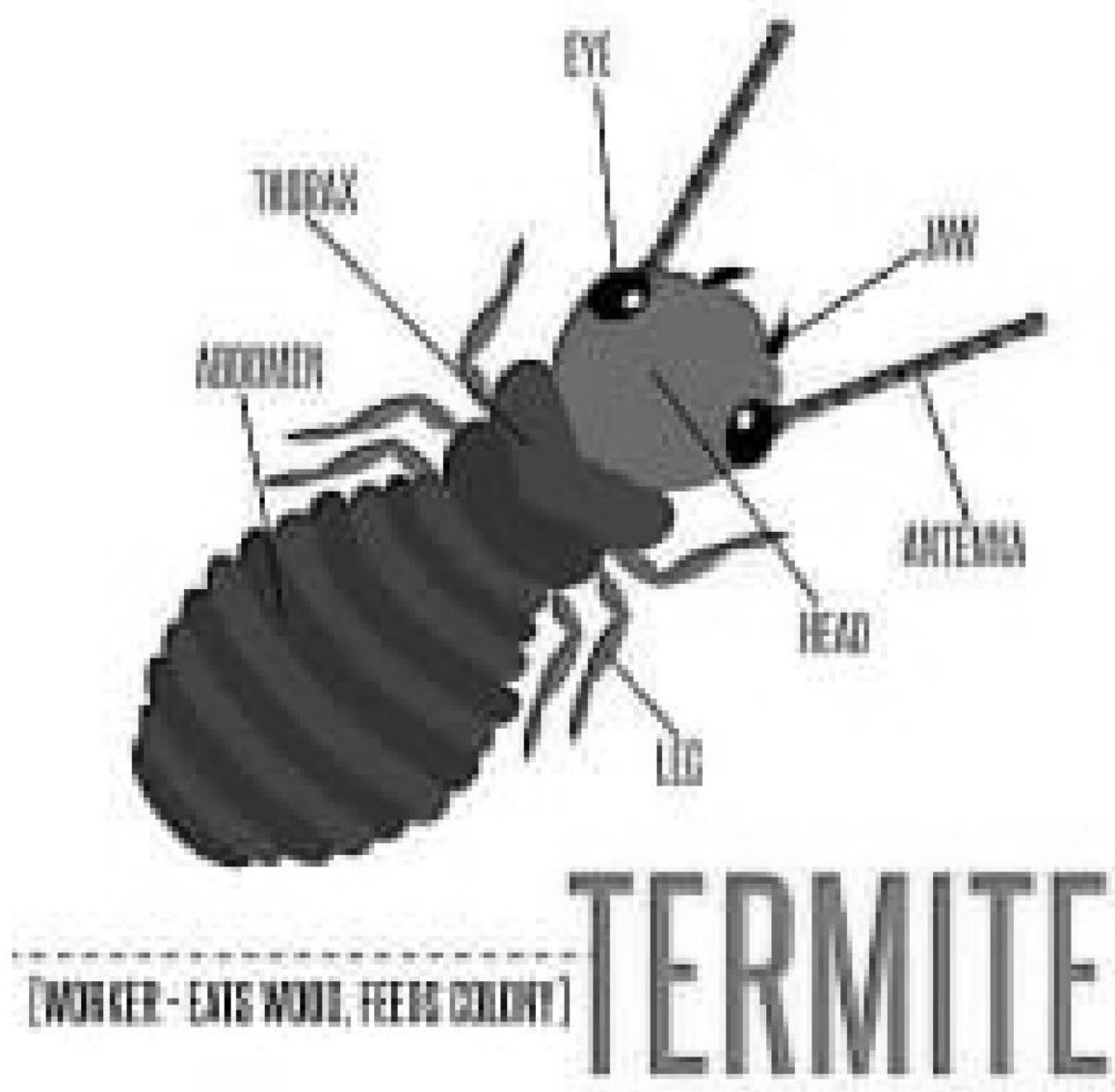
Lecture/Notes · Google Drive harvest



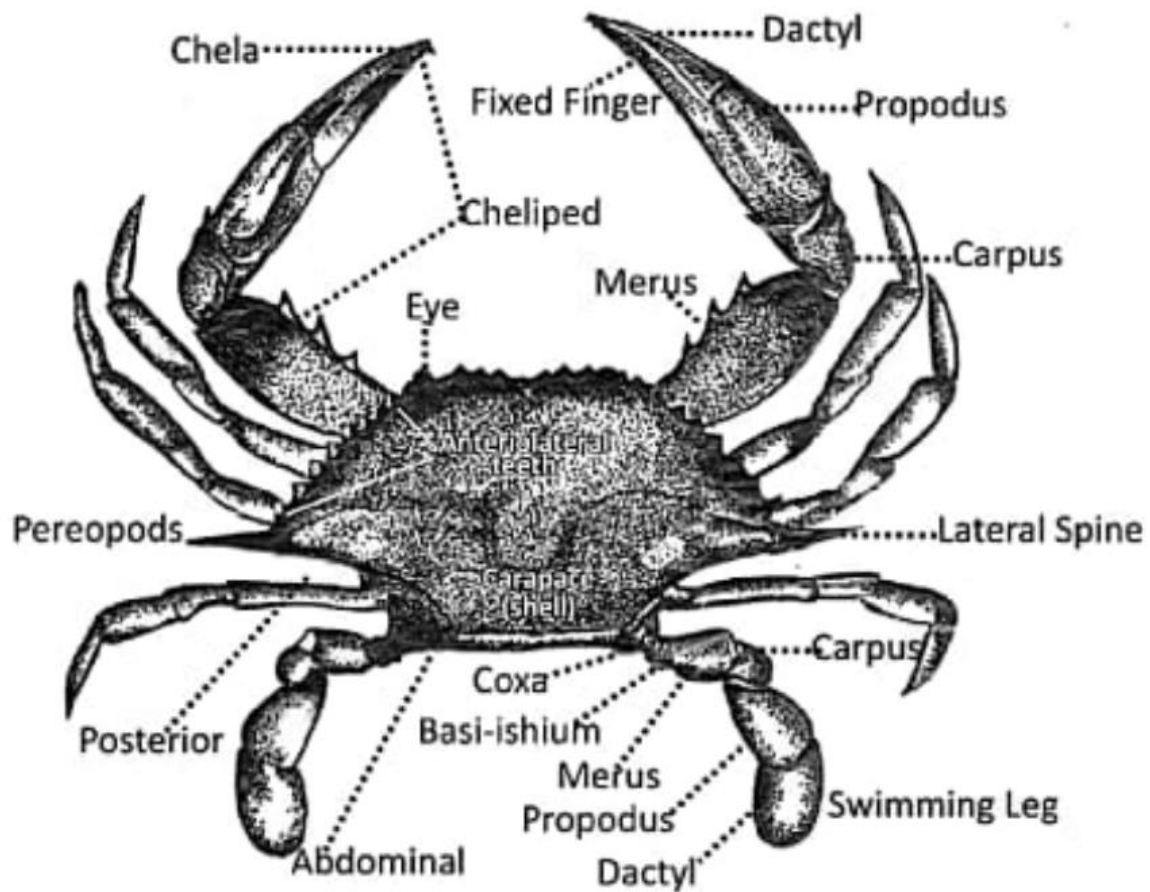
Scanned by CamScanner



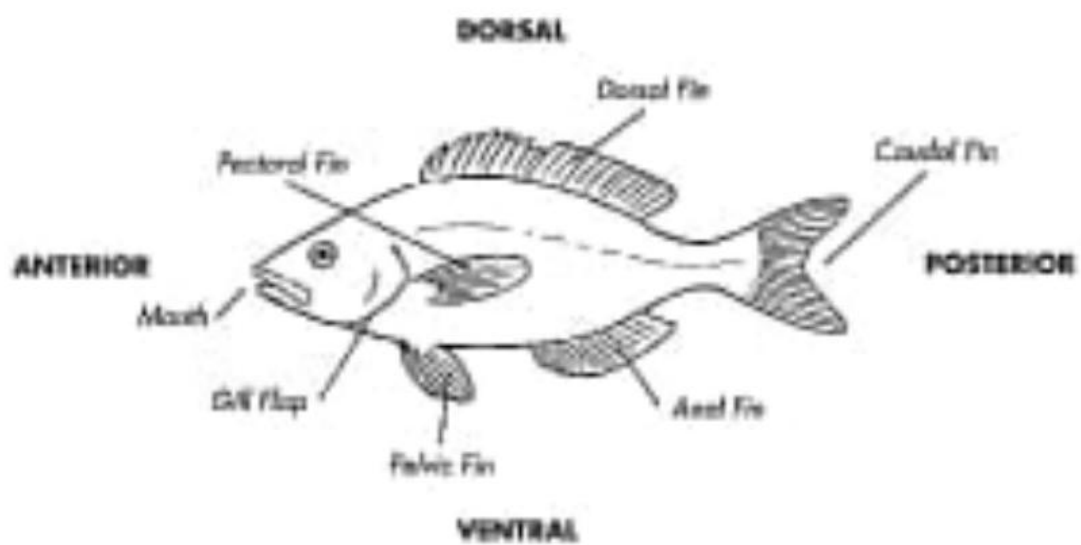
Scanned by CamScanner



Scanned by CamScanner



Scanned by CamScanner



Scanned by CamScanner

PRACTICAL III - THE ARTHROPODS

Introduction:

PHYLUM - ARTHROPODA

1. They have bilateral symmetry and metameric segmentation.
2. They are triploblastic.
3. Possession of jointed appendages.
4. Highly cephalised, exoskeleton of cuticle containing protein, lipid, chitin and often calcium carbonate secreted by underlying epidermis and shed at intervals.
5. Coelom is reduced in adult, most of body cavity consisting of haemocoel (sinuses or spaces in the tissue) filled with blood.
6. Sexes are separate, respiration by body surface, gills, trachea (air tubes) or book lungs.

Examples:

- a) Subphylum - Chelicerata
Class Arachnida e.g. Horseshoe crabs, scorpions, spiders, ticks, aquatic chelicerates like *Limulus* etc.
- b) Subphylum - Crustacea e.g. prawn, crayfish, crab, shrimps, waterfleas etc.
- c) Subphylum - Uniramia (Insects and Myriapods)
 - i) Class - Diploda (Millipedes) e.g. *Julus*, *Spirobolus* etc.
 - ii) Class - Chilopoda (Centipede) e.g. *Cermatia*, *Lithobius*, *Geophilus*, etc.
 - iii) Class - Pauropoda e.g. *Pauropus*
 - iv) Class - Insecta e.g. Beetles, butterflies, dragonflies, bugs, flies, termites, bees, wasps, etc.

Exercise:

Observe, draw and label all the arthropod specimens supplied. Some of them are millipedes, centipedes, scorpions, mites, spiders, mosquitoes, bugs, beetles, flies, ants, crab, crayfish, cotton stainers, butterflies etc.

PHYLUM - ECHINODERMATA

1. Body unsegmented (non-metameric) with radial, pentamerous symmetry
2. No head or brain
3. Locomotion by tube feet
4. Coelom is extensive
5. Sexes are separate (except a few hermaphrodites) with large gonads.

Examples: Sea Stars, *Asterias*, *Plaster*, Brittle Stars, Basket Stars, Starfish, Sea urchins, Sea biscuits, Sand dollars, Sea cucumbers, Sea lilies, feather stars, *Arbacia*, *Mellina*, *Parastichopus*, *Cucumaria*, etc.

Scanned by CamScanner

ClOgumABA Bukola MME FAT/12/1136

BIO 104 (General Biology Practical II)

Marie No. FAT/12/1136 Dept. FAT Date of exam...

2- Similarities between Specimen A and B

i. Both have distinct head

ii. Both have Eyespot

Similarities between specimen A and E

i. Both have legs for movement

ii. Both have head

Similarities between specimen C and B

i. Both have distinct head

ii. Both have eyespot

Similarities between specimen C and E

i. Both have pairs of appendages

ii. Both have pairs of appendages

Similarities between specimen D and B

i. Presence of bristles between both skeleton specimen

ii. Both have head

Similarities between specimen D and E

i. Both have head [Erythrotoxin]

ii. Both have mouth

Similarities between specimen

Differences between specimen A and B

Specimen A	Specimen B
i. Presence of claw	Absence of claw
ii. Presence of compound eyes	Presence of simple eyes

33

Scanned by CamScanner

Difference Between Specimen A and E	
Specimen A	Specimen E
Presence of claws	Absence of claws
Absence of abdomen	Presence of abdomen

Difference Between Specimen B and C	
Specimen B	Specimen C
Absence of wings	Presence of wings
Unsegmented body	Segmented body

Difference Between Specimen D and E	
Specimen D	Specimen E
Presence of wings	Absence of wings
Segmented body	Unsegmented body

Difference Between Specimen B and D	
Specimen B	Specimen D
Presence of tail	Absence of tail
Presence of anus	Absence of anus

Scanned by CamScanner

OLUGUNGBA BUKOLA MAR 1 FAT/17/1936
 bf-18

Matric No. FAT/17/1936 Dept. FAT Date 01/08/2018
 BIO 104 (General Biology Practical II)

Difference Between Specimen D and E

Specimen D	Specimen E
① Present of segmented body	① Absent of segmented body
② Posses numerous walking legs.	② Posses few legs.

③ Compare and Contrast Specimen F and G

Difference between Specimen F and G

Specimen F	Specimen G
① Presence of gill cover	① Absence of gill cover
② Presence of scale	② Absence of body scale -

Similarities of Specimen F and G

① They both have tail

② They both have dorsal fin

Scanned by CamScanner

Marie N.

Bf=1

Physiological Adaptation of Specimen A

Specimen A (Coral)

- ① Possession of claws of mouthparts for catching prey.
- ② Possession of swimming leg which is used for swimming while.

Specimen C (Termite)

- ① Possession of labrum and mandible for chewing and piercing.
- ② Possession of antennae for sensitivity.

Specimen D (Millipede)

- ① Possession of labrum and mandible for chewing.
- ② Possession of eyes for visualizing both predators and prey.

Specimen F (Tilapia fish)

- ① Possession of scales to prevent loss of heat.
- ② Possession of tail for steady movement/steering in water.

Specimen B (Cray fish)

- ① Possession of streamlined body.
- ② Possession of eye for detecting light.

Specimen E (Tick)

- ① Possession of pedipalps to control abhor.
- ② Possession of salivarium for covering its body.

Specimen G (Dog fish)

- ① Possession of tail, for propelling backward and forward.
- ② Possession of streamlined body for easy movement in water.

36

CP-2284

Scanned by CamScanner

- (1) SIMILARITIES BETWEEN SPECIMEN A & C Ref: 4/2
- i Possession of two pairs of legs / limb
 - ii Presence of head

DIFFERENCES BETWEEN SPECIMEN A & C

SPECIMEN A	SPECIMEN C
i Presence of scales	• Feather is present
ii Presence of trunk	• Trunk is absent.

b SIMILARITIES BETWEEN B & C

- i Both possess head which comprise,
- ii Both possess eye, mouth.

DIFFERENCES BETWEEN SPECIMEN B & C

SPECIMEN B	SPECIMEN C
i Absence of neck	• Presence of neck
ii Presence of tail	• Absence of tail.

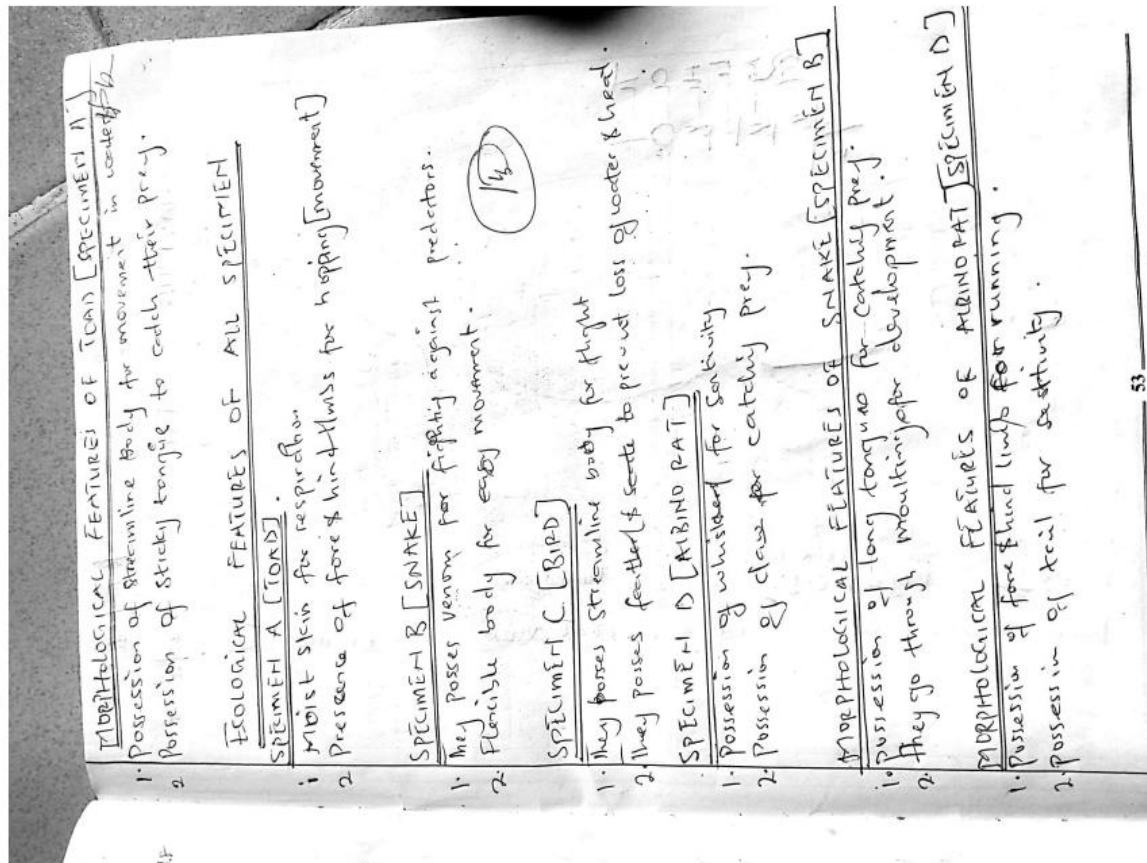
SIMILARITIES BETWEEN SPECIMEN D & C

- i They both possess two pairs of legs
- ii They both have necks

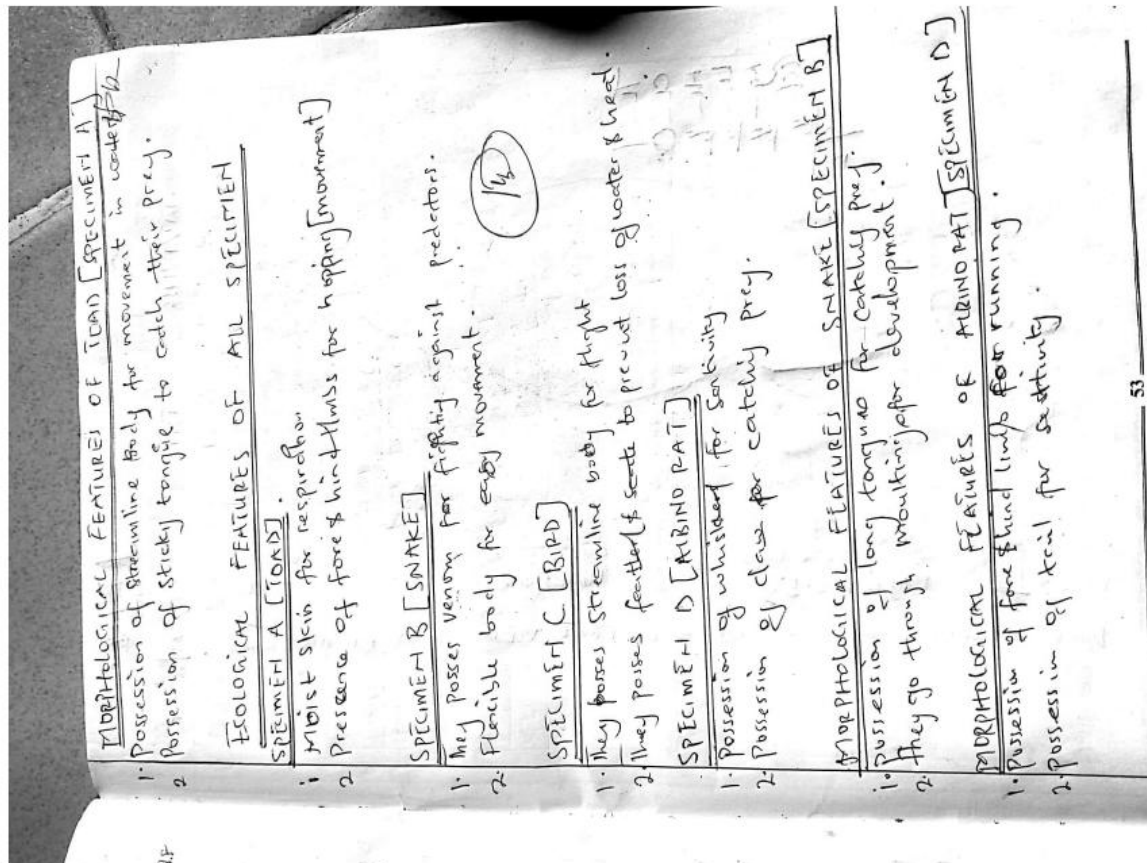
DIFFERENCES BETWEEN SPECIMEN D & C

SPECIMEN D	SPECIMEN C
i Presence of fur	• No fur - Presence of feather
ii Possession of tail	• Absence of tail.

Scanned by CamScanner



Scanned by CamScanner



DIFFERENCE
IMPORTANCE B/W CONTOUR / V
BY DIFFERENCE

CONTOUR / VANE FEATHER

1. long rachis

2. Presence of posterior

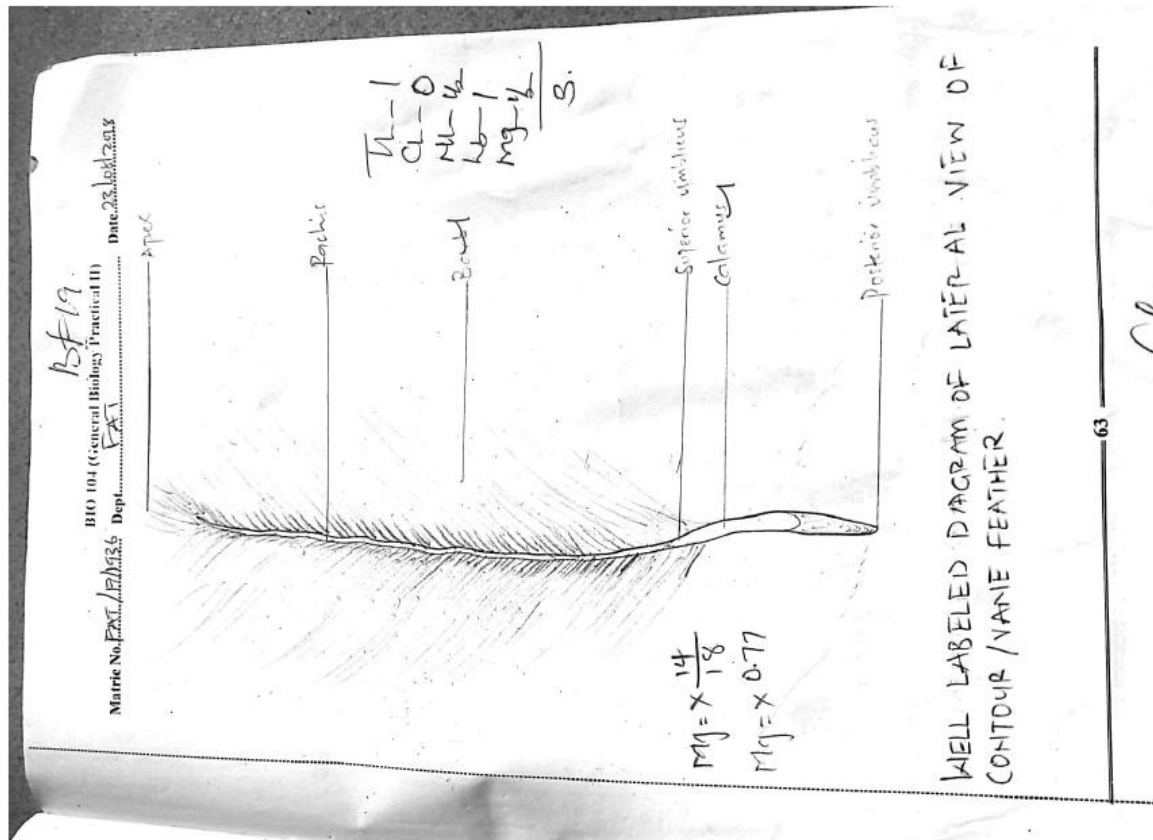
Superior Umbilicus

IMPORTANCE OF COH

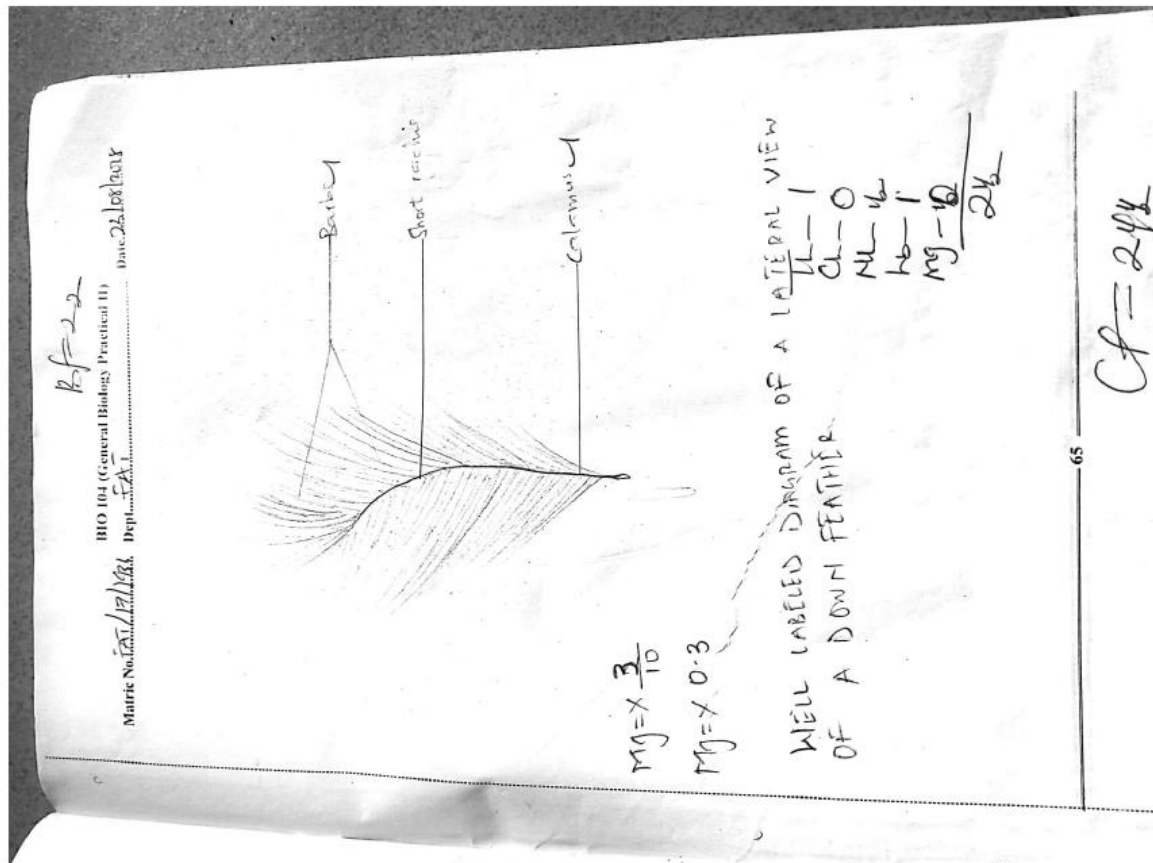
- I. For temperature regulate
- II. It is used to guide

IMPORTANCE OF DO

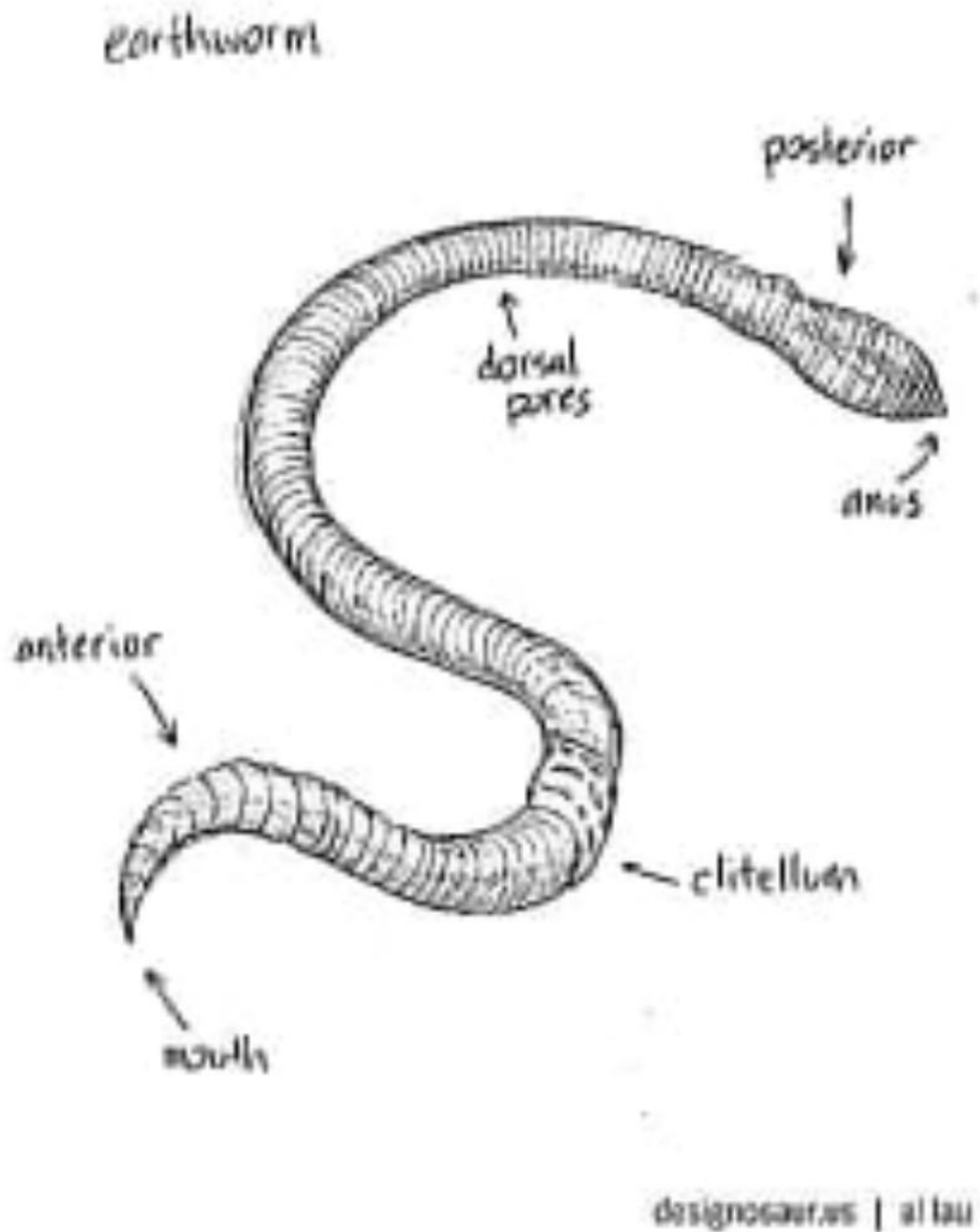
Scanned by CamScanner



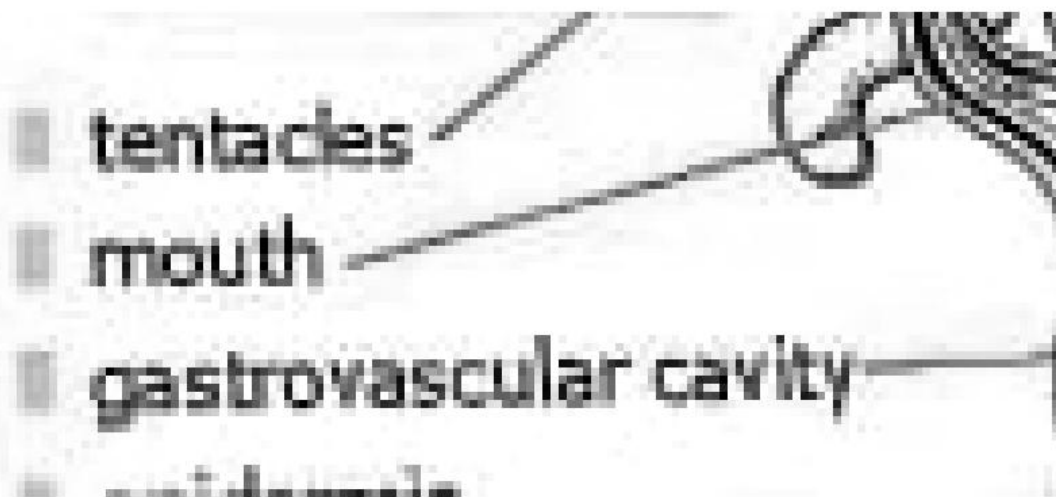
Scanned by CamScanner



Scanned by CamScanner



Scanned by CamScanner



Scanned by CamScanner

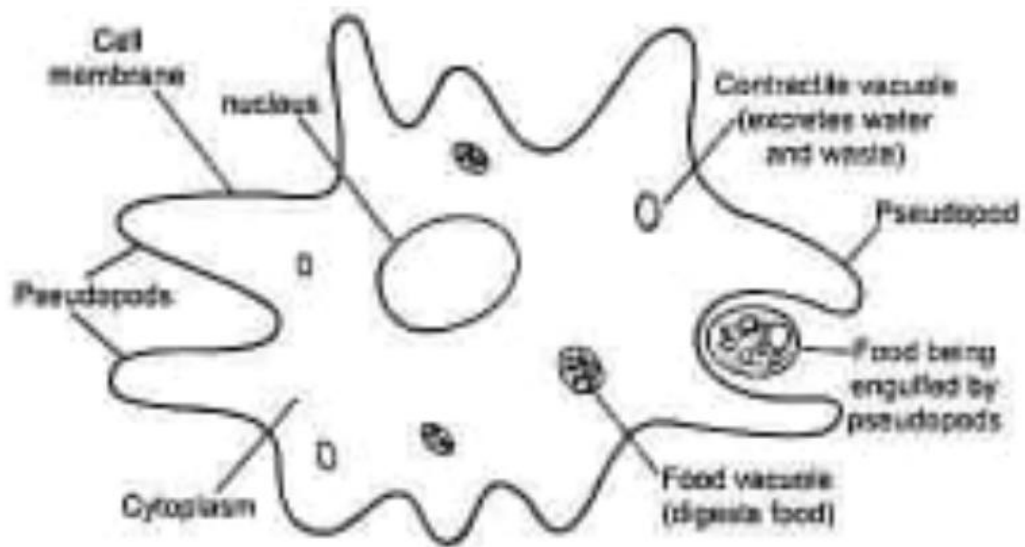
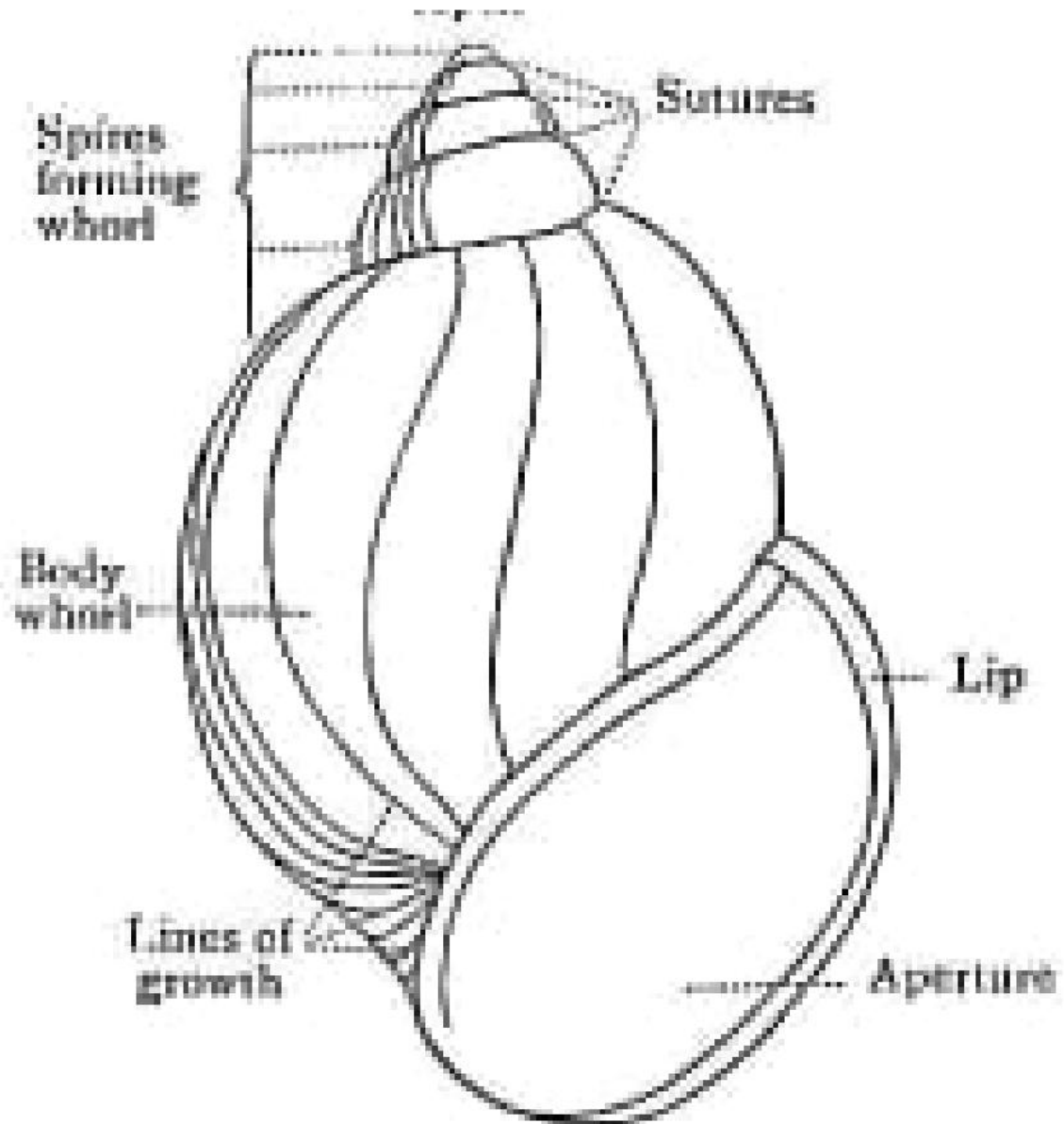
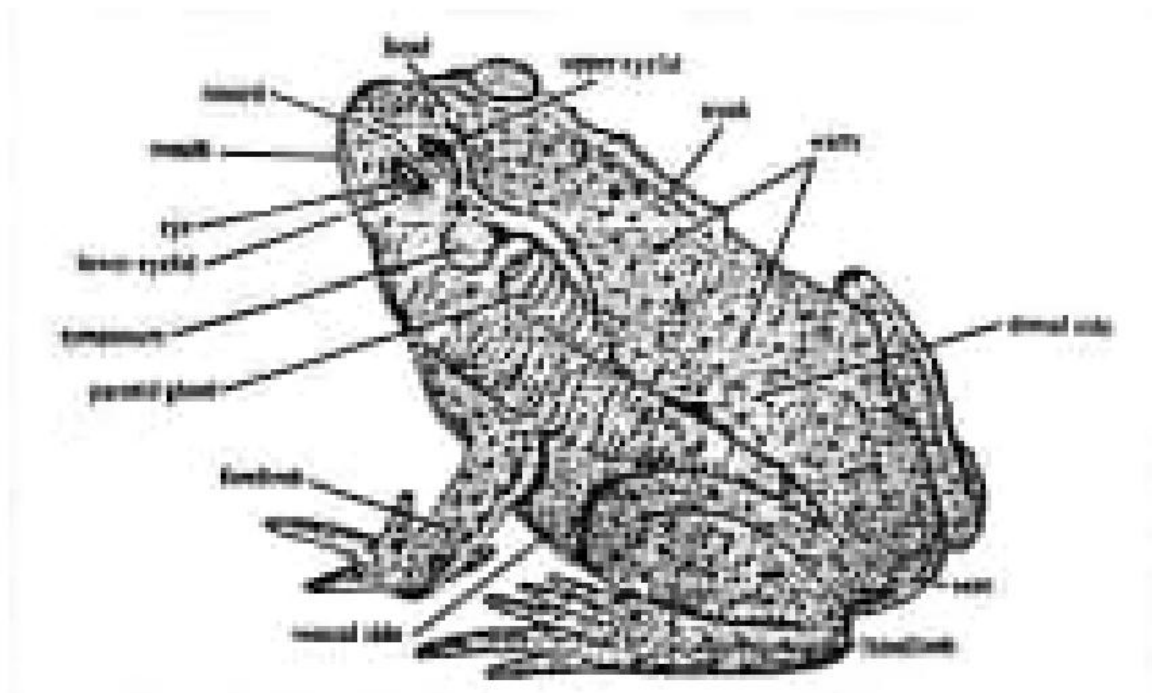


Fig. 9.7 Amoeba proteus

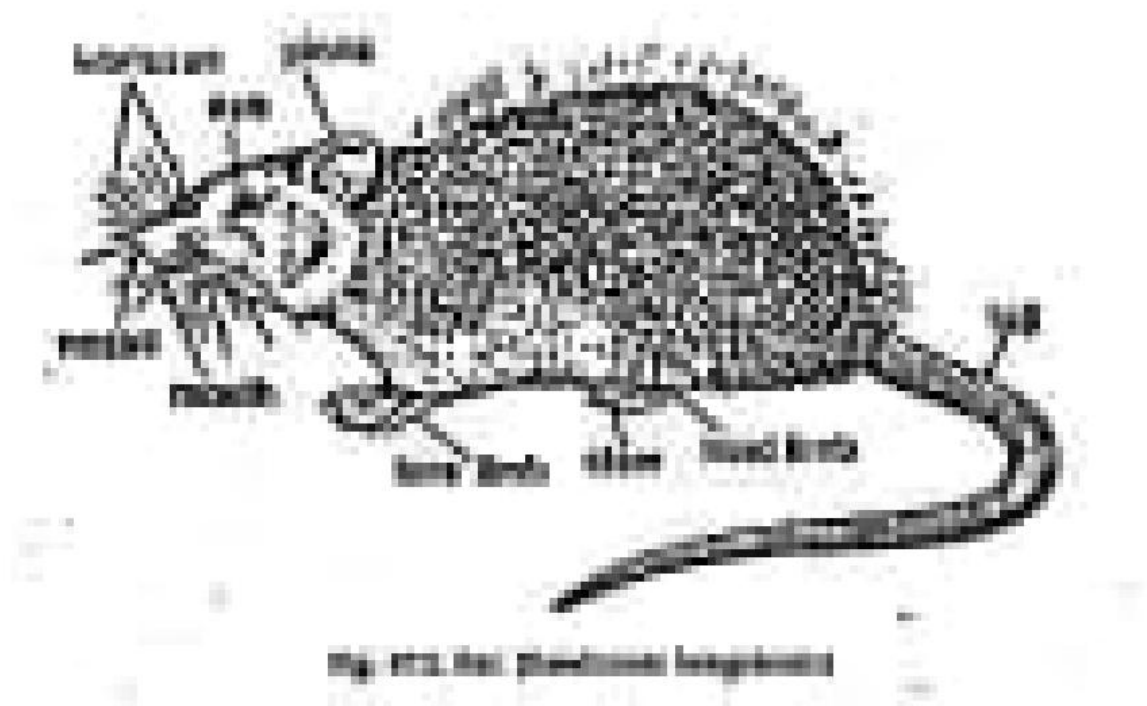
Scanned by CamScanner



Scanned by CamScanner



Scanned by CamScanner



Scanned by CamScanner

B#194

BIO 104 (General Biology Practical II)

Matric No. FA/17/028Dept. FATDate 06/09/2019Ecological features of Specimen C

1. They are found in ^{body} of sheep and goat
- 2.

Ecological features of Specimen E [Land snail]

1. Possession of shell for protection.
2. Possession of tentacle eye for sensitivity

(1/5)

Ecological features of Specimen D [Earthworm]

1. Possession of moist skin for easy movement in the soil.
2. Possession of segmented body for movement.

Scanned by CamScanner

Bf-164

Matric No. FAT/17/1936

BIO 104 (General Biology Practical II)

Dept. FAT

Date: 06/09/2018

Similarities between Specimen A and Specimen E

1. They are unsegmented body.
2. They are both unicellular organisms.

Similarities between Specimen A and Specimen E

1. They are unsegmented body.
- 2.

Different Between Specimen A and Specimen E

Specimen A	Specimen E
1. They are coelomates i.e. present of body cavity. 2. It is a unicellular organism.	1. They are acoelomate i.e. body cavity is not present 2. They are multicellular organism

Different between Specimen A and Specimen E

Specimen A	Specimen E
1. They have no sense organ 2. They are not bilaterally segmented	1. They have simple sense organ 2. They are bilaterally segmented.

Similarities between Specimen B and Specimen C

1. They both have mouth
2. They both reproduce asexually

BFA74

BIO 104 (General Biology Practical II)
 Matric No. BAT/17/1936 Dept. FAT Date 06/08/2018

Different between Specimen D and Specimen C

Specimen D	Specimen C
1. They have segmented bodies	They do not have a segmented body
2. They have two germ layers	They have three (3) germ layers

Different between Specimen D and Specimen E

Specimen D	Specimen E
1. They have segmented body	They don't have segmented body
2. It is segmented body	It moves with the aid of their 15 used for movement mouth.

Ecological Features of Specimen A

1. They move with the aid of pseudopodia.
2. They are found in moist areas. (2)

Ecological features of Specimen B.

1. They move with waves of their tentacles.
2. They are mostly found in fresh water and salt water.

Scanned by CamScanner

PRACTICAL I-ROTOZOANS AND COELENTERATES

Introduction:

PHYLUM PROTOZOA

1. They are unicellular, some colonial and some with multicellular stages in their life cycles.
2. They are mostly microscopic, although some are large enough to be seen with naked eye.
3. Locomotion by pseudopodia, flagella, cilia and direct cell movements.
4. Nutrition of all types – autotrophic, heterotrophic, saprozoic
5. Free-living, mutualism, commensalism, parasitism are all represented in the group.

Examples: *Ameoba*, *Entamoeba*, *Paramecium*, *Vorticella*, *Monocystis*, *Eimeria*, *Plasmodium*, *Toxoplasma*, *Babesia*, *Opalina*, *Trichomonas*, *Trypanosoma*, *Leishmania* etc.

Unicellular, eukaryotic

Class: Mastigophora (Flagellate) e.g. *Euglena* sp, *Chlamydomonas* sp

Class: Rhizopoda (Sarcodina) e.g. *Ameoba* sp

Class: Ciliophora e.g. *Paramecium* sp

Class: Sporozoa e.g. *Plasmodium* sp

THE COELENTERATES

This group comprises the cnidarians and ctenophorans among others. They are aquatic, radially symmetrical, diploblastic and have no blood system e.g. *Hydra*, sea anemone, hydroids, jelly fish.

PHYLUM – CNIDARIA

1. Two basic types of individuals: polyp and medusa.
2. Radial symmetry or biradial symmetry around a longitudinal axis with oval and arboreal ends.
3. Asexual reproduction by budding (in polyps) or sexual reproduction by gametes (in Medusae and some Polyps).
4. Presence of special stinging cell organelles called nematocysts in either epidermis or gastrodermis or in both.
5. Body with two layers, epidermis and gastrodermis with mesogloea (Diplo/blastic).

Examples: *Hydra*, *Obelia*, *Tubularia*, *Aurelia*, *Acropora*, *Tubipora*, etc.

PHYLUM – CTENOPHORA

1. Symmetry biradial
2. Nematocysts absent (except in one species), but adhesive cells (colloblasts) present
3. No polymorphism
4. Luminescence common

PRACTICAL II - WORMS AND MOLLUSCS

Introduction:

PHYLUM - PLATYHELMINTHES

1. Bilateral symmetry, definite polarity of anterior and posterior ends.
2. Three germ layers (triploblastic).
3. No internal body space other than digestive tube (Acoelomate).
4. Presence of simple sense organs – eyes / spots in some.
5. Class Turbellaria are mostly free living. Classes Monogenea, Trematoda and Cestoda are entirely parasitic.

Examples: *Planaria*, *Digenia*, *Gyrodactylus*, *Dactylogymus*, *Fasciola*, *Clonorchis*, *Schistosoma*, *Paragonimus* (lung flukes), *Taenia solium*, *Taenia sp*, *Echinococcus multilocularis* etc.

PHYLUM NEMATODA

1. Unsegmented round worms with slender cylindrical bodies.
2. They possess pseudocoel.
3. They are mostly fluid feeders.
4. They have no circulatory or respiratory system.
5. Sexes are usually separate.

Examples: *Ascaris lumbricoides* (roundworm), *Enterobius*, *Ancylostoma* and *Necator* (hookworms), *Onchocerca*, *Dracunculus* (guinea worm), *Trichinella* etc

PHYLUM ANNELIDA

1. Body metamerically segmented, symmetry-bilateral.
2. Chitinous setae often present, Setae absent in leeches.

3. Coelom (Schizocoel) well developed and divided by septa except in leeches.
4. Hermaphroditic or separate sexes.
5. Larvae if present are trochophore type.

Examples: Earthworms, *Lumbricus*, *Stylaria*, *Nereis*, Leech or *Hirudo*, *Macrobdella* etc.

PHYLUM MOLLUSCA

1. Body bilaterally symmetrical (bilateral asymmetry in some).
2. Body unsegmented often with definite head.
3. Coelom limited mainly to area around the heart and perhaps lumen of gonads and part of kidneys.
4. Ventral body wall specialized as a muscular foot, variously modified but used chiefly for locomotion.
5. Sensory organs to touch, smell, taste, equilibrium and vision (in some): eye highly developed in Cephalopods.

Examples: *Neoplina*, *Chitons*, *Dentalium*, Snails and slugs such as *Archachatina marginata*, *A. fulica*, *Helix*, *Aplysia*, *Apodonta*, *Venus*, *Tagelus*, *Teredo*, *Anadara senilis*, *Egaria*, *Octopus*, *Sepia* etc.

Exercises:

1. **The Platyhelminthes (flatworms)**
You are provided with prepared slides of *Taenia piriformis*. Note the section provided and draw. What is the economic importance of *Taenia sp*?
2. **The Nematodes (roundworms)**
 - Draw and label the prepared slides of *Ascaris sp* provided.
 - Note the section/part
 - What is the main difference between a nematode and a platyhelminth?
3. **The Annelids**
 - Draw and label the specimen of *Lumbricus sp.* (The earthworm) provided.
 - What are the differences between a round worm and an annelid?
4. **Molluscs**
Observe and draw the supplied specimen of *Archachatina marginata*.

Scanned by CamScanner

PRACTICAL III - THE ARTHROPODS

Introduction:

PHYLUM - ARTHROPODA

1. They have bilateral symmetry and metameric segmentation.
2. They are triploblastic.
3. Possession of jointed appendages.
4. Highly cephalised, exoskeleton of cuticle containing protein, lipid,
3. Possession of jointed appendages.
4. Highly cephalised, exoskeleton of cuticle containing protein, lipid, chitin and often calcium carbonate secreted by underlying epidermis and shed at intervals.
5. Coelom is reduced in adult, most of body cavity consisting of haemocoel (sinuses or spaces in the tissue) filled with blood.
6. Sexes are separate, respiration by body surface, gills, trachea (air tubes) or book lungs.

Examples:

- a) Subphylum – Chelicerata
Class Arachnida e.g. Horseshoe crabs, scorpions, spiders, ticks, aquatic chelicerates like *Limulus* etc.
- b) Subphylum – Crustacea e.g. prawn, crayfish, crab, shrimps, waterfleas etc.
- c) Subphylum – Uniramia (Insects and Myriapods)
 - i) Class – Diplopoda (Millipedes) e.g. *Julus*, *Spirobolus* etc.
 - ii) Class – Chilopoda (Centipede) e.g. *Cermatia*, *Lithobius*, *Geophilus*, etc.
 - iii) Class – Pauropoda e.g. *Pauropus*
 - iv) Class – Insecta e.g. Beetles, butterflies, dragonflies, bugs, flies, termites, bees, wasps, etc.

Exercise:

Observe, draw and label all the arthropod specimens supplied. Some of them are millipedes, centipedes, scorpions, mites, spiders, mosquitoes, bugs, beetles, flies, ants, crab, crayfish, cotton stainers, butterflies etc.

PHYLUM - ECHINODERMATA

1. Body unsegmented (non-metameric) with radial, pentamerous symmetry
2. No head or brain
3. Locomotion by tube feet
4. Coelom is extensive
5. Sexes are separate (except a few hermaphrodites) with large gonads.

Examples: Sea Stars, *Asterias*, *Piaster*, Brittle Stars, Basket Stars, Starfish, Sea urchins, Sea biscuits, Sand dollars, Sea cucumbers, Sea lilies, feather stars, *Arbacia*, *Mellita*, *Parastichopus*, *Cucumaria*, etc.

Scanned by CamScanner

PRACTICAL V-AMPHIBIANS

Introduction:

1. They are Ectothermic tetrapods.
2. Live in water and also adapted for life on land.
3. Skin moist and slimy (containing mucous glands) and lacks scales.
4. Mouth usually large with small teeth in upper or both jaws, two nostrils open into anterior part of mouth cavity.
5. Blood vascular system consists of a three chambered heart, two atria and one ventricle, and a double circulation through the heart.
6. Respiration by lungs, skin and gills in some; either separately or in combination.

Examples: Toad (*Bufo regularis*), frogs, Salamanders, Caecilians, etc.

Basically two major types

- (1) The limbless type of the caecum caecilian – worm-like and
- (2) Those with limbs which can be subdivided into four (4) groups:
 - Early/Extinct amphibians (Rachitons, cacops)
 - Urodeles / Canadata (Newts and Salamanders)
 - Apodans / Shark-like
 - Anura / Salienta (Frogs and Toads)

Exercises:

- a) Make a drawing of the specimen of a live toad / preserved toad and label parts fully.
- b) Examine eggs and larva (tadpole). Draw external gill stage and the internal gill stage.
- c) Dissect a toad/frog and draw the alimentary canal, respiratory system and reproductive organs

PRACTICAL VI-REPTILES

Introduction:

1. Body varied in shape, compact in some and elongated in others.
2. They possess shelled eggs, they have no gills and larva stage.
3. Sexes are separate and fertilization is internal.
4. They have dry skin, their body is covered with an exoskeleton of horny dermal scales with addition sometimes of bony dermal plates.
5. They are ectothermic and many behaviourally thermoregulate.
6. Heart consists of three chambers; Crocodiles have four chambered heart.

Examples: Turtles, Tortoise, Lizard (e.g. *Agama agama*), Snakes, Crocodiles, etc.

Scanned by CamScanner

Three (3) main groups of the living or modern day reptiles:

- Chelonian = Tortoise and Turtles
- Crocodilian = Alligators and Crocodiles
- Squamata = Lizards and Snakes

A fourth group known as the Rhynchocephalia consists of mainly extinct members which were lizard-like.

Exercises:

- a) Examine the features of the Lizard, *Agama agama*.
- b) Draw the side view/ dorsal lateral view and label.
- c) Recall the structure of the fishes and the toad. Note any feature(s) of resemblance or difference to the fish and the toad.

THE LIZARD (*Lacerta yirulis*)

EXTERNAL FEATURES

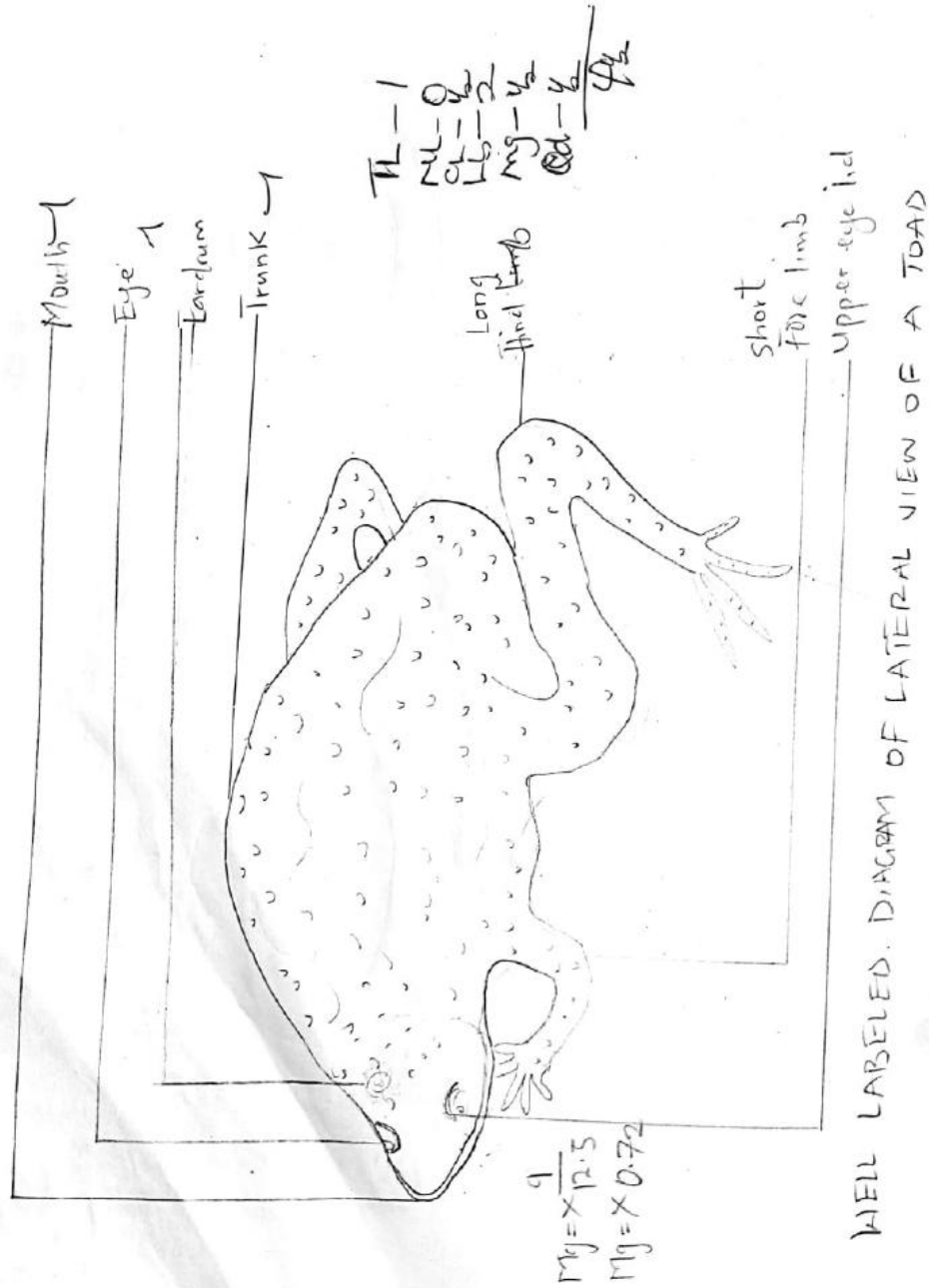
NOTE: That the general appearance of the body is very much like that of the salamander. The limbs are a little longer but hardly raise the belly off the ground. The surface of the body is covered with horny scales arranged in a regular pattern. On the head the scales are large and closely applied to the roofing bones of the skull, but the position of these scales does not correspond with that of the underlying bone. The scales on the under surface are larger than the scales on the upper surface. The body behind the head region is slightly constricted to form the neck. The tail is long and tapering, being about twice the length of the head and trunk together. The mouth is tenninal as in the salamander. The paired external nares lie on either side at the tip of the snout. The eyes lie on either side of the head and are provided with movable eyelids which can completely cover over the eye. A nictitating membrane freely exposed on the surface. (In the skin the tympanic membrane is covered by a fold of skin bearing scales). Open the mouth and observe the small teeth on the upper and lower jaws. On the floor of the mouth lies the bifid tongue which is attached posteriorly. On the fore and hind limbs there are five long digits each ending in strong claws. The fourth digit on the hind limb is much longer than the rest just behind the hind limb lies the cloaca which appears as a transverse slit. A large transverse scale, the preanal plate, lies in front of the cloaca;-

GRASS SNAKE. (*Tropidonotus natrix*)

EXTERNAL FEATURES

Observe the elongation shape of the body and note the entire absence of limbs. This elongation of the body is produced by drawing out the trunk region, the tail region behind the cloaca is no more developed than it is in Lacertilia. Find the cloaca a short distance from the posterior end of the body. It is a transverse slit, bounded in front by two scales which fold over it and make the opening rather inconspicuous. The general surface of the body is covered by epidermal scales arranged in a regular pattern. The scales on the dorsal surface are small except in the head region where they are large and fit closely to the skull. As in the lizard, these large scales on the head do not correspond in

BIO 104 (General Biology Practical II)
 Matric No. FAT/17/1936 Dept. FAT Date 22 October



50

CF-44

Scanned by CamScanner

SIMILARITIES BETWEEN SPECIMEN A & C Ref: 4/6

- i Possession of two pairs of legs / limb
- ii Presence of head

DIFFERENCES BETWEEN SPECIMEN A & C

SPECIMEN A	SPECIMEN C
i Presence of scales	Feather is present
ii Presence of trunk	Trunk is absent.

b SIMILARITIES BETWEEN B & C

- i Both possess head. which comprise,
- ii Both possess eye, mouth.

DIFFERENCES BETWEEN SPECIMEN B & C

SPECIMEN B	SPECIMEN C
i Absence of neck	presence of neck
ii Presence of tail	Absence of tail.

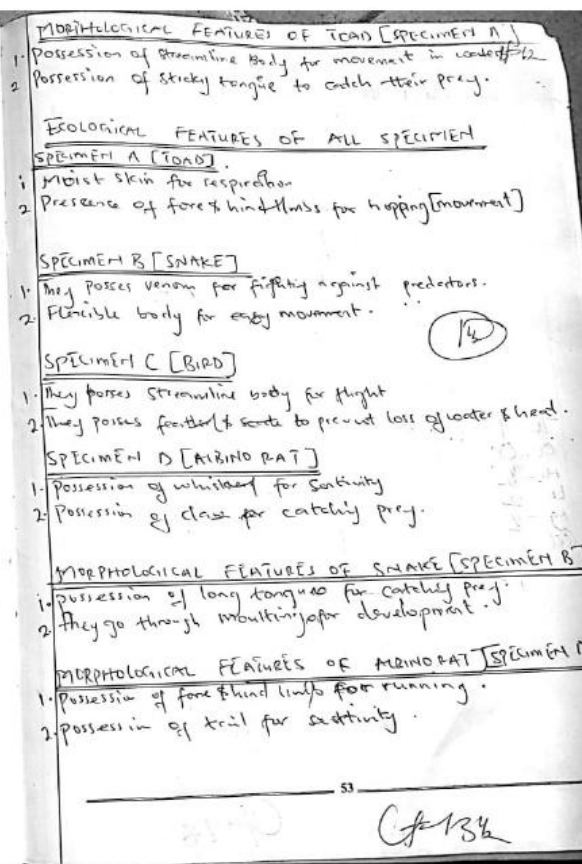
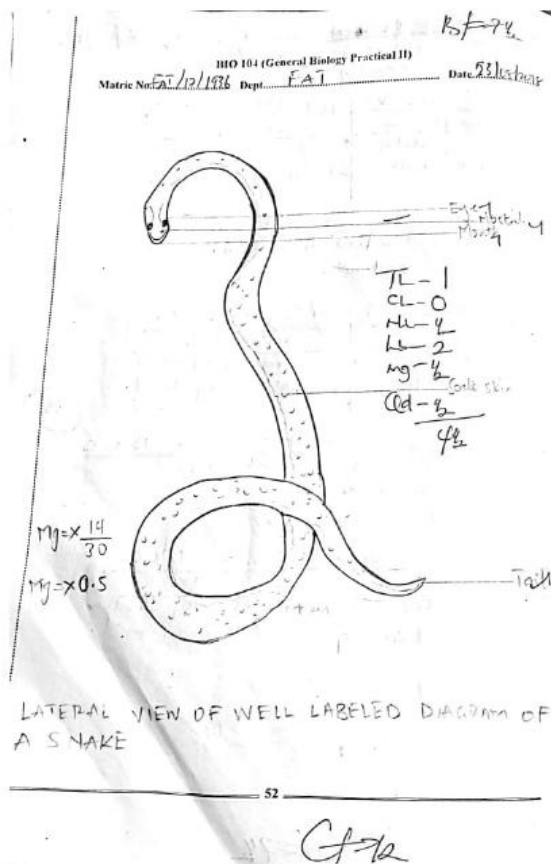
SIMILARITIES BETWEEN SPECIMEN D & C

- i They both possess two pairs of legs
- ii They both have necks

DIFFERENCES BETWEEN SPECIMEN D & C

SPECIMEN D	SPECIMEN C
i Presence of fur	no fur - Presence of feather
ii Possession of tail	Absence of tail.

Scanned by CamScanner



Scanned by CamScanner

MORPHOLOGICAL FEATURES OF TOAD [SPECIMEN A]

1. Possession of streamline body for movement in water.
2. Possession of sticky tongue to catch their prey.

ECOLOGICAL FEATURES OF ALL SPECIMEN SPECIMEN A [TOAD]

1. Moist skin for respiration
2. Presence of fore & hind limbs for hopping [movement]

SPECIMEN B [SNAKE]

1. They possess venom for fighting against predators.
2. Flexible body for easy movement.

(1/2)

SPECIMEN C [BIRD]

1. They possess streamline body for flight
2. They possess feather & scale to prevent loss of water & heat.

SPECIMEN D [ALBINO RAT]

1. Possession of whiskers for sensitivity
2. Possession of claws for catching prey.

MORPHOLOGICAL FEATURES OF SNAKE [SPECIMEN B]

1. Possession of long tongue for catching prey.
2. They go through moulting after development.

MORPHOLOGICAL FEATURES OF ALBINO RAT [SPECIMEN D]

1. Possession of fore & hind limbs for running.
2. Possession of tail for sensitivity.

Cf 134

DIFFERENCE IN IMPORTANCE BTW CONTOUR/VANE AND DOWN FEATHER BY ITS INTERPRETATION

CONTOUR / VANE FEATHER	DOWN FEATHER
1. Long rachis	Short rachis
2. Presence of posterior and superior umbilicus	Absence of umbilicus

IMPORTANCE OF CONTOUR / VANE FEATHER.

1. For temperature regulation
11. It is used to ~~aid~~ flight.



IMPORTANCE OF DOWN FEATHER.

1. It is also used for flight.
11. It is used to cover their skin for protection.

Scanned by CamScanner

PRACTICAL VII – THE BIRDS

Introduction:

CLASS: AVES

1. They are warm blooded
2. The body is usually spindle shaped with four divisions: head, neck, trunk, and tail. Neck disproportionately long for balancing and food gathering.
3. Sexes are separate, females with left ovary and oviduct only.
4. Possession of fully ossified skeleton with air cavities.
5. Epidermal covering of feathers and leg scales, thin integument of epidermis and dermis, no sweat glands.
6. Limbs paired with the fore limbs usually modified for flying.

Examples: Ostrich (*Struthio camelus*), Eagle, Hawk, Vulture, Falcon, Condors, Buzzards, Quail, Grouse, Pheasant, Turkey, Domestic fowl, Cranes, Rails, Woodcocks, Sand pipers, Pigeon, Dove, Parrot, Parakeet, Roadrunners, Owl, Goatsuckers, Night hawks, Mouse birds, King fishers, Hornbills, Wood peckers, Honey guides, etc.

There are two main types of the modern birds:

1. Flightless birds e.g. Ostrich, Emu, Kiwi etc.
2. Flying birds e.g. Eagle, Dove, Vulture, etc.

The Pigeon: External Features

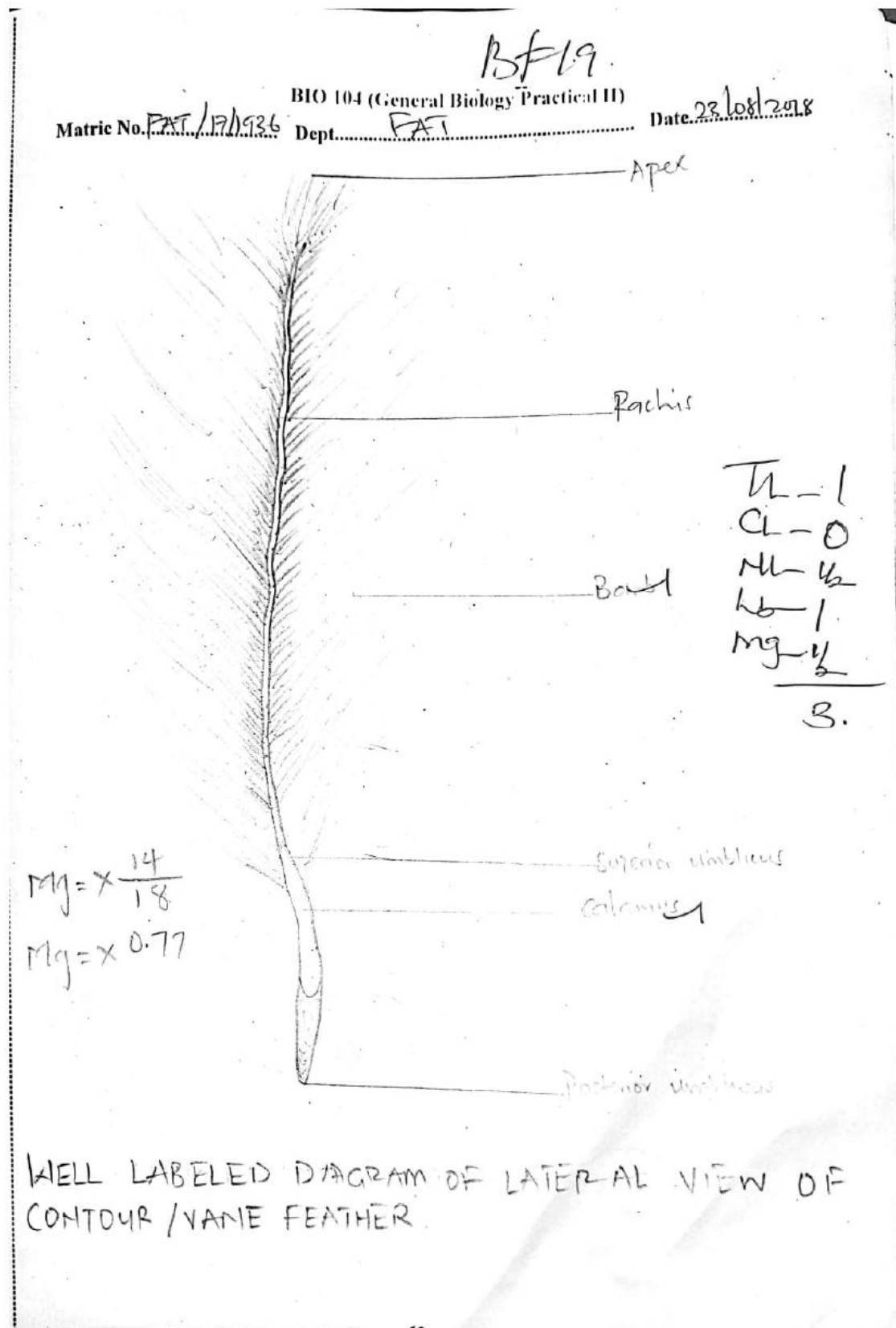
General

The general shape of the body of a bird is entirely different from any vertebrate hitherto studied. The modifications are due to the adoption of bipedal gait and the habit of flight. The body is carried well off the ground as the animal stands on its two hind limbs. When the bird is standing upright, the center of gravity tends to be well in front of the two hip joints but over the toes.

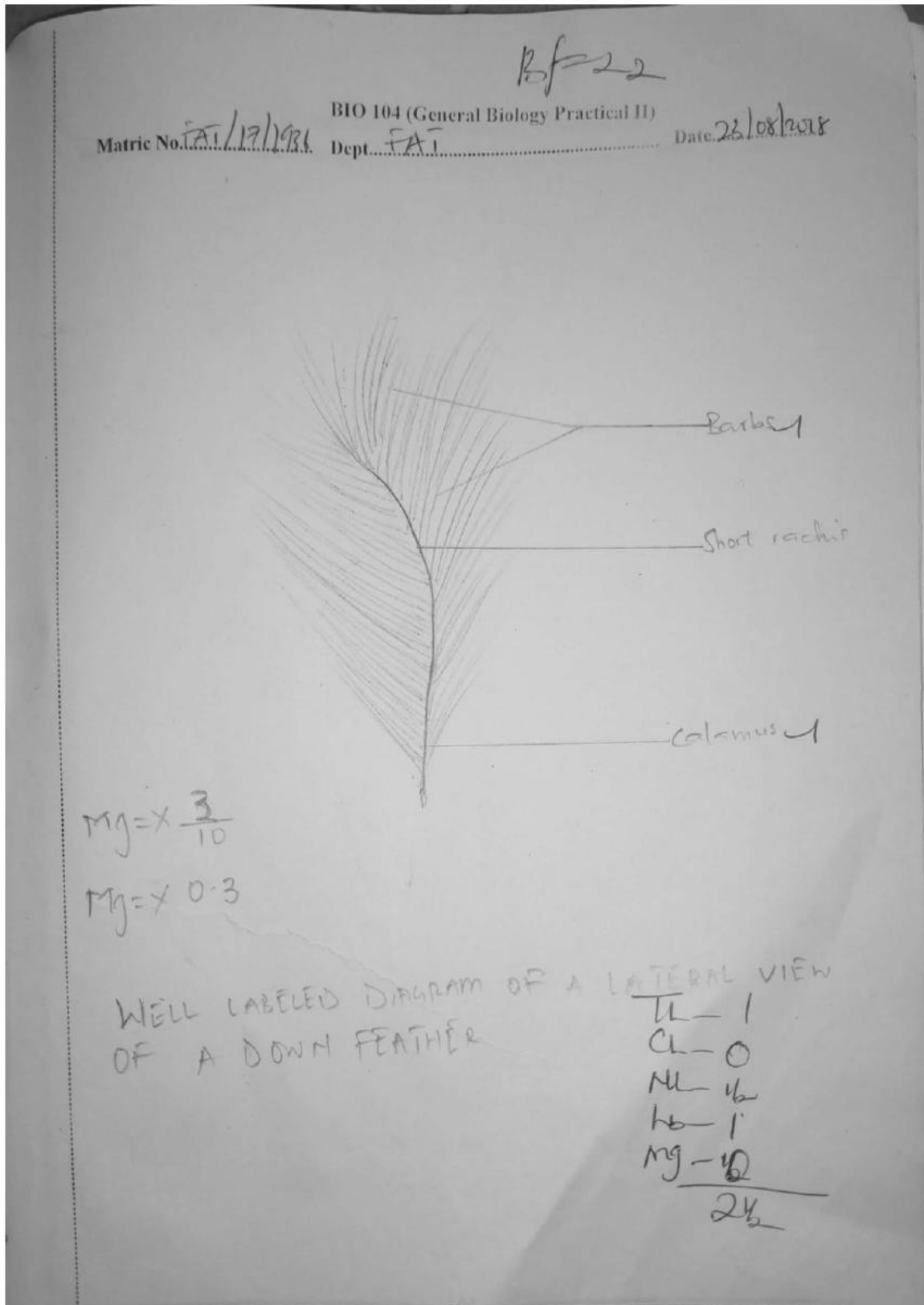
The head is separated from the trunk by well-marked neck. The trunk is deep dorsoventrally. The tail region is very short. The mouth is situated at the anterior end of the body. The horny covering of the jaws form the beak.

The external nares lie at the posterior end of the upper beak. Just above the nares is a swollen patch of skin, usually white in colour called the **cere**.

Immediately behind the mouth are the eyes. This is provided with two eyelids; there is also a flap of transparent skin which can be drawn over the eye from the upper anterior border, the nictating membrane. The nictating membrane can easily be pulled down by a pair of forceps.



Scanned by CamScanner



Scanned by CamScanner

Paper 8 of 10 — BAMBOY 100L 2nd semBIO 104BIO 104 CBE BAMBOY Questions with Correct Answers Marked.pdf

Lecture/Notes · Google Drive harvest

BIO 104 CBE – MCQ Questions with Correct Answers Marked

1. In an ocean floor where there is little or no penetration which cause darkness, the organisms adapt through?

- A) a large mass
- B) a small body size
- C) well developed sense organs ✓
- D) quick movements

2. Which is not a primary consumer?

- A) grasshopper
- B) lizard
- C) rabbit
- D) snake ✓

3. Thermistor is used to measure _____ in an aquatic environment

- A) pressure
- B) current
- C) depth
- D) temperature ✓

4. Fish belongs to class _____

- A) Pisce
- B) Pisces ✓

5. Nuchal crest is _____ when frog is in an aggressive state

- A) lowered
- B) strong
- C) weak
- D) raised ✓

6. Poison on toad are

- A) relatively few on the gland
- B) abundant on all the glands ✓

C) there is no poison gland

7. Egg of lizard is covered by _____

- A) coat
- B) layer
- C) shell ☒
- D) gel

8. Snakes mostly live on

- A) Terrestrial ☒
- B) Aquatic
- C) Terrestrial and Aquatic
- D) None

9. Is an example of Mastigophora

Answer: Euglena ☒

10. What's the function of Phytoplankton

- A) Tertiary consumer
- B) Secondary consumer
- C) Primary consumer
- D) Primary producer ☒

11. Does Snake have Teeth?

Answer: Yes ☒

12. What type of Dentition does Lizard have.....

- A) Homodont Dentition ☒
- B) Heterodont Dentition

13. The Hind Limb of Toad is

- A) Short and stout ☒
- B) Long and lankier
- C) Long and stout

14. What do reptiles use for respiration

- A) Lungs ☒
- B) Nostril
- C) Gills
- D) None

15. Diagram of Euglena
Answer: Euglena ☒

16. Diagram of Paramecium
Answer: Paramecium ☒

17. Diagram of Amoeba
Answer: Amoeba ☒

18. The tympanum in frog is located where
- A) Between the head and the body gland
 - B) Between the parotoid gland and hind limb
 - C) Between the wart
 - D) Between the eyes and parotid gland ☒

19. Tilapia Fish reproduce by....
- A) Asexual reproduction
 - B) Sexual reproduction ☒
 - C) Egg laying
 - D)

20. Which of the following organisms are cartilaginous?
- A) Skates ☒
 - B) Rays ☒
 - C) Mackerel
 - D)

21. Secchi disc is used to check?
- A) Turbidity ☒
 - B) Temperature

22. In an ocean floor where there is little or no penetration which cause darkness, the organisms adapt through?

- A) a large mass
- B) a small body size
- C) well developed sense organs ☒
- D) quick movements

23. Lizards are hermaphrodite.

True / False ☒ (False)

24. Phytoplanktons serve as..... in aquatic food chain.

Answer: Primary producer ☒

25. Tilapia is an example of - animal

- A) Exoskeleton
- B) Hydrostatic
- C) Vertebrate ☒

26. Which of the following has a slippery skin?

- A) Frog ☒
- B) Toad
- C) Newt
- D) None of the above

27. Fish reproduce through

- A) Sexual reproduction ☒
- B) Asexual reproduction
- C) Egg laying
- D) None of the above

28. The role of phytoplankton in the aquatic food chain?

- A) Secondary consumer
- B) Primary consumer
- C) Primary Producer ☒
- D) Decomposer

29. The immature stage of a toad is

- A) Terrestrial like the adult
- B) Terrestrial with tail
- C) Aquatic without tail
- D) Aquatic with tail ☒

30. The black spot on a toad is called

Answer: Wart ☒

31. What is the scientific name of Earthworm

- A) *Lumbricus terrestris* ☒
- B) *Lumbricus gigantica*
- C) *Actina*
- D) *Neresis*

32. The poison on toad

- A) Relatively few in glands
- B) Abundant in glands ☒
- C) Absent in glands
- D) No poison glands in toads

33. The nuchal crest in lizard is responsible for

- A) Movement of head
- B) Reproduction ☒
- C) Locomotion

34. Diagram of *Euglena* – what is the function of part a (flagellum)

- A) Reproduction
- B) Respiration
- C) Locomotion ☒
- D) Irritability

35. Diagram of *Amoeba* – what protozoan is this

- A) *Paramecium*
- B) *Amoeba* ☒
- C) *Euglena*
- D) *Chlamydomonas*

36. Diagram of Paramecium – what is the labelled part I

Answer: Anal pore ☒

37. Which of the following is true about snakes

- A) Do not have legs for walking ☒
- B) Body is streamlined for walking
- C) Have legs for walking

38. Which of the following is a rhizopoda

- A) Amoeba ☒
- B) Paramecium
- C) Plasmodium
- D) Euglena

39. What is the specimen in the diagram above

- A) Garden snail ☒
- B) Slug
- C) Holothuroid
- D) Actina

40. Tympanum in toad is located

- A) On the hind limb
- B) Between the forearm and arm
- C) Between the eyes and parotid gland ☒
- D) On the head

41. Which of the following feather is abundant in birds

- A) Contour ☒
- B) Down
- C) Filoplume
- D) Quill

42. Function of Gular Fold in Lizard?

Answer: Display / defense (attack enemies) ☒

43. Habitat where snakes mostly reside?

Answer: Terrestrial ☒

44. Natural habitat of Euglena?

Answer: Aquatic ☒

45. Function of phytoplankton in the sea?

Answer: Producers ☒

46. Function of crown in birds?

Answer: Thermoregulation ☒

47. What phylum does Euglena belong to?

Answer: Protista ☒

48. Streamline body of earthworm is adapted to what kind of habitat?

Answer: Narrow burrows and soil ☒

49. Which of the following is an example of Mastigophora?

A) Amoeba

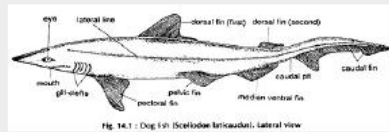
B) Paramecium

C) Plasmodium

D) Euglena ☒

Paper 9 of 10 — BAMBOY 100L 2nd semBIO 104BIO 104 Dogfish.pdf

Lecture/Notes · Google Drive harvest



the external features of dogfish
with the help of suitable diagrams.

Shape, Size and Colour:

Dogfish (*Scoliodon*) has a long, laterally compressed spindle-shaped body tapering at both ends. The full grown specimen measures from 30 to 60 cm in length. The colour of the body is dark grey above and pale white beneath, while the portions of the caudal fin are more or less dark. Body surface is rough due to backwardly directed spines of placoid scales embedded in the dermis.

Division of Body:

The body is divisible into head, trunk and tail, though there are no distinct boundaries between these regions.

(i) Head:

The head is strongly compressed dorso-ventrally and is produced in front into a wedge-shaped snout or rostrum.

(ii) Trunk:

The trunk is almost elliptical in transverse section. Its thickest part lying in front of the middle of the body. The trunk gradually tapers behind into the tail.

(iii) Tail:

The tail is laterally compressed and is bent upwards at a small angle and fringed with a caudal fin. Such a tail is known as heterocercal tail.

Fins:

Dogfish (*Scoliodon*) is provided with two sets of fins which are flattened expansions of the skin supported by cartilaginous rods and horny fin rays- these are unpaired or median fins and paired lateral fins.

(i) Median Fins:

The median fins are two dorsal

JOMO

fins, a ventral or anal fin and a caudal fin. The first dorsal fin is large and triangular in shape and is situated a little in front of the middle of the body. It has a basal lobe. The second dorsal fin is also triangular in outline but is very small and is situated midway between the first dorsal and the tip of the tail. The caudal fin extends along the dorsal and ventral surfaces of the tail in the median line and forms a dorsal and ventral lobe.

The dorsal lobe is very much reduced and forms only a low ridge along the greater part of the upper surface but the ventral lobe is well developed and is divided into two parts, the anterior part being much larger and more extensive than the posterior. The ventral or anal fin is situated in the mid-ventral line about 5.0 cm in front of the caudal fin, more or less opposite the second dorsal. Each of the two dorsals and the ventral fin are produced behind into long and narrow fleshy processes called basal lobes.

(ii) Lateral Fins:

The lateral fins are paired pectoral and pelvic fins. The large pectoral fins originate from the ventro-lateral margins of the body immediately behind the gill-clefts.

The pelvic fins are much smaller than the pectoral fins and arise close together from the ventral surface at the junction of the trunk and tail. In the male *Scoliodon*, the medial part of each pelvic fins is produced into a dorsally grooved, stiff rod-like clasper (intromittent organ) used during copulation. All the fins are directed backwards a feature which is advantageous in forward progression.

(iii) Eyes:

On each side of the head is a large circular eye. Each eye has two poorly-formed immovable eye-lids and nictitating membrane located antero-ventrally. The pupil is narrow and vertical.

Body Apertures:

The following important apertures are present on the body surface:

(i) Mouth:

On the ventral side of the head is a wide crescentic mouth. It is bounded by upper and lower jaws. Each jaw is armed with one or two rows of sharp backwardly directed teeth, having smooth not-serrated edges for holding and tearing the prey. In catching its prey the shark brings its jaws into play by raising the snout and thrusting the mouth forward.

(ii) Nostrils:

On the anterior side of the mouth are present two crescentic apertures the nostrils leading into olfactory sacs which do not open into the mouth cavity. A small fold of skin from the anterior edge partially covers each nostril. The nostrils are only olfactory and have no respiratory function.

(iii) Gill-Clefts:

Behind the eyes are situated a series of vertical slits, five on each side, called branchial or gill-clefts. These apertures lead into the gill-pouches and thence into pharynx. These are respiratory in function.

(iv) Cloacal Aperture:

The cloacal aperture is an elongated median opening at the root of the tail between two pelvic fins. It leads into the cloaca into which the intestine and urinary and genital ducts open. Anus lies anteriorly and in the cloaca and a cone-shaped papilla bearing urinary pore in female and urino-

JOMO

genital pore in male located behind the anus.

(v) Abdominal Pores:

There is a pair of openings situated on the elevated papillae on either side of the cloaca. These are the abdominal pores. Through these pores the coelom communicates with the exterior.

(vi) Caudal Pit:

At the root of the tail, just in front of the caudal fins, is a small dorsal and a ventral shallow depressions, the caudal pits. The caudal pits are the characteristic feature of the genus Dogfish (*Scoliodon*).

(vii) Lateral Line and Pores:

A faint line runs on either side of the body extending from the head to the posterior end of the tail, this is called the lateral line. It marks the position of an underlying canal which runs along each side of the body and contains special receptor organs. The lateral line canal extends anteriorly into the head where it branches into several canals; at intervals these canals open to the exterior through minute pores.

(viii) Ampullary Pores:

On the head and snout open several groups of minute ampullary pores of the receptors called ampullae of Lorenzini. When pressed they exude mucus.

Sexual Dimorphism:

In the male the inner margins of the pelvic fins bear a pair of rod-shaped copulatory organs called claspers or myxopterygia, they are modified parts of the pelvic fins.

Each clasper is a stiff rod-like appendage having a groove on its dorsal surface which leads into a cavity, the siphon, beginning at the base of the clasper.

Skin:

The body is invested by an outer

leathery covering called skin or integument. In Dogfish (Scoliodon), the skin or integument consists of two layers- an outer ectodermal epidermis and an inner mesodermal dermis.

1. Epidermis:

The epidermis is composed of many layers of epithelial cells amongst which are interspersed numerous unicellular mucous glands secreting mucus for lubricating the body surface. Innermost layer of cells rest on a basement membrane and called stratum germinativum.

2. Dermis:

The dermis comprises dense areolar connective tissue mixed with smooth muscle fibres, blood capillaries, pigment cells and nerves. Dermis or corium is divided into an outer layer having few loose fibres called the stratum laxum, and an inner layer having compact fibres called stratum compactum. Just below the epidermis are found pigment cells, the melanophores or chromatophores.

The dermis is firmly attached to the underlying muscles. In a fresh specimen the skin is slimy, but in preserved specimens the slimy mucus is generally removed and the skin becomes rough. This roughness of the skin is due to the presence of closely lying minute dermal denticles called placoid scales which are arranged in regular oblique rows and form the exoskeleton of the shark covering the entire surface of the body and even parts of the fins.

Functions of Skin:

In Dogfish (Scoliodon), the skin serves the following functions:

- (i) Like a wrapper, it protects internal organs against mechanical injuries,

- (ii) Secretion of slimy mucus makes body surface slippery and difficult to catch by predators, minimises friction in locomotion and resists entry of microorganisms into body.
- (iii) Its colouration imparts camouflage, as the fish blends with silvery surface of water when seen from below and with the dark bottom when seen from above, by predators,
- (iv) Skin receptors help in reacting to changes in the surrounding.

Paper 10 of 10 — BAMBOY 100L 2nd semBIO 104BIO 104 PQs by BAMBOY.pdf

Lecture/Notes · Google Drive harvest

BIO 104 CBE -BAMBOY questions with answers marked

1. The rat belongs to which class?
 - a) Mammalia ✓
 - b) Reptilia
 - c) Amphibia
 - d) Aves

2. Which organ in a rat is primarily responsible for detoxification of harmful substances?
 - a) Heart
 - b) Liver ✓
 - c) Kidney
 - d) Spleen

3. Which of the following is true about the rat's digestive system?
 - a) It lacks a small intestine
 - b) The caecum is large and functions in cellulose digestion ✓
 - c) It has a single-chambered stomach similar to humans
 - d) It has gills for respiration

4. Snakes belong to which class?
 - a) Mammalia
 - b) Reptilia ✓
 - c) Amphibia
 - d) Aves

5. Which organ in snakes is responsible for sensing heat in prey?
 - a) Eyes
 - b) Pit organs ✓
 - c) Lungs
 - d) Liver

6. Which structure in crabs helps in respiration?
 - a) Trachea
 - b) Book lungs
 - c) Gills ✓
 - d) Spiracles

7. Millipedes belong to which class?
 - a) Chilopoda
 - b) Diplopoda ✓
 - c) Insecta

d) Arachnida

8. The main defense mechanism of millipedes is:

- a) Biting
- b) Secreting toxic fluids ✓
- c) Jumping
- d) Camouflage

9. How many pairs of legs are typically found in a single body segment of a millipede?

- a) 1 pair
- b) 2 pairs ✓
- c) 3 pairs
- d) 4 pairs

10. Birds belong to which class?

- a) Mammalia
- b) Reptilia
- c) Aves ✓
- d) Amphibia

11. The hollow bones in birds primarily aid in:

- a) Digestion
- b) Flight ✓
- c) Reproduction
- d) Thermoregulation

12. The function of the gizzard in birds is:

- a) Filtering blood
- b) Grinding food ✓
- c) Oxygen absorption
- d) Egg production

13. Flatworms belong to which phylum?

- a) Nematoda
- b) Platyhelminthes ✓
- c) Annelida
- d) Arthropoda

14. Which organ in flatworms is used for excretion?

- a) Flame cells ✓
- b) Nephridia
- c) Malpighian tubules
- d) Gills

15. Flatworms are generally:

- a) Coelomates
- b) Acoelomates ☒
- c) Pseudocoelomates
- d) Diploblastic

16. Roundworms belong to which phylum?

- a) Annelida
- b) Nematoda ☒
- c) Platyhelminthes
- d) Arthropoda

17. The body cavity of roundworms is called:

- a) Coelom
- b) Pseudocoelom ☒
- c) Acoelom
- d) None

18. Roundworms have a complete digestive system, which means:

- a) Only a mouth exists
- b) Only an anus exists
- c) Both mouth and anus are present ☒
- d) Neither mouth nor anus

19. Tilapia is a:

- a) Mammal
- b) Amphibian
- c) Fish ☒
- d) Reptile

20. Tilapia uses which organ for respiration?

- a) Lungs
- b) Gills ☒
- c) Skin
- d) Spiracles

21. The lateral line in Tilapia functions in:

- a) Breathing
- b) Detecting vibrations and movement in water ☒
- c) Swimming speed
- d) Digestion

22. Dogfish is a type of:

- a) Bony fish

- b) Cartilaginous fish ✓
- c) Amphibian
- d) Reptile

23. Dogfish lacks:

- a) Fins
- b) Jaws
- c) Bones ✓
- d) Gills

24. The function of the spiral valve in dogfish is:

- a) Respiration
- b) Increasing digestive surface area ✓
- c) Reproduction
- d) Locomotion

25. Amoeba moves using:

- a) Cilia
- b) Flagella
- c) Pseudopodia ✓
- d) Fins

26. The structure in Amoeba responsible for osmoregulation is:

- a) Nucleus
- b) Food vacuole
- c) Contractile vacuole ✓
- d) Cytoplasm

27. Amoeba reproduces by:

- a) Binary fission ✓
- b) Budding
- c) Spore formation
- d) Fragmentation

28. Trypanosome belongs to which phylum?

- a) Sarcomastigophora ✓
- b) Euglenophyta
- c) Ciliophora
- d) Amoebozoa

29. Trypanosome is known to cause:

- a) Malaria
- b) Sleeping sickness ✓
- c) Amoebiasis

d) Cholera

30. Trypanosomes move using:

- a) Pseudopodia
- b) Cilia
- c) Flagella ✓
- d) Undulating membrane only

31. Euglena is:

- a) Animal-like
- b) Plant-like
- c) Both plant and animal-like ✓
- d) Fungal-like

32. The light-sensitive organ in Euglena is called:

- a) Nucleus
- b) Eyespot ✓
- c) Chloroplast
- d) Flagellum

33. Euglena stores food in the form of:

- a) Glycogen
- b) Starch ✓
- c) Lipids
- d) Protein

34. Frogs belong to which class?

- a) Mammalia
- b) Reptilia
- c) Amphibia ✓
- d) Aves

35.. Frogs respire through:

- a) Gills only
- b) Lungs only
- c) Skin, lungs, and mouth lining ✓
- d) Spiracles only

36. The structure responsible for vocalization in frogs is:

- a) Nostrils
- b) Vocal sacs ✓
- c) Larynx
- d) Tympanum

37. Natural habitat of Euglena?
- A) Aquatic ☒
 - B) Terrestrial
 - C) Desert
 - D) None of the above
38. Function of phytoplankton in the sea?
- A) Decomposer
 - B) Consumers
 - C) Producers ☒
 - D) Parasites
39. Function of crown in birds?
- A) Protection
 - B) Thermoregulation ☒
 - C) Camouflage
 - D) Breeding display
40. What phylum does Euglena belong to?
- A) Protozoa
 - B) Protista ☒
 - C) Plantae
 - D) Animalia
41. Streamline body of earthworm is adapted to what kind of habitat?
- A) Narrow burrows and soil ☒
 - B) Aquatic environment
 - C) Desert sand
 - D) Inside host body
42. Which of the following is an example of Mastigophora?
- A) Amoeba
 - B) Paramecium
 - C) Plasmodium
 - D) Euglena ☒
43. In an ocean floor where there is little or no penetration which cause darkness, the organisms adapt through?
- A) a large mass
 - B) a small body size
 - C) well developed sense organs ☒
 - D) quick movements
44. Which is not a primary consumer?

- A) Grasshopper
- B) Lizard ☒
- C) Rabbit
- D) Snake

45. Thermistor is used to measure _____ in an aquatic environment

- A) Pressure
- B) Current
- C) Depth
- D) Temperature ☒

46. Fish belongs to class _____

- A) Pisce
- B) Pisces ☒
- C) Amphibia
- D) Chondrichthyes

47. Nuchal crest is _____ when frog is in an aggressive state

- A) Lowered
- B) Strong
- C) Weak
- D) Raised ☒

48. Poison on toad are

- A) Relatively few on the gland
- B) Abundant on all the glands ☒
- C) There is no poison gland
- D) Spread on skin surface only

49. Egg of lizard is covered by _____

- A) Coat
- B) Layer
- C) Shell ☒
- D) Gel

50. Snakes mostly live on

- A) Terrestrial ☒
- B) Aquatic
- C) Terrestrial and Aquatic
- D) None

51. What's the function of phytoplankton?

- A) Tertiary consumer
- B) Secondary consumer

- C) Primary consumer
- D) Primary producer ☒

52. Does Snake have teeth?

- A) Yes ☒
- B) No

53. What type of dentition does lizard have?

- A) Homodont dentition ☒
- B) Heterodont dentition
- C) Diphyodont dentition
- D) Polyphyodont dentition

54. The hind limb of toad is

- A) Short and stout
- B) Long and lankier
- C) Long and stout ☒
- D) Reduced and weak

55. What do reptiles use for respiration?

- A) Lungs ☒
- B) Nostril
- C) Gills
- D) None

56. Diagram of Euglena

- A) Euglena ☒
- B) Paramecium
- C) Amoeba
- D) Chlamydomonas

57. The tympanum in frog is located where?

- A) Between the head and the body gland
- B) Between the parotoid gland and hind limb
- C) Between the wart
- D) Between the eyes and parotid gland ☒

58. Tilapia fish reproduce by.....

- A) Asexual reproduction
- B) Sexual reproduction ☒
- C) Egg laying
- D) Budding

59. The nuchal crest in lizard is responsible for

A) Movement of head ☒

B) Reproduction

C) Locomotion

D) Protection

60. Diagram of Euglena – what is the function of part a (flagellum)?

A) Reproduction

B) Respiration

C) Locomotion ☒

D) Irritability

61. Lizards are hermaphrodite.

A) True

B) False ☒

62. Tilapia is an example of - animal

A) Exoskeleton

B) Hydrostatic

C) Vertebrate ☒

D) Invertebrate

63. Which of the following has a slippery skin?

A) Frog ☒

B) Toad

C) Newt

D) None of the above

64. Fish reproduce through

A) Sexual reproduction ☒

B) Asexual reproduction

C) Egg laying

D) Budding

65. The immature stage of a toad is

A) Terrestrial like the adult

B) Terrestrial with tail

C) Aquatic without tail

D) Aquatic with tail ☒

66. The black spot on a toad is called

A) Nuchal crest

B) Wart ☒

C) Tympanum

D) Parotoid gland

67. What is the scientific name of Earthworm?

- A) *Lumbricus terrestris* ✓
- B) *Lumbricus gigantica*
- C) *Actina*
- D) *Neresis*

68. Diagram of Amoeba – what protozoan is this?

- A) *Paramecium*
- B) *Amoeba* ✓
- C) *Euglena*
- D) *Chlamydomonas*

69. Which of the following is a rhizopoda?

- A) *Amoeba* ✓
- B) *Paramecium*
- C) *Plasmodium*
- D) *Euglena*

70. Which of the following is true about snakes?

- A) Do not have legs for walking ✓
- B) Body is streamlined for walking
- C) Have legs for walking
- D) Use appendages for crawling

71. What is the specimen in the diagram above?

- A) Garden snail ✓
- B) Slug
- C) Holothuroid
- D) *Actina*

72. Which of the following feather is abundant in birds?

- A) Contour ✓
- B) Down
- C) Filoplume
- D) Quill

73. Function of Gular Fold in Lizard?

- A) Display / defense ✓
- B) Respiration
- C) Feeding
- D) Movement

74. The red spot found on the head of tilapia fish functions mainly for

- A) Sexual attraction ☒
- B) Respiration
- C) Excretion
- D) Thermoregulation

75. The contractile vacuole in Amoeba regulates

- A) Osmoregulation ☒
- B) Respiration
- C) Excretion
- D) Locomotion

76. Which of the following animals exhibits neoteny?

- A) Toad
- B) Salamander ☒
- C) Lizard
- D) Frog

77. The gular fold in male lizard is most prominent during

- A) Mating display ☒
- B) Feeding
- C) Resting
- D) Respiration

78. Which of the following structures in Euglena is photosensitive?

- A) Chloroplast
- B) Pellicle
- C) Eyespot (stigma) ☒
- D) Flagellum

79. In Paramecium, locomotion is achieved by

- A) Flagella
- B) Pseudopodia
- C) Cilia ☒
- D) Tentacles

80. The excreta of birds is called:

- a) Urea
- b) Ammonia
- c) Uric acid ☒
- d) Sweat

81. The body of a snake is covered with _____.

- a) Feathers
- b) Scales ☒

- c) Fur
- d) Shell

82. The ear of a lizard is _____.

- a) Absent
- b) Covered with fur
- c) A tympanic membrane ☒
- d) A long external ear

83. The ear of a lizard has a _____.

- a) Tympanic membrane
- b) Pinna
- c) Feathers
- d) External auditory canal

84. The rat is considered a _____ in relation to human activity.

- a) Predator
- b) Pest ☒
- c) Symbiont
- d) Host

85. The crab is an example of a _____.

- a) Mollusc
- b) Arthropod ☒
- c) Annelid
- d) Amphibian

86. Earthworms belong to the phylum _____.

- a) Arthropoda
- b) Mollusca
- c) Annelida ☒
- d) Echinodermata

87. Earthworms improve soil fertility mainly through _____.

- a) Predation
- b) Burrowing and decomposition ☒
- c) Carrying seeds
- d) Producing toxins